CRANSWICK MANUFACTURING STANDARD

CRANSWICK

Great British Taste

ISSUE 02

DIRECTOR'S STATEMENT

"GOOD IN EQUALS GOOD OUT"

THE SIMPLE MANTRA THAT INSPIRED A GROUP OF FARMERS TO START CRANSWICK IN 1975, STRIVING FOR GREAT BRITISH TASTE. THEIR BEGINNINGS WERE HUMBLE BUT THEIR AMBITIONS WERE ANYTHING BUT.

In the 1980's, Cranswick moved into food production, bringing one of the first 'farm to fork' offerings. Today, we're one of the largest food producers in Britain, remaining true to our ethos of creating authentically made British food, without compromise.

Through shrewd investment, organic growth and good old-fashioned Yorkshire pragmatism, we've built a business focused on flavour, authentic production and quality. Our technical vision is to champion world class food quality, safety & sustainability across our entire supply chain with integrity, trust and innovation.

We aim to guarantee the quality, safety, legality and authenticity of our food, alongside exceptional animal welfare standards with simple, practical, consistent & effective systems & processes across our group in order to exceed customer expectations in all that we do.

I am delighted to support this second edition of the CMS which is a step on from the first, incorporating revised editions of both third party and customer requirements, our own best practice standards and a new and improved section for our ready to eat sites, all of which we believe lead the way within our industry.

By ensuring that our sites operate to this standard, we can cultivate the best food safety culture amongst our colleagues, relating to the importance of and the confidence in the product safety systems, processes and procedures used by the site which will in turn guarantee that we will produce food that is safe, legal and of the best quality at all times.

I would like to thank the team that has worked hard to produce this document, the team that will audit our sites to assist them in achieving compliance, and of course, the site teams that will work hard to ensure that we deliver these standards every day."

JACKIE CARTER
GROUP TECHNICAL DIRECTOR

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THE CRANSWICK MANUFACTURING STANDARD (CMS) SCOPE

HOW THE CRANSWICK MANUFACTURING STANDARD (CMS) IS LAID OUT:

Each main Cranswick Manufacturing Standard section or subsection of the requirements begins with a statement of intent. This states the expected outcome of compliance with the requirements of the section. This forms part of the audit process and all sites must comply with the statements of intent.

Below the statements of intent in the tables are more specific and detailed clauses that if are achieved appropriately will help achieve the outcome of the statement of intent. All requirements form the part of the audit process.

COLOUR CODING OF REQUIREMENTS:

Throughout the Cranswick Manufacturing Standard, a colour coding system is used to determine key areas and where the requirements will be audited. The following colour coding is in place:

🔴	Audit of production facilities and good manufacturing practice
🔵	Audit of records, systems and documentation
🔵🔴	Requirements assessed in both areas

FUNDAMENTAL REQUIREMENTS:

Within the Cranswick Manufacturing Standard there are several requirements which have been allocated as 'Fundamental Requirements'. These are highlighted at the start of the sections Statement of Intent. The "Fundamental Requirements" relate to systems that are crucial to the establishment and operation of an effective food safety and quality manufacturing operation. The requirements that are fundamental are listed below:

- **1** Senior Management Commitment, Leadership and Continual Improvement (Whole Section)
- **2** HACCP (Whole Section)
- **3.3** Internal Audits
- **3.4** Supplier and Raw Material Approval and Performance Monitoring
- **3.6** Incident, Non-Conformity, Corrective Action, Root Cause Analysis and Preventative Action
- **3.8** Traceability
- **4.3** Layout Product Flow and Segregation
- **4.9** Chemical and Physical Product Contamination Control: Raw Material Handling, Preparation, Processing, Packing and Storage Areas
- **4.11** Housekeeping and Hygiene
- **5.3** Management of Allergens and Speciation Control
- **6.1** Control of Operations

6.2 CCP Process Control
6.3 Label and Pack Control
6.4 Quantity – Weight, Volume and Number Control
7.1 Training: Raw Material Handling, Preparation, Processing, Packing and Storage
8.1 Layout, Product Flow and Segregation in High-Risk, High-Care and Ambient High-Care Zones
8.5 Housekeeping And Hygiene in High-Risk and High-Care Zones
9.1 HACCP (Traded Products)

A substantial failure to comply with the statement of intent of a fundamental requirement will result in a red rated audit when a major non-conformance is raised against the fundamental. When a Critical Non-Conformance is raised the audit will become an automatic purple rated audit and the Critical Non-Conformance Flow Diagram in the Group Internal Audit Procedure will need to be followed. For further internal audit requirements please refer to the Group Internal Audit Procedure.

ADDITIONAL REQUIREMENTS:

The requirements in sections 1-7 shall be applied to manufacturing operations. Any site that has High-Risk, High-Care or Ambient High-Care facilities must meet the requirements of section 8.

Where a site also handles traded products (i.e, products not manufactured or processed on the site but bought in and sold by the site), there are additional requirements for these products as detailed in section 9. The site must meet the requirements of the associated section.

1 SENIOR MANAGEMENT LEADERSHIP, CULTURE AND CONTINUAL IMPROVEMENT

1.1 SENIOR MANAGEMENT COMMITMENT, LEADERSHIP AND CONTINUAL IMPROVEMENT

FUNDAMENTAL: The site's senior management shall demonstrate that they are fully committed to Cranswick policies and the implementation of the requirements in this standard. There shall be resources in place to facilitate the continual improvement of the management systems required by this standard.

CLAUSE	REQUIREMENTS
1.1.1	Cranswick shall issue policies that communicate the company's position on areas which affect the business. These shall be communicated to each site and published on the company website. Each site shall display current versions of company policies in staff areas with Site Directors countersigning Tier 1 policies. Sites shall ensure that relevant policies are also made available to visitors and contractors.
1.1.2	Cranswick shall create group strategies which supports its position outlined in each policy. Each site is required to develop its own strategy with clearly defined objectives that meet the requirements and objectives set out within the company's policies and strategies. The site strategy shall have defined activities with timescales and have measurable outcomes which can be monitored to ensure that they are effective. These objectives shall be: • documented and include targets or clear measures of success • clearly communicated to all staff • monitored and results reported at least quarterly to site senior management and all staff Activities shall be reviewed at appropriate planned intervals.
1.1.3	The site's senior management shall demonstrate leadership and commitment and continuing improvement with respect to the site's: • Food Safety, Authenticity and Quality Management System • Food Safety and Quality Culture • Occupational Health & Safety Management System • Environmental Management System • Energy Management System • Sustainability initiatives that support the Environmental and Energy Management Systems This shall be achieved by: • taking responsibility and accountability for the prevention of work-related injury and ill heath, as well as the provision of safe and healthy workplaces and activities • allocating the appropriate resources needed to establish, implement, maintain and continually improve the management systems listed above and to meet the requirements of this standard • supporting other relevant management roles to demonstrate their leadership as it applies to their areas of responsibility • developing, leading and promoting a culture on site that supports the intended outcomes of this standard (see section 1.2).

1.1.4 The site shall display information relating to the confidential reporting system used by Cranswick to enable staff to report concerns relating to product safety, quality, authenticity and legality, health and safety, and other work-related issues. Posters shall be displayed in staff areas in languages that reflect the workforce employed by the site.

Where customer reporting systems are available, these shall also be displayed.

The mechanism (e.g. the relevant telephone number) for reporting concerns must be clearly communicated to staff and briefed out every 6 months. The reporting system shall also be tested every 6 months.

The company's senior management shall have a process for assessing any concerns raised. Records of the assessment and, where appropriate, actions taken, shall be documented.

1.1.5 The company's senior management shall have a system in place to ensure that the site is kept informed of and reviews:

- scientific and technical developments
- legal requirements
- industry guidelines or codes of practice
- new risks to authenticity of raw materials
- emerging risks
- all relevant legislation in the country where the product will be sold (where known).

1.1.6 The site shall have access to all relevant customer manufacturing standards and codes of practices via online portals and shall work collaboratively with its customers to ensure it meets the terms and conditions of supply. Where updates are issued, these shall be effectively communicated to relevant stakeholders.

Additionally, the site shall have genuine, original hard copy or electronic versions of the current ISO 14001, ISO 45001, ISO 50001 standards and the BRCGS for Food Safety available, Interpretation Guidelines and position statements. It shall be aware of any changes to the Standard or protocol that are published on the BRCGS website.

1.1.7 The site shall ensure that recertification audits occur on or before the audit due date indicated on the certificate.

The site is required to be registered for unannounced BRCGS recertification audits and shall opt into the unannounced scheme within 3 months from the last audit.

1.1.8 The most senior production or operations manager on site shall participate in the opening and closing meetings of all third-party audits/regulatory inspections. Relevant departmental managers or their deputies shall be available as required.

A member of senior management team on site shall be available during an internal or external audit for a discussion on effective implementation of the food safety and quality plan.

1.1.9 The site's senior management shall ensure that the root causes of any non-conformities against the Standard identified at the previous audit have been effectively addressed to prevent recurrence.

1.1.10 The BRCGS logo and references to certification status shall only be used in accordance with the conditions of use detailed in the audit protocol section (Part III, section 6.7) of the BRCGS for Food Safety.

1.1.11 It is the responsibility of senior management to ensure all legally required registrations are completed.

Where required by legislation, the site shall maintain appropriate registrations with the relevant authorities.

1.2 CULTURE

FUNDAMENTAL: The site's senior management shall define and maintain a culture on site that meets the definitions of culture prescribed by this standard.

CLAUSE	REQUIREMENTS

1.2.1 The site's senior management shall define and maintain a clear plan for the development and continuing improvement of a food safety and quality culture. This shall include:

- defined activities involving all sections of the site that have an impact on product safety
- as a minimum, these activities shall be designed around:
- clear and open communication on product safety
- training
- feedback from employees
- the behaviours required to maintain and improve product safety processes
- performance measurement on activities related to the safety, authenticity, legality and quality of products
- an action plan indicating how the activities will be undertaken and measured, and the intended timescales
- a review of the effectiveness of completed activities. The plan shall be reviewed and updated at least annually, at a minimum.

1.2.2 The site shall have mechanisms in place for consultation and participation with its employees in the development, planning, implementation, performance evaluation and actions for improvement of the site's Health and Safety management system.

1.2.3 The site shall engage with its employees to identify areas of waste and improvement opportunities in the site's processes to improve the site's environmental and energy performance.

1.3 MANAGEMENT REVIEW

FUNDAMENTAL: The site shall have a meeting programme in place to ensure that the performance of each management system is monitored continually with appropriate action taken where issues have been identified.

CLAUSE	REQUIREMENTS

1.3.1 Management review meetings attended by the site's senior management shall be undertaken at appropriate planned intervals, annually as a minimum, to review the site performance against the Standard and objectives set in clause 1.1.2. The review process shall include the evaluation of:

- previous management review action plans and timeframes
- the results of internal, second party and/or third-party audits
- performance of the relevant management system, including trends in non-conformities and corrective actions, results from monitoring systems designed to measure performance
- any objectives that have not been met, to understand the underlying reasons. This information shall be used when setting future objectives and to facilitate continual improvement
- resource requirements.

Review meetings shall evaluate the performance of each management system listed in clause 1.1.3. Where these meetings are held separately, the agenda items above shall be covered.

SECTION 1: SENIOR MANAGEMENT LEADERSHIP, CULTURE AND CONTINUAL IMPROVEMENT

1.3.1 cont.

Food safety and quality management:

- any customer complaints and the results of any customer feedback
- any incidents (including both recalls and withdrawals), non-conformances, corrective actions, out-of-specification results and non-conforming materials
- the effectiveness of the systems for HACCP, food defence and authenticity and the food safety and quality culture plan

Health and safety management:

- changes in external and internal issues that are relevant to the management system (e.g. needs and expectations of interested parties, legal requirements, risks and opportunities)
- any incidents, non-conformances, corrective actions
- opportunities to improve integration of the management system.

Environmental management:

- external and internal issues that are relevant to the environmental management system
- the needs and expectations of interested parties, including compliance obligations
- significant environmental aspects
- extent to which environmental objectives have been achieved
- relevant communications from interested parties, including complaints.

Energy management:

- changes in external and internal issues and associated risks and opportunities that are relevant to the energy management system.

Records of the meeting shall be documented with each item covered in sufficient detail and used to revise the objectives thereby encouraging continual improvement. The decisions and actions agreed within the review process shall be effectively communicated to appropriate staff, and actions implemented within agreed timescales.

1.3.2 The site shall have a demonstrable meeting programme which enables food safety, legality, integrity and quality issues to be brought to the attention of senior management. These meetings shall occur at least monthly.

1.4 ORGANISATIONAL STRUCTURE, RESPONSIBILITIES AND MANAGEMENT AUTHORITY

FUNDAMENTAL: The company shall have a clear organisational structure and lines of communication to enable effective management of operations to meet the requirements of this standard.

CLAUSE	REQUIREMENTS
1.4.1	The site shall have an organisation chart demonstrating the management structure. The responsibilities for the management of activities which ensure food safety, quality, authenticity and legality, health and safety, environment and energy performance shall be clearly allocated and understood by the managers responsible. It shall be clearly documented who deputises in the absence of the responsible person. If the site does not have the appropriate in-house knowledge of food safety, authenticity, legality or quality, the appropriate Divisional Technical Controller and the Head of Food Safety and Quality shall be notified immediately. If required, external expertise may be used, but day-to-day management of the food safety system shall remain the responsibility of the site. Evidence of competence for external expertise should be available for review.
1.4.2	The site shall have a designated, suitably competent representative responsible for the implementation of each management system (defined in 1.1.3) and their adherence to the relevant certification requirements.
1.4.3	The site's senior management shall ensure that all employees are aware of their responsibilities relating to food safety, authenticity and quality, health and safety, energy performance and sustainability. Documented and agreed job descriptions (or working instructions where appropriate) shall be in place. Where documented work instructions exist for activities undertaken, the relevant employees shall have access to these and be able to demonstrate that work is carried out in accordance with the instructions.
1.4.4	The site's senior management shall ensure that all employees are aware of the need to report any risks or any evidence of unsafe or out-of-specification product, equipment, packaging or raw materials, environmental incidents, unsafe acts or hazardous situations or other issues that may affect the sites ability to operate safely and with integrity to a designated manager to enable the resolution of issues requiring immediate action.

SECTION 1: SENIOR MANAGEMENT LEADERSHIP, CULTURE AND CONTINUAL IMPROVEMENT

2 HACCP

2.1 HACCP TEAM, SCOPE & TERMS OF REFERENCE

FUNDAMENTAL: The site shall have a fully implemented and effective HACCP plan incorporating the Codex Alimentarius HACCP principles.

CLAUSE	REQUIREMENTS
2.1.1	The HACCP plan shall be developed and managed by a multi-disciplinary food safety team that includes those responsible for quality assurance, technical management, hygiene, NPD, supply chain/procurement, production operations, engineering and other relevant functions. A member of the site Senior Leadership team shall also be present on the team. The team leader and a deputy shall have an in-depth knowledge of Codex HACCP principles (or equivalent) and be able to demonstrate competence, experience and training. All team leaders and deputies shall be trained to HACCP level 4. The team members shall have specific knowledge of HACCP and relevant knowledge of products, processes and associated hazards. All team members shall be trained to HACCP level 3 as a minimum. In the event of sites not having the appropriate in-house knowledge, external expertise may be used, but day-to-day management of the food safety system shall remain the responsibility of the site. Evidence of competence for external expertise should be available for review. All of this shall be documented within the HACCP study.
2.1.2	The scope of the HACCP plan shall define the products and processes covered ensuring that all products that are manufactured on site have been detailed either separately or within a suitable product group. The HACCP Plan shall cover the entire manufacturing process from raw material intake and storage (inc. packaging), through to dispatch of finished product. Any outsourced processing shall also be highlighted. The scope shall define the physical, chemical, radiological, biological and allergen hazards. Speciation risks shall also be adequately defined. For microbiological risks highlighted; intrinsic food safety controls for each target organism, reference to relevant data detailing the characteristics of the organism and justifications for any control measures shall be detailed. Where toxin or spore forming micro-organisms have been identified, monitoring and control measures shall be included.
2.1.3	The scope shall detail the factors relevant to product safety such as: • composition (raw materials, ingredients, allergens, recipe) • origin of ingredients • physical or chemical properties (e.g. pH, aW) • treatment & processing (e.g. cooking, cooling, freezing, curing and other forms of preservation) • packaging systems (e.g. MAP, vacuum) • storage & distribution (e.g. chilled, frozen or ambient) • target shelf life under defined storage and usage conditions • instructions for intended use • potential for known customer misuse, e.g. storage, preparation • target consumer groups (including any vulnerable groups, e.g. infants, elderly, allergen sufferers) • any known alternative uses.

2.1.4	All relevant information needed to conduct the hazard analysis shall be collected, maintained, documented and updated. The HACCP plan shall be based on comprehensive information sources, which are referenced and available on request. This shall include the following: • the latest scientific literature • historical and known hazards associated with specific food products • relevant codes of practice • recognised guidelines • food safety legislation relevant for the production and sale of products • customer requirements. • a copy of any existing site HACCP plans (e.g., for products already in production at the site) • a map of the premises and equipment layout • a water distribution diagram for the site • indication of any areas (zones) (high-risk or ambient high-care) The scope should include reference to Codex and Campden BRI Guideline no. 42, a copy of both shall be available for reference by the HACCP team.
2.1.5	The scope shall make reference to a pre-requisite programme (see clause 2.2.1)

2.2 PRERE QUISITE PROGRAMMES

FUNDAMENTAL

CLAUSE	REQUIREMENTS
2.2.1	The site shall establish and maintain environmental and operational programmes necessary to create an environment suitable to produce safe and legal food products (prerequisite programmes). This shall cover the following as a minimum: • purchasing • raw material supplier approval • transportation arrangements • traceability • allergen controls (latex shall be considered within the site's HACCP scope) • speciation controls • quality assurance programmes • foreign body controls (start-up inspections, audits, etc.) • maintenance programmes for equipment and buildings • pest management • waste controls • cleaning and sanitising • environmental monitoring • laboratory analysis • utilities and services (water, air handling systems) • personal hygiene requirements • staff training • health supervision • site security. Hazards associated with each prerequisite programme shall be identified. The likelihood and severity of occurrence shall be considered. This shall be documented. Control measures and monitoring procedures for the prerequisite programmes must be clearly documented and shall be included within the development and reviews of the HACCP plan.

2.2.2 The site may establish operational pre-requisites within the prerequisite programme to control a specific hazard at a specific process step or for a sequence of process steps. The likelihood and severity of occurrence shall be assessed and documented. Control measures and monitoring procedures for operational prerequisites must be clearly documented and shall be included within the development and reviews of the HACCP plan.

2.3 PROCESS FLOW DIAGRAMS

FUNDAMENTAL

CLAUSE	REQUIREMENTS

2.3.1 Flow diagrams shall be prepared to accurately depict each product, product category or process. This shall set out all aspects of the food process operation within the HACCP scope, from raw material receipt through to processing, storage and distribution. This shall include the following, although this is not an exhaustive list:

- plan of premises and equipment layout
- raw materials, including introduction of utilities and other contact materials (e.g. water (including ice & steam), primary and secondary packaging)
- sequence and interaction of all process steps
- outsourced processes and subcontracted work (including contracted storage and distribution)
- potential for process delay
- daily and seasonal variations
- rework and recycling
- low-risk/high-risk/high-care area segregation
- finished products, intermediate/semi-processed products, by-products and waste.
- office operations

Process parameters (e.g. temperature, maturation times, critical limits) shall be highlighted on flow diagrams.

2.3.2 The HACCP team shall verify the accuracy of the flow diagrams by on-site audit and challenge at least annually or when changes are made. Daily and seasonal variations shall be considered and evaluated. Records of verified flow diagrams shall be maintained.

2.4 HAZARD ANALYSIS & RISK MANAGEMENT

FUNDAMENTAL

2.4.1 RAW MATERIAL HAZARD ANALYSIS & RISK ASSESSMENT

CLAUSE	REQUIREMENTS

2.4.1.1 The HACCP team shall identify and record all the potential hazards that are reasonably expected to be present in each raw material group highlighted within the scope. Consideration shall be given to the following types of hazards:

- microbiological
- physical contamination
- chemical and radiological contamination
- fraud (e.g. substitution or deliberate/intentional adulteration)
- malicious contamination of products
- speciation risks
- allergen risks.

Supplier approval controls outlined in section 3.4 relating to chemical and radiological contamination, fraud and allergens can be acknowledged either in the risk assessment or within the scope.

2.4.1.2 The HACCP team shall conduct a hazard analysis to identify hazards which need to be prevented, eliminated or reduced to acceptable levels. Consideration shall be given to the following:

- likely occurrence of hazard
- severity of the effects on consumer safety
- vulnerability of those exposed
- survival and multiplication of micro-organisms of specific concern to the product
- presence or production of toxins, chemicals or foreign bodies
- contamination of raw materials, intermediate/semi-processed product, or finished product
- any risks associated with raw materials which are subject to legislative control.

Where elimination of the hazard is not practical, justification for acceptable levels of the hazard in the finished product shall be determined and documented.

2.4.1.3 The HACCP team shall consider the control measures necessary to prevent or eliminate a food safety hazard or reduce it to an acceptable level.

2.4.2 PROCESS STEP HAZARD ANALYSIS & RISK ASSESSMENT

CLAUSE	REQUIREMENTS

2.4.2.1 The HACCP team shall identify and record all the potential hazards that are reasonably expected to occur at each step in relation to the process and facilities. Consideration shall be given to the following types of hazards:

- microbiological
- physical contamination
- chemical and radiological contamination
- fraud (e.g. substitution or deliberate/intentional adulteration)
- malicious contamination of products
- speciation risks
- allergen risks
- preceding and following steps of the process.

The list of hazards shall provide typical examples found in that step of the process and not just generic descriptions (e.g. metal from blades and clips, listeria spp.,).

The risks and controls for malicious contamination and fraud from actions carried out by employees are expected to be reviewed within the site's security and food defence risk assessments. The site may acknowledge this within the process step risk assessment as long as all production and storage areas have been reviewed.

2.4.2.2 For each process step, the HACCP team shall conduct a hazard analysis to identify hazards which need to be prevented, eliminated or reduced to acceptable levels. Consideration shall be given to the following:

- likely occurrence of hazard
- severity of the effects on consumer safety
- vulnerability of those exposed
- survival and multiplication of micro-organisms of specific concern to the product
- presence or production of toxins, chemicals or foreign bodies
- contamination of raw materials, intermediate/semi-processed product, or finished product.

Where elimination of the hazard is not practical, justification for acceptable levels of the hazard in the finished product shall be determined and documented.

2.4.2.3 The HACCP team shall consider the control measures necessary to prevent or eliminate a food safety hazard or reduce it to an acceptable level. Where the control is achieved through existing prerequisite programmes, this shall be referenced appropriately. The adequacy of the programme to control the specific hazard shall be validated through a combination of scientific literature and/or a defined measurement process. Consideration may be given to using more than one control measure.

2.5 DETERMINE THE CRITICAL CONTROL POINTS

FUNDAMENTAL

CLAUSE	REQUIREMENTS
2.5.1	The control points determination for each hazard shall be based on hazard analysis and reviewed to identify those that are critical. The group CCP decision tree (see appendix 1) shall be used for each process step to determine this. If a hazard is identified at a step where control is necessary for safety but the control does not exist, the product or process shall be modified at that step, or at an earlier step, to provide a control measure.

2.6 ESTABLISH & VALIDATE CRITICAL LIMITS

FUNDAMENTAL

CLAUSE	REQUIREMENTS
2.6.1	HACCP shall be validated according to the worst-case scenario by conducting a scientific study based on product and processing variables, prior to any changes which may affect product safety, to ensure that the plan will effectively control the identified hazards before implementation.
2.6.2	For each CCP, the appropriate critical limits shall be defined in order to identify clearly whether the process is in or out of control. Critical limits shall be: • measurable wherever possible (e.g. time, temperature, pH) • supported by clear guidance or examples where measures are subjective (e.g. photographs). Details on how the critical limits have been determined must be documented; these may include industry best practice, legislation or validation studies undertaken by the site.
2.6.3	The HACCP team shall validate each CCP. Documented evidence shall show that the control measures selected and critical limits identified are capable of consistently controlling the hazard to the specified acceptable level. Specific requirements for the following processes identified as a CCP are detailed in section 2.7: • deep chill • cooking and cooling • modified atmosphere/vacuum packing • foreign body detection/removal (x-ray, metal detection, sieving). All process validations shall be completed annually as a minimum, or when; • significant engineering work has taken place which may affect the performance of equipment • there is a significant change to the process • where there is an elevated trend of failures or complaints that indicate a loss of control.

2.7 CCP PROCESS VALIDATIONS

FUNDAMENTAL

2.7.1 DEEP CHILL

CLAUSE	REQUIREMENTS
2.7.1.1	The site shall have a procedure for establishing that the chiller can maintain product temperature below 3°C throughout its storage life. The validation method shall include the monitoring of air temperature during normal production hours with periodic product temperature checks. The monitoring period shall also include air and product temperatures during and after defrost cycles of the chiller to ensure that product temperature can be maintained. Consideration shall be given to the following: • a study to determine the warmest areas of the chiller. These points shall be used for the validation • maximum load of the chiller • ambient temperature of surrounding areas where entrances to the chiller are situated. If these areas are not temperature controlled, then validations should take place in the summer months to challenge worst-case.

2.7.2 COOK/COOL

CLAUSE	REQUIREMENTS
2.7.2.1	The site shall have a procedure for determining the heat distribution profile of cooking and cooling equipment to determine where hot and cold spots are located. Once identified, this shall be the placement for any in-cook probes. This must be reviewed annually and repeated every 3 years as a minimum or after any significant engineering work which may affect the heat distribution of equipment.
2.7.2.2	The site shall have a procedure for the validation of cooking and cooling. The procedure shall define where probes are to be placed, justification shall be detailed for this. The procedure shall also define the frequency of review in accordance with clause 2.6.2 and where an increase in adverse microbiological results has been identified. The site shall document evidence that the methodology has taken the following into consideration: • scientific literature, legislation, Campden BRI Guidance (or other recognised research associations) and customer food safety policies • identification of potential organisms of concern • process parameters such as worst-case scenarios (i.e. weight, size/shape, in-going temperature) • placement of probes.
2.7.2.3	Where p-values are used to calculate a thermal process, the correct d-value shall be used for the organism of concern. When calculating the p-value, only temperatures above 60°C are to be included and shall exclude the cool down phase of the cook cycle (i.e. once the product has left the oven, start of cool phase for water immersion cooking equipment). Use of any validation which does not meet this criteria must have written agreement from the relevant customer. Cooked products shall be cooled to <5°C within 4 hours. Cooling parameters for bulk cooked meats (>5kg) shall be observed where required by the customer. Sites must be able to provide evidence of adherence to this.

2.7.2.4 A report summarising the validation shall include the following:

- cooker/chiller used
- programme settings (e.g. temperature, belt speed)
- cumulative p-value
- measurement of variables (e.g. weight/size of product, in-going temperature)
- diagram showing product layout and probe positioning
- verification records of probes used
- the safety margin to be included within the process control shall be highlighted.

The following documentation shall also be included in in the report:

- a copy of the process records
- the raw data from the logger with annotations which indicate the start/finish times of the cook/cool programmes
- a graph which depicts the cook/cool curve
- shelf life study of the product (this shall meet the requirements outlined in clause 5.1.5).

2.7.2.5 The site shall have a competent person on site to oversee the cook/cool validation process and to formally sign off all validations. This person shall have attended the Campden BRI Safe Cooking Course or equivalent.

2.7.3 MODIFIED ATMOSPHERE PACKING & VACUUM PACKING

CLAUSE	REQUIREMENTS
2.7.3.1	The site shall establish parameters and a clear methodology for MAP and vacuum packing which must have been validated though a shelf life study of the product (see clause 5.1.5). This must comply with Campden BRI Guideline no. 11 The manufacture of vacuum and modified atmosphere packed chilled foods: a code of practice.

2.7.4 FOREIGN OBJECT DETECTION AND REMOVAL

CLAUSE	REQUIREMENTS
2.7.4.1	The site shall complete an assessment to determine whether detection/removal equipment will improve the protection of final products from foreign body contamination. The assessment shall also consider the most effective location for equipment (e.g. after product has been packed, immediately before being placed into a bulk container).
2.7.4.2	The site shall complete an assessment to determine the optimum detection/removal limit. For sieving and filtration this will require trials using a range of different mesh sizes to establish the smallest possible mesh size without compromising the integrity of the material being inspected. For metal detection/x-ray this will require determining different product categories based on relevant factors such as pack size, salt content, temperature of product (i.e. frozen or chilled) packaging type then trialling different test piece sizes to identify what the detection equipment can consistently detect. The site shall ensure that customer requirements are achieved. Where this is not possible then written approval must be obtained.

2.8 ESTABLISH A MONITORING SYSTEM FOR EACH CCP

FUNDAMENTAL

CLAUSE	REQUIREMENTS
2.8.1	A monitoring procedure shall be established for each CCP to ensure compliance with critical limits. The monitoring system shall be able to detect loss of control of CCPs and, wherever possible, provide information in time for corrective action to be taken. The critical limits and process parameters shall match what has been detailed within the CCP control summary and worst-case limits established within the process validation. As a guide, consideration may be given to the following, although this is not an exhaustive list: • online measurement • offline measurement • continuous measurement (e.g. thermographs, pH meters etc). Where discontinuous measurement is used, the system shall ensure that the sample taken is representative of the batch of product.
2.8.2	Critical limits and key process parameters shall be made available for reference to CCP operatives.
2.8.3	Records associated with the monitoring of each CCP shall include the date, time and result of measurement and shall be signed by the person responsible for the monitoring and verified, when appropriate, by an authorised person. Where records are in electronic form, there shall be evidence that records have been checked and verified. Records shall be unambiguous and legible.

2.9 ESTABLISH A CORRECTIVE ACTION PLAN

FUNDAMENTAL

CLAUSE	REQUIREMENTS
2.9.1	The HACCP team shall specify and document the corrective action to be taken in a timely manner when monitored results indicate a failure to meet a control limit, or when monitored results indicate a trend towards loss of control. This shall include the action to be taken by nominated personnel with regard to any products that have been manufactured during the period when the process was out of control. The corrective action for each CCP shall be specified within each CCP procedure. This shall include: • immediate remedial action to be taken – escalation process notifying the appropriate personnel • quarantine procedures – identification of implicated products and safe handling and storage until the safety status is established. The justification for how product has been determined to be safe shall be recorded for every instance. Root cause investigation shall also be completed.

2.10 ESTABLISH VERIFICATION PROCEDURES

FUNDAMENTAL

CLAUSE	REQUIREMENTS
2.10.1	Procedures of verification shall be established to confirm that the HACCP plan, including controls managed by prerequisite programmes, continues to be effective. Examples of verification activities include: • internal audits (see section 3.3) • review of records where acceptable limits have been exceeded • review of complaints by enforcement authorities or customers • review of incidents of product withdrawal or recall. Results of verification shall be recorded and communicated to the HACCP team.

2.11 HACCP DOCUMENTATION & RECORD KEEPING

CLAUSE	REQUIREMENTS
2.11.1	Documentation and record-keeping shall be sufficient to enable the site to verify that the HACCP and food safety controls, including controls managed by prerequisite programmes, are in place and maintained.

2.12 HACCP REVIEW

CLAUSE	REQUIREMENTS
2.12.1	The HACCP team shall review the HACCP plan and prerequisite programmes at least annually and prior to any changes which may affect food safety. As a guide, these may include the following, although this is not an exhaustive list: • change in raw materials or supplier of raw materials • change in ingredients/recipe • change in processing conditions, process flow or equipment • change in packaging, storage or distribution conditions • change in consumer use • emergence of a new risk (e.g. known adulteration of an ingredient or other relevant, published information, such as the recall of a similar product) • review following a significant product safety incident (e.g. a product recall) • new developments in scientific information new developments in scientific information associated with ingredients, process or product • an elevated trend in CCP failures or complaints that indicate a loss of control with the HACCP plan or prerequisite programmes. The annual HACCP review shall take place after the HACCP Verification Audit. Appropriate changes resulting from the review shall be incorporated into the HACCP plan and/or prerequisite programmes, fully documented and the validation recorded. Where appropriate, the changes shall also be reflected in the site's, procedures and food safety objectives.
2.12.2	Actions from meetings shall be assigned to relevant members of the HACCP team with agreed timescales. These shall be signed off once completed.

3 MANAGEMENT SYSTEM (FOOD SAFETY & QUALITY)

3.1 FOOD SAFETY AND QUALITY MANUAL

The site's processes and procedures to meet the requirements of this Standard shall be documented to allow consistent application, facilitate training, and support due diligence in the production of a safe product.

CLAUSE	REQUIREMENTS
3.1.1	The Cranswick Manufacturing Standard and the Group Integrated Management System (IMS) shall form the basis of the site's quality management system. All site procedures, working methods and practices shall be collated in the form of a printed or electronic quality manual which covers the requirements of this standard. The site's quality manual shall be fully implemented and relevant components shall be readily available to relevant staff.
3.1.2	All procedures and work instructions shall be clearly legible, unambiguous, in appropriate languages and sufficiently detailed to enable their correct application by appropriate staff. This shall include the use of photographs, diagrams or other pictorial instructions where written communication alone is not sufficient (e.g. there are issues of literacy or foreign language). All staff authorised to use/complete a controlled document (printed or electronic) must be able to understand the relevance, what it is for and how to use it.
3.1.3	The company's processes and procedures to meet the requirements of this Standard are documented within the Integrated Management System (IMS). Where changes on site require the IMS to be updated, the site shall ensure that this is communicated to Group within an appropriate timescale.

3.2 DOCUMENT CONTROL, RECORD COMPLETION AND MAINTENANCE OF RECORDS

The site shall operate an effective document control system to ensure that only the correct versions of documents, including recording forms, are available and in use. The site shall maintain genuine records to demonstrate the effective control of product safety, legality and quality.

CLAUSE	REQUIREMENTS
3.2.1	The site shall have a procedure to manage documents which form part of the food safety and quality system. This shall include: • a list of all controlled documents indicating the latest version number • the method for the identification and authorisation of controlled documents • a record of the reason for any changes or amendments to documents • the system for the replacement of existing documents when these are updated. Group documents shall be controlled in accordance with GTS-041 Group Document Control Procedure. Where documents are stored in electronic form these shall also be: • stored securely (e.g. with authorised access, control of amendments, or password protected) • backed up to prevent loss.

3.2.2 Records shall be legible, maintained in good condition and retrievable. They must be completed at the time of the check and not before or after the event. A suitable procedure must be in place to manage any mistakes that are made, this should include a line neatly crossing through the error with the initials of the person making the change.

No correction fluid and over label is allowed to be used to alter any mistakes made.

Any alterations to records shall be authorised and justification for the alteration shall be recorded. Where records are in electronic form these shall also be:

- stored securely (e.g. with authorised access, control of amendments, or password protected)
- suitably backed up to prevent loss.

All staff authorised to use/complete a controlled document (printed or electronic) must be able to understand the relevance, what it is for and how to use it.

3.2.3 Records shall be retained for a defined period with consideration given to:

- secure storage
- any legal or customer requirements
- the shelf life of the product.

This shall take into account, where it is specified on the label, the possibility that shelf life may be extended by the consumer (e.g. by freezing).

The site shall retain records (electronic and/or paper format) in accordance with customer requirements, as a minimum this shall be for the shelf life of the product plus 12 months, however requirements which exceed this shall be adhered to.

Non process related documents shall be retained as per Cranswick Data Retention Policy.

3.2.4 Where records are in the electronic form they shall also be:

- stored securely e.g., with authorised access, control of amendments or password protection
- suitably backed up to prevent loss
- compliant to group IT security processes

In the event of the electronic system not being available, the site shall have a suitable system to ensure all documents are correctly completed, accessible, maintained and stored.

3.3 INTERNAL AUDITS

FUNDAMENTAL: The company shall be able to demonstrate that it verifies the effective application of the food safety plan and the implementation of the requirements of this standard and other retailer requirements for sites across the group.

The site shall be able to demonstrate that it continually verifies the effective application of its quality management system through its own internal audit programme.

CLAUSE	REQUIREMENTS
3.3.1	The site shall be subject to an internal audit programme, the scope of which shall be based against this standard and other relevant customer requirements. Each internal audit shall have a defined scope and consider a specific activity or a section of the HACCP plan. Internal audits shall be completed by trained members of the Group Compliance team (or nominated person) who are independent from the scope of the audit. The internal audit programme shall be scheduled throughout the financial year. The frequency of each section will be based on the risks associated with the activity and previous audit performance. (See Group Internal Audit Procedure)

3.3.2 The site shall have its own internal audit programme which covers CCPs and other key pre-requisites (see list below). The scope of this programme shall be to review compliance against this standard and customer requirements and to verify implementation of site procedures within the factory.
The schedule shall ensure that all shift patterns are captured within the audit programme.
The procedure shall include the following:

- clear definition of the audit scope for each activity
- the frequency for each audit
- auditors to be independent from the activity being audited
- report writing
- responsibility for the close out of non-conformities, investigation/root cause analysis and the verification that actions have been implemented.

The site's internal audit programme shall cover the following areas as a minimum:

- CCPs
- foreign body control
- sieving
- building and fabrication
- allergen control
- high care/high risk barrier integrity
- pest management
- label and pack control
- on-site catering contractors.

The frequency of these audits shall be based on risk but as a minimum must be completed every 3 months for production areas with other non-production areas (e.g. roof void, external stores) to be completed every 6 months as a minimum.

(For Examples of Group Audits, Site Based Audits and Site Based Inspections see Appendix 2)

3.3.3 The site shall complete a process audit for each product category annually as a minimum (see glossary for definition). The scope of the audit is to establish compliance to the specification and shall include confirmation of:

- the accuracy of specification information and its consistency with internal documents such as internal recipe sheets, production instructions and line sheets
- adherence to the recipe with regards to quantity, components used and methodology
- adherence to procedures relevant to the production run
- the adequacy of internal quality attribute sheets in use
- adherence to food safety and quality parameters

3.3.4 The site shall complete hygiene inspections which confirm compliance to cleaning instruction methodologies and finished standards. Finished standards shall be assessed through visual inspection and may also include swab sampling for microbiological analysis by an accredited laboratory (see clause 5.6.2.1).

Hygiene inspections for Low Risk areas shall be completed monthly with all areas covered over a 3-monthly period. Hygiene inspections for all High Care/High Risk areas shall be completed monthly.

3.3.5 The sites shall demonstrate a commitment to both internal audit programmes through the following evidence:

- completion of both site and group internal audit schedules
- close out of actions within timescale
- attendance of opening and closing meetings from relevant members of the management team for Group internal audits. The management team shall also ensure that there is sufficient engagement from staff, where required, during the factory inspection
- communication of audit performance with the senior management team.

3.3.6 In addition to the internal audit programme, there shall be a separate programme of documented inspections to ensure that good manufacturing practices and housekeeping standards are being maintained in production and non-production areas (including staff facilities). The following areas shall be reviewed (where appropriate):

- building and fabrication
- foreign body control
- personal hygiene
- allergen control
- pest activity
- traceability and identification of materials.

The frequency of these inspections shall be based on risk but shall be no less than once per month in open product areas.

3.3.7 Audit and documented inspection reports shall identify conformity as well as non-conformity and include objective evidence of the findings. Results of the inspection shall be reported to the relevant responsible person. Corrective and Preventative actions and timescales for their implementation shall be agreed. Preventative action shall be based on root cause analysis. Action taken shall be verified.

Audits and inspections results shall be trended and regularly reviewed, as a minimum annually during management review meetings (see clause 1.3.1).

3.4 SUPPLIER AND RAW MATERIAL APPROVAL AND PERFORMANCE MONITORING

3.4.1 MANAGEMENT OF SUPPLIERS OF RAW MATERIALS AND PACKAGING

FUNDAMENTAL: The site shall ensure that all suppliers have been formally approved by the Group Technical Supply Chain Assurance Team (GTS) to ensure that any potential risks from raw materials (including primary packaging) to the safety, authenticity, legality and quality of the final product are understood and managed.

There shall be a collaborative approach with GTS for monitoring the performance of suppliers.

CLAUSE	REQUIREMENTS
3.4.1.1	The approval process for raw materials and packaging, and the suppliers of these materials shall be managed by GTS. The approval process shall meet the requirements outlined within GTS-027 Supplier Approval Procedure. This approval process shall also include the use of agents and brokers and shall extend to the original manufacturer. The site shall be responsible for notifying GTS of the intended customer so that, where applicable, the relevant sourcing policy can be adhered to.
3.4.1.2	The site shall ensure that all raw materials and packaging are purchased from suppliers which have undergone the GTS supplier approval process, this includes contingency suppliers. While samples may be obtained before/during the approval process for new product concepts, the supplier must have completed the approval process prior to the purchase of materials intended for use in finished products. If a supplier has been requested by a customer which has subsequently not been approved by GTS then the site shall inform the customer and obtain written confirmation of derogation. That supplier shall only be approved for the specified customer, materials shall not be used within products for other customers until full approval from GTS is obtained.

3.4.1.3	Ongoing supplier performance review shall be managed by GTS, this shall meet the requirements outlined within Group Supplier Approval Procedure. The site shall ensure that all issues identified on receipt of raw materials and packaging (see 3.4.2.1) are communicated to GTS via approved online platform so that the information may be used as part of the supplier review.
3.4.1.4	The site shall have a copy of an up to date list of approved suppliers which has been issued by GTS. This shall be readily available to the relevant staff (e.g. intake operatives, purchasing personnel).
3.4.1.5	Where site operate their own Management of Suppliers of Raw Materials and Packaging System, the site shall have defined procedure in place. Procedure shall comply with relevant legal and customer requirements. Procedure shall be communicated to GTS team.

3.4.2 RAW MATERIAL AND PACKAGING ACCEPTANCE, MONITORING AND MANAGEMENT PROCEDURES

Controls on the acceptance of raw materials (including primary packaging) shall ensure that these do not compromise the safety, legality or quality of products or where appropriate, any claims of authenticity.

CLAUSE	REQUIREMENTS
3.4.2.1	The site shall have a procedure for the acceptance of raw materials and primary packaging on receipt, this shall include: • responsibility for completing checks • confirmation that the material has come from an approved supplier (this shall make reference to the approved supplier list) • vehicle inspection (tamper/security checks, temperature and physical condition of trailer, evidence of pest activity) • checks to confirm that the material matches the description on the vehicle manifest/invoice • allergen status of materials (including certificates of conformance for 'free from' materials e.g. gluten free) • product temperature checks (where applicable) • physical condition of the material and its packaging (signs of damage, contamination, tampering) • shelf life checks • material inspection to establish conformance to specification requirements and any other safety/quality checks determined by risk assessment (see clause 3.4.2.2) • identification of materials pending inspection and subsequent release/rejection • action to be taken for materials that do not meet specification, are damaged/contaminated or show signs they have been tampered with. • RSPO Palm Oil checks (where applicable)
3.4.2.2	The site shall undertake a risk assessment for each raw material (including products where part of the processing has been outsourced) to determine what additional inspection and/or testing is required and the frequency of inspection. Consideration shall be given to the significance of a raw material to the safety, legality, authenticity and quality of the final product. Raw material and primary packaging inspection criteria may include: • raw material analysis (e.g. microbiological, chemical, organoleptic) • specification/internal QAS checks • certificate of analysis • certificate of conformance. Sampling shall be carried out by trained personnel. There shall be sampling procedures in place to prevent contamination of the sample being taken and other materials stored in the area.

3.4.2.2 A list of raw materials (including primary packaging) and the requirements to be met for acceptance shall be made available in the intake area. The parameters for acceptance and frequency of testing shall be clearly defined, implemented and reviewed. The use of written descriptions, visual aids and objective measures shall be considered.

The risk assessment for each raw material shall be reviewed:

- when there is a change in raw material, the processing of a raw material or the supplier of a raw material
- if a new risk emerges
- following a product recall or withdrawal, where a specific raw material has been implicated
- at least every 3 years.

3.4.2.3 Sites that purchase ingredients shall participate in the chemical testing plan in accordance with th e schedule detailed in Chemical Testing Plan.

Sites that purchase meat from outside the group shall participate in the speciation testing schedule issued by GTS. See Speciation Sampling Procedure for more details on sampling and testing.

All samples shall be sent to an external laboratory for analysis. The laboratory shall meet the requirements set out in clause 5.6.2.1. Certificates of analysis shall be submitted to GTS.

3.4.2.4 The site shall ensure that approved changes to raw material (including primary packaging) are communicated to intake personnel. Internal specification documents shall be updated to reflect any changes, the updated documents shall be made available to intake operatives to ensure that the changes are being met by the supplier.

3.4.2.5 Where the site is in receipt of live animals, there shall be an inspection by a suitably competent individual at lairage and post mortem to ensure that the animals are fit for human consumption.

3.4.3 MANAGEMENT OF SUPPLIERS OF SERVICES

The company shall approve and monitor the performance of suppliers and services that have been outsourced to ensure the service is appropriate and any risks presented to food safety, legality, authenticity and quality have been evaluated to ensure effective controls are in place.

The site shall assume responsibility for the management of service providers not covered by GTS. (see appendix 3)

CLAUSE	REQUIREMENTS
3.4.3.1	The approval and monitoring of the following service providers shall be managed by GTS in accordance with GTS-027: • Offsite storage facilities of ingredients, packaging or products • Laundry • Catering at non-Cranswick plc premises • Food contact gases • Transport for distribution This approval and performance monitoring process shall be risk-based and take into consideration: • risk to the safety and quality of products • compliance with any specific legal requirements • potential risks to the security of the product (i.e. risks identified in the vulnerability and food defence assessments). The outcome of the performance monitoring shall be reviewed and documented.

3.4.3.2 The site shall have procedures for the approval and monitoring of suppliers of services not managed by GTS. Such services may include, as appropriate:

- pest control
- contracted cleaning
- contracted services and maintenance of equipment
- off-site packing of products
- laboratory testing
- vending machine providers (excluding sandwich manufacturers which are managed by GTS)
- onsite catering contractors
- waste management.

This approval and performance monitoring process shall be risk-based and take into consideration:

- risk to the safety and quality of products
- compliance with any specific legal requirements
- potential risks to the security of the product (i.e. risks identified in the vulnerability and food defence assessments).

The outcome of the performance monitoring shall be reviewed and documented.

3.4.3.3 Contracts or formal agreements shall exist with the suppliers of services that clearly define service expectations and ensure that the potential food safety risks associated with the service have been addressed.

3.4.4 MANAGEMENT OF OUTSOURCED PROCESSING

Where any process step in the manufacture of a product is outsourced to a third party or undertaken at another site, this shall be managed to ensure it does not compromise the safety, legality, quality or authenticity of the product.

CLAUSE	REQUIREMENTS
3.4.4.1	The approval process for outsourced processors shall be managed. The approval process shall meet the requirements outlined within Group Supplier Approval Procedure. The site shall ensure that appropriate approval process is in place and approval documentation is maintained. GTS team shall be informed.
3.4.4.2	The site shall be able to demonstrate that, where part of the production process or any part of the final packing is outsourced and undertaken off-site, this has been declared to the customer and, where required, approval granted.
3.4.4.3	The site shall ensure that outsourced processors have been approved by GTS. Outsourced processors shall be monitored to ensure that they effectively manage risks to product safety and quality and are operating effective traceability processes.
3.4.4.4	Any outsourced processing operations shall: • be undertaken in accordance with established contracts which clearly define any processing and/or packing requirements and product specification • maintain product traceability.
3.4.4.5	The site shall establish inspection and test procedures for products where part of the processing has been outsourced, including visual, chemical and/or microbiological testing. The frequency and methods of inspection or testing shall depend on risk assessment.
3.4.4.6	Where any processes are outsourced, including production, manufacture, processing or storage, the risks to the product safety, authenticity and legality shall form part of the site's HACCP plan. (see clause 2.1.2).

3.5 SPECIFICATIONS

Specifications shall exist for raw materials (including primary packaging and products where part of the processing has been outsourced), finished products and any product or service which could affect the integrity of the finished product.

CLAUSE	REQUIREMENTS
3.5.1	Specifications for raw materials and primary packaging and products where part of the processing has been outsourced shall be adequate and accurate and ensure compliance with relevant safety and legislative requirements. The specifications shall include defined limits for relevant attributes of the material which may affect the quality or safety of the final products (e.g. chemical, microbiological or physical standards).
3.5.2	Accurate, up-to-date specifications shall be available for all finished products. These may be in the form of a printed or electronic document, or part of an online specification system. They shall include key data to meet customer and legal requirements and assist the user in the safe usage of the product. Where required, these shall be uploaded onto the relevant customer portal.
3.5.3	Where the site is manufacturing customer-branded products, it shall seek formal agreement of the finished product specifications. Where specifications are not formally agreed then the company shall be able to demonstrate that it has taken steps to ensure formal agreement is in place
3.5.4	Specification review shall be sufficiently frequent to ensure that data is current or every 3 years as a minimum, taking into account product changes, suppliers, regulations and other risks. Reviews and changes shall be documented.

3.6 INCIDENT, NON-CONFORMITY, CORRECTIVE ACTION, ROOT CAUSE ANALYSIS AND PREVENTATIVE ACTIONS

FUNDAMENTAL: The site shall be able to demonstrate that it uses the information from identified failures in the food safety and quality management system to make necessary corrections and prevent reoccurrence.

CLAUSE	REQUIREMENTS
3.6.1	The site shall have systems in place to handle and correct failures identified in the food safety and quality management system. There shall be a procedure in place for managing all failures generated by the site (e.g. non-conforming product, customer complaints and system failures). This shall include immediate actions to control or eliminate the hazard presenting a risk to product safety, legality or quality and corrective/preventative measures to prevent reoccurrence. Responsibilities shall be assigned to individuals and not departments. Non-conformities raised by internal and third-party audits shall be subject to the same procedures. (For Food Safety Non-conformance Escalation and Action Chart, see appendix 4)
3.6.2	The site shall have systems in place to handle and correct failures identified with the health and safety, and environmental and energy management systems. There shall be procedures in place for managing incidents or non-conformities relating to IMS. This shall include immediate actions to control or eliminate the hazard and manage the consequences, including mitigating any adverse environmental impacts. Incidents and non-conformities shall be investigated in accordance with clause 3.6.3. Additionally, investigations and determining appropriate corrective actions for health and safety related incidents and non-conformities shall involve the participation of staff and other relevant interested parties. The site shall ensure that any temporary or permanent changes that impact health and safety performance have been assessed with action taken to mitigate any adverse effects.
3.6.3	All non-conformities shall be thoroughly investigated. The following information shall be recorded: • clear description of the non-conformity (including the quantities of any affected product) • assessment of the potential consequences of the non-conformity by a suitably competent and authorised person • corrective action taken to address the immediate issue, this shall include the date/time action was taken and the person responsible • investigation to establish the circumstances leading to the non-conformity • root cause analysis where appropriate (see clause 3.6.4) • appropriate timescales for preventative action and the person responsible • a documented assessment of any temporary or permanent changes to working conditions, process and/or equipment where appropriate • verification that the preventative actions have been implemented and are effective. The actions shall be included in a regular review of activities and systems. If timescales for actions are not met, the reasons shall be recorded.

SECTION 3: MANAGEMENT SYSTEM (FOOD SAFETY & QUALITY)

3.6.4 The site shall have a procedure for the completion of root cause analysis. As a minimum, root cause analysis shall be used to implement ongoing improvements and to prevent reoccurrence of non-conformities when:

- analysis of non-conformities for trends shows there has been a significant increase in a type of non-conformity
- a non-conformity places the safety, legality or quality of a product at risk.

The site shall define what constitutes a trend to initiate root cause analysis procedures. The use of a multidisciplinary team shall be considered where appropriate.

Root cause analysis procedure shall include:

- a set of parameters that will initiate the use of an appropriate tool (i.e. the site must define when the analysis will be completed if it is not already prescribed within this standard)
- the methods that the site will use to conduct the analysis
- the records which detail the analysis and any subsequent preventive actions
- the methods for the verification of the completed actions.

Only personnel that have been trained against the site's root cause analysis procedure shall be assigned responsibility for a root cause analysis investigation.

Personnel responsible for verifying root cause analysis has been carried out appropriately shall have received formal training in root cause analysis in addition to the site's procedure.

3.6.5 The site shall trend and analyse non-conformity data to establish if repeat issues are occurring. Where repeat issues are identified, the site shall review the close out for each incident including the root cause analysis and preventative actions to establish why repeat issues are occurring and what additional actions need to be implemented to prevent reoccurrence.

3.7 CONTROL OF NON-CONFORMING PRODUCT/EQUIPMENT

The site shall ensure that any non-conforming product (including rejected product returned to site) and equipment is effectively managed to prevent unauthorised release/use.

CLAUSE	REQUIREMENTS
3.7.1	There shall be procedures for managing non-conforming materials, product or equipment. These procedures shall include: • the requirement for staff to identify and report potentially non-conforming material, product or equipment • method for clear identification of non-conforming material, product or equipment (e.g. direct labelling or the use of IT systems) • 'on hold' protocols to ensure secure storage and to prevent accidental release or use (e.g. physical or computer-based isolation) • escalation process to the relevant managers, departments and customers where required (food safety and quality issues shall be reported via appropriate channels in accordance with customer requirements, communication shall be documented where possible) • defined responsibilities for decision-making on the use or disposal of material/product appropriate to the issue (e.g. destruction, reworking, downgrading to an alternative label or acceptance by concession) • records of the decision on the use or disposal of material/product, or action taken to rectify equipment • records of destruction where a material/product is destroyed for food safety reasons. Incidents defined as reportable under Group Risk Incident Procedure shall be submitted onto the risk register within 5 working days of occurrence.

3.8 TRACEABILITY

FUNDAMENTAL: The site shall be able to trace all raw material product lots (including primary packaging) from its suppliers through all stages of processing and dispatch to its customers and vice versa.

CLAUSE	REQUIREMENTS
3.8.1	The site shall have a documented traceability procedure designed to maintain traceability throughout the site's processes. As a minimum, this shall include: • how the traceability system works • the labelling and records required • the different type of traceability tests and the frequency of completion (see clause 3.8.3). Where applicable, the traceability system shall meet the legal requirements in the country of sale or intended use.
3.8.2	Identification of raw materials (including primary packaging), intermediate/semi-processed products, part-used materials, finished products and materials pending investigation shall be adequate to ensure traceability.
3.8.3	The site shall have a schedule in place for testing the traceability system across the range of product groups it manufactures to ensure traceability can be determined from the supplier of raw material (including primary packaging) to the finished product and vice versa, including quantity check/mass balance. The traceability test shall also be able to identify where finished products have been despatched to, including re-distributed product (see Group Donations Procedure). For slaughter sites, traceability shall go back to farm and, where required, provide evidence of farm and transport certification. The frequency of these tests shall match the frequency defined within the traceability procedure and meet any other retailer or scheme requirements where applicable. The raw materials/finished products to be challenged within the schedule shall be based on risk with consideration to the likelihood of raw materials carrying a claim being at risk of substitution. As a minimum, the schedule of traceability tests shall include: • forwards – raw material to finished product • backwards – finished product back to intake • mass balance of a raw material (to be completed annually. Shall be completed on key, or 'at risk' raw materials) • packaging mass balance • mock recall – see clause 3.10.6. For sites that process raw materials carrying a pack claim, mass balance exercises shall be completed every 3 months. Raw materials selected for mass balance exercises shall be based on risk. The site shall define an acceptable tolerance for mass balance exercises and be able to justify any variances within a mass balance test (e.g. drip loss). An investigation along with appropriate corrective actions shall be documented for any results outside the expected tolerance. Mass balance shall be verified against other systems e.g., Detectors, verifiers (e.g., X-ray machines, label verification pack counts). Finished product mass balance shall include any printed packaging or labels that have food safety or legal information on it. The traceability test shall include a summary of the documents that should be referenced during the test, and clearly show the links between them. Traceability should be achievable within 4 hours, where this has taken longer, a summary of findings must be included with actions taken for improvement.

3.8.4 Where rework or any reworking operation is performed, traceability shall be maintained.

3.9 COMPLAINT HANDLING

Customer complaints shall be handled effectively, and information used to reduce recurring complaint levels.

CLAUSE	REQUIREMENTS
3.9.1	The site shall have a procedure for the handling of complaints. The procedure shall identify what steps are to be taken depending on the nature of complaint and in accordance with customer requirement. The root cause analysis shall be completed (see clause 3.6.3) and appropriate timescales for investigations and responses to the customer shall be defined. All complaints shall be captured on record and be given a unique reference number. The record shall include: • method complaint was received (i.e. phone email, media) • the product information (customer, product name, use by date and any other identification numbers provided) • the nature of the complaint. • the grade of the complaint (Standard, Serious, Critical) The handling of customer complaints must be overseen by a nominated member of senior management team.
3.9.2	Complaints shall be categorised and investigated in accordance with the site's procedures. Evidence of records reviewed and factory inspections which took place shall form part of the investigation report. Involvement from Operations, Hygiene, Technical and Engineering staff shall be considered where appropriate. The outcome of the investigation shall be reported to relevant personnel and departments. Corrective actions shall be effective to prevent reoccurrence. There shall be systems in place for briefing corrective action with the relevant departments. The site shall communicate the outcome of complaint investigations in accordance with their customer requirements.
3.9.3	Complaint data shall be monitored for trends. Where trends have been identified, the site shall identify and implement an action plan for improvement and to avoid reoccurrence. Complaint data and action plans shall be communicated to all staff. Trending shall determine the level of complaints relative to each product, production line as well as to production volumes. Where there are seasonal issues identified by trending, there shall be a process in place to capture this and proactively address for future seasons. The site shall set targets for complaints and, where possible, have this agreed with the relevant customer. Performance against these targets shall be continually monitored. Targets shall be reviewed annually.
3.9.4	The site shall have a complaint reduction team that includes those responsible for quality assurance, technical management, operations, engineering, NPD, procurement and other relevant functions. The complaint reduction team shall formally review complaint data on a monthly basis to actively reduce the number of complaints received. The team shall: • identify trends and areas of concern from complaint data • identify appropriate actions and agree and timescales • review complaint investigations and corrective action close out • monitor performance against complaint targets. The outcome of the complaint reduction reviews shall be communicated to staff.

3.10 INCIDENT MANAGEMENT, CRISIS AND BUSINESS CONTINUITY

The site shall have the necessary systems in place to effectively implement the Group Crisis & Business Continuity Manual in the event of an incident.

CLAUSE	REQUIREMENTS
3.10.1	The site shall complete a business continuity risk assessment for events which may impact on the site's ability to operate. The controls to mitigate the impact of each event shall be defined.
3.10.2	The site shall have the most up to date version of the Group Crisis & Business Continuity Manual. The site shall have a nominated Crisis Team with a Crisis Team Leader. All members of the team shall have access to either a hard copy or electronic copy of the crisis manual. Hard copies of the manual shall be held in the Crisis Team Leader's office, the Technical Office and in the Site Security office (if applicable).
3.10.3	The site shall maintain the emergency contact list with site specific emergency details (e.g. members of the Site Crisis Management Team and their deputies, customers, regulatory bodies, emergency services, suppliers, local authority, utility suppliers). This shall include out of hours contact details.
3.10.4	The site shall have an emergency box which contains the following: • an up-to-date version of the Group Crisis & Business Continuity Manual • site crisis procedures • site plans • emergency contact details. The location of the emergency box may be in one of the locations listed in clause 3.10.1.
3.10.5	In the event of a significant incident/emergency, the site shall follow the guidelines set out within the Group Crisis & Business Continuity Manual. The site shall ensure that all communication and key activities which took place during a crisis are recorded on the event log. After the event, the team shall review how the crisis was managed and identify where any improvements to preparation and response can be made. Records of actions should be maintained.
3.10.6	• The product recall and withdrawal procedures shall be tested, at least annually, in a way that ensures their effective operation. Results of the test shall be retained and shall include timings of key activities. The results of the test and of any actual recall shall be used to review the procedure and implement improvements as necessary. An out of hours test shall be completed every 2 years.
3.10.7	In the event of a significant food safety incident, including a product recall or regulatory food safety non-conformity (e.g. a regulatory enforcement notice), the certification body issuing the current BRCGS certificate for the site shall be informed within 3 working days.
3.10.8	Incidents which are defined as reportable under Group Risk Incident Procedure shall be submitted onto the risk register within 5 working days of occurrence.

4 SITE STANDARDS

4.1 EXTERNAL STANDARDS AND SITE SECURITY

The production site shall be of suitable size, location and construction, and be maintained to reduce the risk of contamination and facilitate the production of safe and legal finished products.

CLAUSE	REQUIREMENTS
4.1.1	Consideration shall be given to local activities and the site environment, which may have an adverse impact on finished product integrity, and measures shall be taken to prevent contamination. Where measures have been put into place to protect the site (from potential contaminants, flooding etc.), they shall be reviewed in response to any changes.
4.1.2	The external areas shall be maintained in good order. Where grassed or planted areas are located near buildings, they shall be regularly tended and well maintained. External traffic routes under site control shall be suitably surfaced and maintained in good repair to mitigate the risk of contamination of the product.
4.1.3	Site boundaries shall be clearly defined and where appropriate have adequate perimeter fencing.
4.1.4	The building fabric shall be maintained to minimise potential for product contamination (e.g. elimination of bird-roosting sites, sealing gaps around pipes to prevent pest entry, ingress of water and other contaminants).
4.1.5	Connecting pipes for bulk tanks or silos shall be maintained in a clean condition, stored off the floor and capped at both ends when not in use.
4.1.6	Policies and systems shall be in place to ensure that only authorised personnel have access to production and storage areas. Staff shall be trained in site security procedures.
4.1.7	Contractors and visitors, including drivers, shall be made aware of the requirements of the areas they are visiting, with special reference to hazards and potential product contamination. Contractors and visitors, including drivers, shall be controlled in accordance with the Visitor and Contractor Controls Procedure. This includes; • reporting system when first arriving and leaving site • communication of site food hygiene rules and procedures with special reference to hazards and potential product contamination and where appropriate High Risk/Care barrier controls. • communication of site and company rules (including but not limited to; mobile phone use on site, the use of recording equipment in production areas, conduct on site) • completion of health questionnaire for visitors and contractors that are entering production areas. All contractors working in product processing or storage areas shall be monitored by a nominated, trained in site security procedures person who shall be responsible for overseeing their work and actions. For contractors who visit regularly, the site may train them to a similar level as employees; records of the training must be retained.

4.2 FOOD DEFENCE

Systems shall protect products, premises and brands from malicious actions while under the control of the site.

CLAUSE	REQUIREMENTS
4.2.1	The site shall have a TACCP team in place to review threats and vulnerabilities to the end to end raw material supply chain, factory processes and site security. The team shall consist of those responsible for technical management, production operations, supply chain/procurement, human resources and IT. The Team Leader shall have training to an intermediate level of TACCP as a minimum, all other members shall have received in-house training as a minimum.
4.2.2	The TACCP team shall undertake a documented risk assessment (threat assessment) of the potential risks to products from any deliberate attempt to inflict contamination or damage. This threat assessment shall include both internal and external threats. The output from this assessment shall be a documented threat assessment plan. This plan shall be kept under review to reflect changing circumstances and market intelligence. It shall be formally reviewed at least annually and whenever: • a new risk emerges (e.g. a new threat is publicised or identified) • an incident occurs, where product security or food defence is implicated • adverse trends are identified.
4.2.3	Where raw materials or products are identified as being at particular risk, the threat assessment plan shall include controls to mitigate these risks. Where prevention is not sufficient or possible, systems shall be in place to identify any tampering. These controls shall be monitored, the results documented, and the controls reviewed at least annually.
4.2.4	Areas where a significant risk is identified shall be defined, monitored and controlled. These shall include external/off-site storage and intake points for products and raw materials (including packaging). Policies and systems shall be in place to ensure that only authorised personnel have access to production and storage areas, and that access to the site by employees, contractors and visitors is controlled. Staff shall be trained in site security procedures and food defence.
4.2.5	The site shall have a CCTV system in place that has the ability to monitor and record areas identified as a potential risk to products from deliberate contamination or damage. External areas and areas where live animals are handled shall also be covered by the system. CCTV systems shall comply with GS-500 Standards and Specification for the Installation of CCTV systems and comply with the group data retention policy.

4.2.6 The site shall have sufficient controls in place to ensure that IT systems, records and documents containing confidential/sensitive information are secure.

Where documents and records are stored off-site, the site shall be able to demonstrate what security measures are in place.

4.2.7 Where these exist, the site is expected to operate systems that comply with the country of sale or intended use and include specific legislative requirements relating to food defence.

4.3 LAYOUT, PRODUCT FLOW AND SEGREGATION

FUNDAMENTAL: The factory layout, flow of processes and movement of personnel shall be sufficient to prevent the risk of product contamination and to comply with relevant legislation.

CLAUSE	REQUIREMENTS

4.3.1 There shall be a map of the site which designates areas where product is at different levels of risk from contamination, these are:

- high-risk areas
- high-care areas
- ambient high-care areas
- low-risk areas
- enclosed product areas
- non-product areas.

See the BRGS on defining the production risk zones.

As a minimum, the map shall define:

- access points for personnel
- access points for raw materials (including packaging), semi-finished products and open products
- routes of movement for personnel
- routes of movement for raw materials (including packaging)
- routes for the removal of waste
- routes for the movement of rework
- location of any staff facilities, including changing rooms, toilets, canteens and smoking areas
- production process flows
- clearly show all transfer points with direction of travel
- drainage plan indicating the direction of travel and the location of any equipment fitted to prevent the back-up of waste water.
- any areas where time segregation is used to complete different activities (for example, time segregation for high-care areas).

4.3.2 The movement of personnel, raw materials, packaging, rework and/or waste shall not compromise the safety of products. The process flow, together with the use of demonstrably effective procedures, shall be in place to minimise the risk of the contamination of raw materials, intermediate/semi-processed products, packaging and finished products.

4.3.3 Premises shall allow sufficient working space and storage capacity to enable all operations to be carried out properly under safe hygienic conditions.

4.3.4 Temporary structures constructed during building work or refurbishment etc. shall be designed and located to avoid pest harbourage and ensure the safety and quality of products. Once completed, they shall be formally approved and audited daily until they have been removed.

4.4 BUILDING FABRIC, RAW MATERIAL HANDLING, PREPARATION, PROCESSING, PACKING AND STORAGE AREAS

The fabrication of the site, buildings and facilities shall be suitable for the intended purpose.

CLAUSE	REQUIREMENTS

4.4.1 Walls shall be finished and maintained to prevent the accumulation of dirt, minimise condensation and mould growth and facilitate cleaning.

Any panels (e.g., back splash panels), signs and equipment mounted to the wall shall be appropriately sealed to avoid accumulation of dirt, condensation and mould growth under the item. (see blueprint)

4.4.2 Floors shall be suitably hard-wearing to meet the demands of the process and withstand cleaning materials and methods. They shall be impervious, be maintained in good repair and facilitate cleaning.

4.4.3 Drainage, where provided, shall be sited, designed and maintained to minimise risk of product contamination and not compromise product safety. Drainage systems for high-risk and high-care shall be independent or designed to flow from high-risk/high-care to low risk. Equipment shall be in place to prevent back flow into high-risk/high-care areas. Drains shall be proofed to prevent the ingress of pests.

Machinery and piping shall be arranged so that, wherever feasible, process waste water goes directly to drain. Where significant amounts of water are used, or direct piping to drain is not feasible, floors shall have adequate falls to cope with the flow of any water or effluent towards suitable drainage.

4.4.4 Ceilings and overheads shall be constructed, finished and maintained to prevent the risk of product contamination.

4.4.5 Evaporator units shall be hygienically designed, positioned, constructed and maintained to minimise the risk of contamination of product, packaging and/or equipment. Condensate drainage shall be piped and flow direct to drain and not drip on product, ingredients, packaging or production equipment.

4.4.6 Where suspended ceilings or roof voids are present, adequate access to the void shall be provided to facilitate inspection for pest activity, unless the void is fully sealed.

Where mezzanine floors are present, adequate access to the mezzanine shall be provided to facilitate inspection of pest activity and cleanliness. Mezzanine floors shall be included in the hygiene schedule and shall be subject to GMP inspection.

4.4.7 Where elevated walkways are adjacent to or pass over production lines, they shall be:

- designed to prevent contamination of products and production lines
- easy to clean
- correctly maintained.

4.4.8 Where there is a risk to product, windows and roof glazing which are designed to be opened for ventilation purposes shall be adequately screened to prevent the ingress of pests.

4.4.9 Doors (both internal and external) shall be maintained in good condition. As a minimum:

- external doors and dock levellers shall be close fitting or adequately proofed
- external doors to open product areas shall not be opened during production periods except in emergencies
- where external doors to enclosed product areas are opened, suitable precautions shall be taken to prevent pest ingress.

Roller doors, rise and fall doors or similar shall be maintained in good condition. Roller doors in high-risk and high-care areas shall not touch the floor to avoid potential transfer of contamination from floor to product, equipment or personnel.

Strip curtains shall be maintained, clean, in good condition and shall not present a contamination risk to product. Where strip curtains are used in open product areas, they shall not touch the floor to avoid potential transfer of contamination from floor to product, equipment or personnel. Strip curtains shall be fitted correctly to prevent pest ingress and allow maintenance of correct area temperature.

4.4.10 Suitable and sufficient lighting shall be provided for the correct operation of processes, inspection of product and effective cleaning.

4.4.11 Adequate ventilation and extraction shall be provided in product storage and processing environments to prevent condensation or excessive dust.

Where condensation still appearing in ventilated areas, additional procedures shall be established to control the condensation build up (e.g., start-up checks).

4.4.12 All services shall be designed from material suitable for the purpose and appropriate to the area where used, intact and allow easy and effective cleaning.

This shall include (where relevant), as a minimum:

- pipework for water/gas/compressed air
- electrical cabling/conduit/sockets
- ventilation ducting, compressors/pumps, fire sprinkler systems.

Electrical cabling shall be adequately mounted and supported to avoid build-up of dirt and to not disturb hygiene activites in the area. Where possible, cabling shall be mounted off the floor.

Conduits and sockets shall be adequately sealed to prevent dirt and condensation build up. Conduits and sockets shall be part of the site's maintenance programme with inspections at a pre-determined frequency.

4.4.13 Facilities for cleaning production equipment must be appropriately located, of suitable construction and easily cleanable. Drainage must be effective and extraction sufficient to remove steam and prevent excessive condensation.

Where manual cleaning is carried out suitable racks, tables or frames must be provided to prevent equipment being washed directly on the floor.

4.4.14 Fabrication repair/improvement action list shall be adequately managed to ensure correct prioritisation and completion in timely manner (see appendix 5). (see clause 8.2.6)

4.5 UTILITIES - WATER, ICE, AIR & OTHER GASES

Utilities used within the production and storage areas shall be monitored to effectively control the risk of product contamination.

CLAUSE	REQUIREMENTS
4.5.1	All water (including ice and steam) used as a raw material in the manufacture of processed food, the preparation of product, hand-washing or for equipment or plant cleaning shall be supplied in sufficient quantity, be potable at point of use or pose no risk of contamination according to applicable legislation. Water which is detailed as "at-risk" (i.e., water used as an ingredient, water used for food contact surfaces or activities e.g., spray cooling) shall be passed through a foreign body filter before use, the mesh size shall be no greater than 1mm. The site shall hold a copy of all filter specifications to demonstrate compliance. Filters shall be positioned after any bulk storage tanks and as close as possible to the point of use. Filters shall be subject to the site's maintenance programme with inspections at a pre-determined frequency. The microbiological and chemical quality of water shall be analysed at least annually. As a minimum, all outlets where water is used as a product ingredient that is not subjected to further heat treatment equivalent to 70°C for 2 minutes by the site or customer shall be tested weekly. The sampling points (including drinking water machines), scope of the test and frequency of analysis for all other points shall be based on risk, taking into account the source of the water, on-site storage and distribution facilities, previous sample history and usage
4.5.2	Where the physical or chemical parameters of the water is critical to the product being manufactured, it shall be subject to suitable treatment to control the relevant parameters (e.g. denitrification). Clearly defined limits shall be detailed within the site's Quality Management System. Treatment shall be monitored and recorded at a frequency determined by risk assessment and through review of historical results.
4.5.3	The site shall have a designated member of the management team (the Site Water Manager) responsible for all matters relating to on-site water quality and safety. The Site Water Manager shall be the main point of contact with the water supplier and any contractors or service providers relevant to the water system. The name and qualifications of the responsible person should be recorded in the site's Quality Management System.
4.5.4	Where water is from a public supply, the site shall hold information relating to the source of the water and the treatment works. An annual summary of microbiological, physical and chemical analysis results for the water supply relevant to the site shall be held and reviewed as part of the site's HACCP review. The water company shall be notified that the site is a food manufacturing facility. The site shall request the water company to notify the Site Water Manager in advance of any changes to the water source that could affect product quality or safety so that appropriate measures can be put in place. If regular changes in the supply source could occur, information about the alternative supplies and any significant differences which could affect product quality or safety shall be established and prior notification of changes agreed with the water supplier.

4.5.5 Where water is abstracted from a borehole, the site shall have an up-to-date "extraction licence" in place. The site shall be able to demonstrate its compliance to the permit. Borehole water shall be wholesome, clean and suitable for its intended use and shall comply with the requirements of 4.5.1.

The borehole shall be constructed in a manner which prevents contamination from local surface run-off or flooding.

The borehole chamber must be in a secure location, protected from contamination and safeguarded against tampering.

A sampling point shall be available on the headworks or on the delivery pipework prior to any treatment process. Water abstracted from a borehole shall be monitored to ensure that usage does not exceed the permit.

The site shall undertake scheduled microbiological testing of borehole water in addition to the routine water sampling inside the factory. The frequency of testing must be based on the sites history of results, risk to product and the daily volume of water abstracted.

Chemical testing shall be undertaken at least twice a year, incorporating as minimum:

- pH, conductivity, nitrates, iron, manganese, total hardnes
- Pesticide screening
- Testing for heavy metals, e.g., Arsenic, cadium, lead, mercury.

4.5.6 An up-to-date schematic diagram shall be available of the water distribution system on site that indicates the direction of flow of the entire system, the location of filtration points, holding tanks, water treatment and water recycling as appropriate. The diagram shall be used as a basis for water sampling and the management of water quality. The schematic plan shall be reviewed annually.

Potable and non-potable water lines must be identified throughout the site. There must be no 'dead ends' on potable water lines.

4.5.7 The site shall have an up to date Legionella Risk Assessment which complies to The Control of Legionella Approved Code of Practice (L8) and covers all water storage, distribution systems, evaporator condensers and cooling towers on site. The risk assessment shall be completed by a member of the Legionella Control Association.

The site shall ensure that recommendations from the risk assessment are actioned in a timely manner.

4.5.8 The site shall have at least one of the following treatment systems in place for water used in high-risk or high-care areas as an ingredient or which comes into direct contact with food products (e.g. during product washing or cooling):

- water filtration to remove protozoan oocysts
- heat treatment of the water to achieve 70°C for 2 minutes (or equivalent)
- chemical treatment (e.g. chlorine or chlorine dioxide)
- UV treatment.

4.5.9 All water filtration and treatment systems shall be subject to suitable checks and controls to ensure they remain in effective operation with records in place to demonstrate that the monitoring and treatment systems are effective. This may include the following, where relevant:

- the design of all filtration systems shall ensure that no unfiltered water can enter the factory (where divert systems are in place these shall be secured and managed by the Site Water Manager)
- regular inspection, cleaning and replacement of filters following a documented hygiene procedure which avoids introducing contamination into the system
- sampling at a pre-determined frequency to verify residual chlorine or chlorine dioxide levels.

4.5.9

- UV equipment to be included on the PPM schedule as recommended by the equipment manufacturer. UV systems shall ideally be fitted with UV output measurement and hours of use counter with readings checked and documented at a pre-determined frequency. There shall also be an alarm to indicate when output levels have dropped.
- where the water contains suspended particles which may collect on the UV lamp, a lamp wiper system (preferably automatic) or a pre filter should be in place. Documented checks of the UV lamp clarity should be recorded.

4.5.10 Air and other gases used as an ingredient or that are in direct contact with products shall be monitored to ensure this does not represent a contamination risk. Compressed air that is in direct contact with the product shall be filtered at point of use. The gauge of filtration (e.g. mesh size) and frequency of checks shall be documented and based on risk assessment, taking into account the source of the air, the type of compressors used and the condition of the pipework (e.g. evidence of corrosion). The compressor suppliers or experts shall be consulted to establish any potential risks and any necessary testing, this may include microbiological analysis.

4.6 EQUIPMENT

All food-processing equipment shall be suitable for the intended purpose and shall be used to minimise the risk of contamination of product.

CLAUSE	REQUIREMENTS

4.6.1 All equipment shall be constructed of appropriate materials.

The design, construction and placement of equipment shall be based on risk, this should be defined by PDI conducted by adequately trained person, to prevent product contamination. For example, the use of the correct seals, impervious surfaces or smooth welds and joints, where they are exposed to product and could otherwise result in foreign body, microbiological or allergen contamination of the product. The design and placement of equipment shall ensure it can be effectively and safely dismantled, cleaned and maintained.

Equipment shall be commissioned in accordance with the Group Commissioning Procedure.

Installation work shall be followed by a documented hygiene sign-off.

4.6.2 There shall be a documented purchase specification for any new equipment detailing the site requirements for the equipment. This may, for example, include:

- any relevant legislation
- where applicable, requirements for food contact surfaces to meet legal requirements
- details of intended use of the equipment and the type of materials it will be handling. Depending on its intended use, new equipment to site (including second-hand equipment) may require authorisation from a multi-disciplinary team.

The supplier should provide evidence that equipment meets these site requirements prior to supply.

4.6.3 Equipment that is in direct contact with food shall be suitable for food contact and meet legal requirements where applicable.

4.6.4 Equipment which is not being used or taken out of service shall be cleaned and stored in a manner which does not pose a risk to the product. Equipment stored in internal production and storage areas shall be kept clean. Product contact equipment which has been stored but is not in daily use shall be re-cleaned prior to use.

4.6.5 Mobile equipment e.g. fork-lift trucks, pallet trucks, scissor lifts and ladders used in open product areas must not pose a risk to the product and must not be used in external site areas. Mobile equipment shall be clean, well maintained and stored in a suitable area when not in use.

Where the use of mobile equipment in open product areas cannot be avoided and poses a risk to the product, the equipment shall be cleaned and disinfected prior to entering production areas.

Battery charging equipment shall not be stored in open product areas unless the batteries are fully sealed/maintenance free.

4.6.6 A procedure shall be in place to manage the movement of static equipment to ensure that food safety is managed and the integrity of the equipment is maintained.

4.6.7 There shall be a documented process for modification to the factory equipment. An engineer, who has hygienic design knowledge, shall design any modification to the factory equipment or environment to ensure it remains hygienically designed and does not pose a risk to product safety.

Modification shall be verified by Site Hygiene Manager and nominated member of the Technical Management team.

4.7 MAINTENANCE

An effective maintenance programme shall be in operation for plant and equipment to prevent contamination and reduce the potential for breakdowns.

CLAUSE	REQUIREMENTS

4.7.1 The site shall have a robust system in place for planned preventative maintenance (PPM) and monitoring the condition of all plant and processing equipment that delivers good equipment maintenance in order to limit wear and tear and to effectively manage food factory fabrication to prevent foreign body or microbiological risks.

The system shall include:

- a schedule which defines the frequency for maintenance and inspection. The frequency shall be established during the commissioning phase in accordance with GE-002
- PPM tasks shall have clear, documented instructions which define the nature of the maintenance and inspection required and do not pose microbiological, chemical or foreign body risks. instructions shall include Health and Safety requirements
- documented evidence of completed maintenance tasks
- procedures to manage tasks not completed on scheduled dates
- monitoring and reporting of the schedule completion rate.

PPM shall be in place for all CCP's (e.g., metal detectors, temperature monitoring equipment) and Barrier Transfer Points (e.g., uv tunnels, alarmed roller doors)

The PPM system shall be monitored to ensure it is up-to-date and reacts to emerging trends in breakdowns and foreign body issues caused by equipment failures/breakdowns.

There shall be a formal review of the system annually as a minimum or more frequently where required by customer. The review shall consider:

- equipment installed and removed from site in the previous 12 months
- completion rate to establish if planning and resources are adequate
- any trends in breakdowns to determine if current PPM frequencies are appropriate
- foreign body issues/complaint trends caused by equipment failure/breakdowns.

4.7.2 In addition to any planned maintenance programme, where there is a risk of product contamination by foreign bodies arising from equipment damage, the equipment shall be inspected at predetermined intervals, the inspection results documented and appropriate action taken.

4.7.3 Wherever possible, maintenance work shall take place in workshops. Where maintenance work must take place in production areas then the site shall arrange for this to be performed out of production hours or away from production activity and open product wherever possible.

The site shall have controls in place to ensure that engineering activities which take place in production areas do not pose a risk to product safety or legality. Such controls shall include:

- engineer working stand ards (e.g. personnel standards)
- toolbox standards, see clause 4.7.9
- work carried out in active areas shall be risk assessed and adequate controls implemented to prevent product contamination
- small parts shall be stored in a manner which prevents them from being lost/misplaced (e.g. sealed containers, magnetic trays). Product containers/trays must not be used as storage containers for parts.
- tools and loose parts shall be attended at all times
- cleaning procedures, see clause 4.7.5
- where equipment has been removed from high-risk or high-care areas, acceptance procedures to ensure the cleanliness and removal of contamination hazards before being transferred back into the area.

Maintenance work shall only be carried out by competent site engineers, approved contractors or employees of the equipment manufacturer.

4.7.4 Where temporary repairs are made, these shall be documented and controlled to ensure that the safety or legality of products is not put at risk. Materials used shall be appropriate for a food production environment.

Temporary measures shall be permanently repaired as soon as practicable and within a defined timescale.

4.7.5 Where water drinking stations are provided in production facilities, drinks shall not be taken away from the area, must be consumed in the vicinity of the drinking station and have hand wash facilities available. Where cups are provided, these shall be single use with a suitable receptacle for disposal after use to be located near the drinking station.

4.7.6 Toilets shall be adequately segregated and shall not open directly into production or packing areas. Toilets shall be provided with hand-washing facilities comprising:

- basins with soap and water at a suitable temperature
- adequate hand-drying facilities
- advisory signs to prompt hand-washing
- hand sanitiser dispensers located at the exit point.

Toilet facilities shall be fit for purpose and suitable for all staff on site. Toilets shall be maintained in a clean and hygienic condition.

4.7.7 Engineering workshops shall be kept clean and tidy with a documented cleaning schedule in place. Controls shall be in place to prevent the transfer of engineering debris to production and storage areas. As a minimum, this shall include:

- swarf mats fitted in the workshop which covers the threshold of exit points leading to production areas
- tools stored off the floor and captive to the workshop or adequately cleaned prior to transferring into production areas
- production equipment repaired in the workshop shall be subject to the same documented sign off detailed in clause 4.7.5
- handwash facilities, see clause 4.8.4
- dedicated overalls for workshops, there shall be adequate segregation of overalls for workshops and overalls for use in production.

Access to workshops shall be restricted to authorised personnel only.

4.7.8 Non-food grade and food grade oils or lubricants shall be clearly identified and segregated from each other. Non-food grade oils and lubricants shall be securely stored to prevent misuse.

Oils and lubricants shall be free from allergens. Data sheets for oils and lubricants shall be readily available in the workshop.

4.7.9 Toolboxes shall be clean, tidy and locked when not in use. Each box shall have an inventory of its contents that is checked at regular intervals, these checks shall be documented. Formal procedures shall be in place and applied where contents are identified as damaged or missing.

Toolboxes which can be placed on food contact surfaces shall be appropriately stored when not in use and shall not come into direct contact with the floor.

Toolboxes shall not contain prohibited items in the food area (e.g., personal food, personal keys, personal mobile phone)

Contractors tools, toolboxes, tool bags etc. shall be subject to the requirements of this clause. A site engineer shall inspect and record all items entering production areas. These items shall be inspected after maintenance work has been completed.

4.7.10 A breakdown procedure shall be in place documenting what to do in the event of a prolonged breakdown with regard to product in process/on line and details of the authorisation procedures required to recommence production.

4.7.11 Wire brushes and scourers shall be in good condition and stored away from the production process and open product.

Wire brushes shall not have wooden handles and shall be a subject to regular documented condition inspection.

4.8 STAFF FACILITIES

Staff facilities shall be sufficient to accommodate the required number of personnel, and shall be designed and operated to minimise the risk of product contamination. The facilities shall be maintained in good and clean condition.

CLAUSE	REQUIREMENTS
4.8.1	Designated changing facilities shall be provided for all personnel, whether staff, visitor or contractor. These shall be sited to allow direct access to the production, packing or storage areas without recourse to any external area. Where this is not possible, a risk assessment shall be carried out and procedures implemented accordingly (e.g. the provision of cleaning facilities for footwear).

4.8.1 Engineering and hygiene protective clothing shall be stored separately.

These facilities shall be well maintained and in a clean and hygienic condition.

4.8.2 Storage facilities of sufficient size to accommodate personal items shall be provided for all personnel who work in raw material handling, preparation, processing, packing and storage areas.

4.8.3 Outdoor clothing and other personal items shall be stored separately from production clothing within the changing facilities. Facilities shall be available to separate clean and dirty production clothing.

Where the site permits the storage of ambient items in lockers, these items shall be adequately sealed to prevent spillages and contamination of protective clothing. Food items are not permitted to be consumed within changing rooms.

4.8.4 Suitable and sufficient hand-washing facilities shall be provided at access to, and at other appropriate points within, production areas. Such hand-washing facilities shall provide, as a minimum:

- advisory signs with photographic standards to prompt hand-washing
- a sufficient quantity of water at a suitable temperature to promote effective hand-washing
- water taps with hands-free operation
- liquid/foam soap
- hand sanitiser
- single-use towels or suitably designed and located air driers
- suitable receptacles for used paper towels where used.

Hand-dryers shall not be located in production areas. The use of hand-dryers at other handwashing stations shall be adequately risk assessed with consideration to aerosols and any potential harbourage points within the hand-dryer unit. Any harbourage points identified shall be included within the site's environmental monitoring system (see 4.11.16).

4.8.5 Where boot washes are in use, these shall be positioned in a manner that does not pose a risk to product. Boot washes shall be cleaned at an appropriate frequency.

4.8.6 Where there is a need for factory / hygiene operatives to work in external and internal areas (e.g., removal of waste), facilities for handwashing, storage of overalls and washing footwear shall be provided and maintained.

4.8.7 Toilets shall be adequately segregated and shall not open directly into production or packing areas.

Toilets shall be provided with hand-washing facilities comprising:

- basins with soap and water at a suitable temperature
- adequate hand-drying facilities
- advisory signs to prompt hand-washing
- hand sanitiser dispensers located at the exit point.

Toilet facilities shall be fit for purpose and suitable for all staff on site. Toilets shall be maintained in a clean and hygienic condition.

4.8.8 Designated smoking areas shall be provided externally at a location that ensures smoke cannot reach production and storage areas. Adequate arrangements for dealing with smokers' waste shall be provided at smoking facilities. Items used for smoking (i.e. cigarettes, rolling tobacco/papers, lighters, electronic cigarettes, liquids) are not permitted to be used or brought into product or storage areas.

4.8.9 Designated smoking areas shall be provided externally at a location that ensures smoke cannot reach production and storage areas. Adequate arrangements for dealing with smokers' waste shall be provided at smoking facilities. Items used for smoking (i.e. cigarettes, rolling tobacco/papers, lighters, electronic cigarettes, liquids) are not permitted to be used or brought into product or storage areas.

4.8.10 All food brought onto manufacturing premises by staff, visitors and contractors shall be appropriately stored in a clean and hygienic state, containers shall be free from glass (including thermos flasks). No food shall be taken into storage, processing or production areas.

The storage and consumption of food shall only take place in designated areas. The site shall provide chilled units for perishable items.

Where eating of food is allowed outside during breaks, this shall be in suitable designated areas with appropriate control of waste.

The controls for the storage and consumption of food items shall be effectively communicated to all staff, visitors and contractors. This shall include any allergen restrictions that the site may have.

4.8.11 Where catering facilities (including vending machines) are provided on the premises, they shall be suitably controlled to prevent contamination of products (e.g. as a source of food poisoning or introduction of allergenic material not permitted on site).

In addition to the quarterly audit (see clause 3.3.2), the activity from on-site catering services (i.e. hot canteens) shall be monitored to ensure that catering activities do not pose a risk to product or staff. Such controls shall include;

- daily temperature checks on cooked food (hot holding) and chilled storage
- environmental swabbing programme

The site shall be aware of the outcome of all local authority inspections, reports shall be shared with the site management team.

4.8.12 Where hot food is prepared on site (canteen kitchen operated by Cranswick staff), this shall be suitably controlled to prevent contamination of products (e.g., as a source of food poisoning or introduction of allergenic material not permitted on site), and the following documentation shall be in place:

- HACCP plan
- Supplier Approval
- Standard Operating Procedures
- Food Safety monitoring documentation
- Hygiene Standards
- GMP
- Pest Management
- Internal Audits
- Training

4.8.13 Inspection of staff areas shall be completed at a pre-determined frequency. The inspection shall include, but is not limited to;

- cleanliness and condition of staff facilities
- temperature checks of vending machines and refrigeration equipment
- adherence to site policy of permitted foods allowed on site and use of glass containers.

4.9 CHEMICAL AND PHYSICAL PRODUCT CONTAMINATION CONTROL: RAW MATERIAL HANDLING, PREPARATION, PROCESSING, PACKING AND STORAGE AREAS

FUNDAMENTAL: Appropriate facilities and procedures shall be in place to control the risk of chemical or physical contamination of product.

4.9.1 CHEMICAL CONTROL

CLAUSE	REQUIREMENTS
4.9.1.1	Processes shall be in place to manage the use, storage and handling of non-food chemicals to prevent chemical contamination. These shall include, as a minimum: • an approved list of chemicals for purchase • availability of material safety data sheets and specifications • confirmation of suitability for use in a food-processing environment • avoidance of strongly scented products • the labelling and/or identification of containers of chemicals at all times • a designated storage area with restricted access to authorised personnel • bulk containers stored on bunded pallets • use by trained personnel only. • procedures to manage any spills • procedures for the safe, legal disposal or return of obsolete or out-of-date chemicals and empty chemical containers. Chemicals shall be used in an appropriate and safe manner.
4.9.1.2	Where strongly scented or taint-forming materials have to be used, for instance for building work, procedures shall be in place to prevent the risk of taint contamination of products.
4.9.1.3	Storage of chemicals for non-food production areas shall be segregated and access restricted. The consequences of spillage of non-food chemicals shall be risk assessed and appropriate controls implemented to prevent product contamination. Highly flammable, hazardous chemicals shall be stored in fire retardant cabinets.

4.9.2 METAL CONTROL

CLAUSE	REQUIREMENTS
4.9.2.1	The site shall have a documented policy for the controlled use and storage of sharp metal implements including knives, cutting blades on equipment, needles and wires. The controls shall include: • only site issued sharps in use • controlled knives, blades, ergo steels, chainmail protective equipment and scissors are uniquely identified • no snap-off blades in use • knives and blades only used for the task for which they were designed • sharps to be accounted for and the condition and cleanliness checked and recorded (start and end of shift as a minimum), the site shall complete a risk assessment to determine where greater frequency is required • storage for knives and blades used for processing must be clean and controlled • blade changes to be recorded • knife and blade sharpening to take place away from production areas, items to be cleaned when returned • sharps must not be permitted in personal lockers • in the event of breakage or loss, all parts to be accounted for and the incident logged, corrective action to be taken to prevent reoccurence

4.9.2.1

- in the event of breakage or loss a system must be in place to identify and control potentially contaminated product.

Use and storage of sharp metal implements shall be risk assessed to understand H&S risk to employees and implement adequate preventative controls.

4.9.2.2 The purchase of ingredients and packaging which use staples or other foreign-body hazards as part of the packaging materials shall be avoided.

Staples, paper clips and drawing pins shall not be used in open product areas.

Where staples or other items are present as packaging materials or closures, appropriate precautions shall be taken to minimise the risk of product contamination.

4.9.3 GLASS, BRITTLE PLASTIC, CERAMICS AND SIMILAR MATERIALS

CLAUSE	REQUIREMENTS

4.9.3.1 The site shall have a policy for the control of glass, plastic and other similar materials in areas where open products are handled or there is a risk of product contamination. This shall include a commitment to actively reduce the amount of glass present in production areas.

The site shall avoid the purchase of raw materials in glass containers. Where this is not possible, they shall be received and decanted into a suitable container in a controlled environment. Glass containers shall be disposed of safely.

4.9.3.2 Any plastic or similar material items used for food contact purposes or in open product areas shall be colour coded to ensure colour differs from the product in order to enable easy detection should breakage occur.

Where possible, food contact plastic or similar material items used for food contact purposes shall be detectable by the foreign-body detection equipment on site.

4.9.3.3 Plastics and similar materials (both hard and soft) shall be of the correct strength / durability for their use and shall be maintained in good condition..

4.9.3.4 Procedures for handling glass, plastic and other similar materials (other than product packaging) shall be in place where open products are handled or there is a risk of product contamination. Non – product areas such as production offices shall also be considered.

These procedures shall include, as a minimum:

- a list of items detailing location, number, type of material and condition
- recorded checks of the condition of items, carried out at a specified frequency that is based on the level of risk to the product with consideration to the type of material and its location in relation to open product and activities that take place in the area. Food contact items daily checks shall be included in prestart up checks.
- details on cleaning or replacing items to minimise the potential for product contamination.

Items which are found to be damaged shall be removed, repaired or replaced within a timescale appropriate to the risk. Where production is to continue with damaged glass or hard plastic, this shall be authorised and formally signed off by a senior manager. The timescale for all actions shall be confirmed by management.

4.9.3.5 Procedures for the management of glass, plastic and other similar material breakages shall be in place. The procedures shall include, as a minimum

- stopping of production, cleaning method and restriction of movement through the affected area
- escalation of breakage to management
- quarantine of materials and area potentially affected
- clean up of breakage and disposal/cleaning of cleaning equipment
- a sample of broken material to be retained and stored in a safe manner
- safe disposal of broken material and contaminated product
- repair or replacement of damaged item
- changing of workwear and inspection of footwear
- inspection of the area prior to authorising production to continue
- recording of the breakage incident and action taken.

Staff shall be trained in the correct procedure for managing breakages.

4.9.3.6 Where they pose a risk to product, glass windows shall be protected against breakage.

4.9.3.7 Where they pose a risk to product, bulbs and strip lights (including those on electric fly-killer devices) shall be protected with a shatter resistant film or housed within a protective environment. Where full protection cannot be provided, alternative management such as wire-mesh screens or monitoring procedures shall be in place.

4.9.4 PRODUCTS PACKED INTO GLASS, PLASTIC OR OTHER BRITTLE CONTAINERS

CLAUSE	REQUIREMENTS

4.9.4.1 The storage of the containers shall be segregated from the storage of raw materials, product or other packaging. There shall be a 'zero tolerance' policy where broken glass or ceramic is not accepted within the production environment.

4.9.4.2 Systems shall be in place to manage container breakages between the container cleaning/inspection point and container closure. This shall include, as a minimum, documented instructions which ensure:

- the removal and disposal of at-risk products in the vicinity of the breakage; this may be specific for different equipment or areas of the production line
- the effective cleaning of the line or equipment which may be contaminated by fragments of the container; cleaning shall not result in the further dispersal of fragments, for instance by the use of high-pressure water or air
- the use of dedicated, clearly identifiable cleaning equipment (e.g. colour-coded) for removal of container breakages; such equipment shall be stored separately from other cleaning equipment
- the use of dedicated, accessible, lidded waste containers for the collection of damaged containers and fragments
- a documented inspection of production equipment is undertaken following the cleaning of a breakage to ensure cleaning has effectively removed any risk of further contamination
- authorisation is given for production to restart following cleaning
- the area around the line is kept clear of broken glass.

4.9.4.3 Records shall be maintained of all container breakages on the line. Where no breakages have occurred during a production period, this shall also be recorded. This record shall be reviewed to identify trends and potential line or container improvements.

4.9.4.4 Plastic packaging for chilled and frozen products shall be specified for use in frozen / chilled environment to avoid plastic becoming brittle and shattering, causing foreign body risk.

4.9.5 WOOD

CLAUSE	REQUIREMENTS
4.9.5.1	Wood should not be used in open product areas except where this is a process requirement (e.g. smoking of materials) or part of the product (e.g. wooden skewers). There shall be adequate controls in place to ensure that these items do not become a source of contamination. Where the use of wood cannot be avoided, the condition of wood shall be continually monitored to ensure it is in good condition and free from damage or splinters which could contaminate products. Wood pallets shall be covered when used in open product areas. Wood used for food contact purposes shall be fit for purpose (e.g., free from damage or splinters, free from taint, wood treatments and approved for food use.). Usage shall be controlled and monitored (e.g., skewers). The site shall have a site plan that highlights the areas where wood is permitted.

4.9.6 OTHER PHYSICAL CONTAMINANTS

CLAUSE	REQUIREMENTS
4.9.6.1	Based on risk, procedures shall be implemented to minimise other types of foreign-body contamination (i.e., types of contamination that are not specifically covered in clause 4.9.2 – 4.9.5 e.g., gaskets, rubber seals, silicone seals, safety goggles)
4.9.6.2	The site shall have controls in place for deboxing and debagging. Procedures shall be in place for each packaging type handled on site and shall include the following: • the use of an approved safety knife to open boxes and bags • photographic standards showing the correct method • materials not opened directly over open product (i.e. kettles or hoppers). Part used bags to be stored in a hygienic manner to prevent contamination.
4.9.6.3	Pens used in open product areas shall be controlled to minimise the risk of physical contamination. Pens shall be without a detachable lid, of contrasting colour to the product and detectable by foreign-body detection equipment.
4.9.6.4	Pens and other items such as plasters and earplugs shall be tested through all of the foreign-body detection equipment used on site prior to issuing, where this is not practical then it shall be risk assessed and tested through the least sensitive equipment. The item shall be placed on product which has previously passed through the detection equipment successfully, the test shall be completed against the lowest sensitivity setting for each piece of equipment. 5 items from each batch shall be placed through the detection equipment in or on a pack of product which has been successfully passed through the equipment. The batch number and results shall be recorded. Containers holding items authorised for use shall be labelled with the date the test was completed.
4.9.6.5	The site shall ensure that it has sufficient containers for operatives to use for tasks such as sprinkling rubs or crumbs onto product. Food contact packaging (i.e. plastic trays, foil tins) shall not be used for this purpose. Additionally, transport trays/crates, food contact dolvas or other food contact containers shall not be used to store other materials (e.g., nuts, bolts, tools, machinery parts, chemicals), until its risk assessed and appropriate controls are in place to prevent cross contamination

4.9.7 CONTROL OF HIGH-RISK EQUIPMENT IN CLOSE PROXIMITY TO OPEN PRODUCTS

CLAUSE	REQUIREMENTS
4.9.7.1	Detailed, documented pre-start up checks of high-risk equipment, in close proximity to open product, shall be in place. Pre-start up checks shall confirm absence of foreign body risks and/or identify any damage to the high-risk equipment with specified immediate corrective action. Any issues that are not immediately resolved must be logged on the non-conformance log and rectified promptly.

4.10 FOREIGN-BODY DETECTION AND REMOVAL EQUIPMENT

The risk of product contamination shall be reduced or eliminated by the effective use of equipment to remove or detect foreign bodies.

4.10.1 SELECTION AND OPERATION OF FOREIGN-BODY DETECTION AND REMOVAL EQUIPMENT

CLAUSE	REQUIREMENTS
4.10.1.1	A documented assessment in association with the HACCP study shall be carried out on each production process to identify the potential use of equipment to detect or remove foreign-body contamination. Typical equipment to be considered may include: • filters • sieves • metal detection • magnets • optical sorting equipment • X-ray equipment • other physical separation equipment (e.g. gravity separation, fluid bed technology). Where the need for foreign body detection equipment has been identified, the site shall consider the use of x-ray equipment. There shall be consideration of historical foreign body issues, composition/size of product and commonly associated contaminants. Where x-ray equipment has not been used, there shall be documented justification as to why. The site shall obtain agreement with customers for any product that does not undergo any foreign-body detection or removal process, this shall be established within the product specification signed by both parties.
4.10.1.2	The type, location and sensitivity of the detection and/or removal method shall be specified as part of the site's documented system. Equipment shall be positioned in line with the standard process flow and located as close as possible to the end of the process. Industry best practice shall be applied with regard to the nature of the ingredient, material, product and/or the packed product. The location of the equipment or any other factors influencing the sensitivity of the equipment shall be validated and justified. Where metal detection or X-ray equipment is not used, justification shall be documented and agreed with the relevant customer. Detection equipment shall meet or surpass any customer requirements for sensitivity limits and shall comply with required failsafe mechanism.

4.10.2 METAL DETECTION AND X-RAY EQUIPMENT

CLAUSE	REQUIREMENTS

4.10.2.1 Metal detectors shall have an aperture size appropriate for the products being inspected.

All foreign body detection systems shall be able to operate at the maximum running speed of the production line and have security features in place to prevent unauthorised adjustments to the sensitivity and access to rejected product. Restarts shall be controlled so that after a fault the system can only be reset and started using a key or password by an authorised person. The equipment shall not be able to be turned off and back on again to reset or clear a fault state.

Passwords and keys shall only be provided to authorised personnel. Where keys are used, these shall not be left in the equipment unattended.

4.10.2.2 All relevant staff shall be adequately trained in the operating principles of the detection equipment, site procedures and a basic level of training on HACCP principles. Only staff that can demonstrate the required level of competency shall be authorised to operate detection equipment. Competency assessments for detection operatives shall be completed annually or when there is a significant change in site procedure.

There shall be a subject matter expert (SME) present on site during production hours. The SME shall be trained to Tier C in the detection equipment used on site. All engineers required to perform maintenance on detection equipment shall be trained to Tier D.

TRAINING REQUIREMENTS

Tier C – SME

This level of training shall be given by the metal detector/x-ray manufacturer, their appointed agent or suitable competent external company in conjunction with the person responsible for technical/quality standards on site. This must be carried out on a 3 yearly basis or more frequently if required and must cover:

- the principles of Maintenance and Performance Validation
- understanding of their equipment's specification
- full understanding of the customer requirements
- new product set-up including how to optimise detection sensitivities

Tier D – Engineering Training

This level of training must be given by the detector/x-ray manufacturer, their appointed agent or suitable competent external company to site Engineers. Tier D Engineers must have an appreciation of the content of the previous level of training (i.e. Tier C) but must also be trained on:

- how to trace faults – root cause analysis
- importance of maintenance and Performance Validation
- manufacturer's maintenance recommendations, lubrication requirements, critical locations on the equipment
- good engineering practices to reduce the risk of product contamination.

4.10.2.3 Detection equipment shall be commissioned at first installation and validated and verified as operating effectively to an optimum setting for the products manufactured without the generation of a high number of false rejects. Initial training upon purchase/installation of detection equipment shall be carried out by the manufacturer or appointed agent with training records kept. All detection equipment shall be included in the site's planned preventative maintenance programme.

All detection equipment shall undergo a performance validation annually. The performance validation shall confirm that the system and its settings are working to optimum levels, all failsafe functions are working correctly and that the detection equipment operator is operating the equipment correctly.

Performance validations shall be completed by the equipment manufacturer, their appointed agent or suitable competent external company. Evidence of competency shall be retained.

Where detection equipment has been relocated, the site shall complete documented inspection to confirm that the system and its settings are still working to the optimum levels established during the performance validation.

4.10.2.4 Foreign body detection equipment shall have the following features installed;

Belt Stop metal detectors shall:

- have an audible and visual alarm that indicates the status of the system
- stop quickly enough for the contaminated product to remain on the belt, if the product passes onto the next belt then that shall also be linked to the metal detector.

Pipeline metal detectors shall have:

- have an audible and visual alarm that indicates the status of the system
- an automatic rejection system which diverts contaminated product
- a lockable reject bin which can only be accessed by authorised staff
- an air pressure monitor linked to the conveyor drive which stops the belt when it detects a fault with the air pressure for the reject mechanism.

Conveyorised detection equipment shall have:

- have an audible and visual alarm that indicates the status of the system
- an automatic rejection system
- a lockable reject bin which can only be accessed by authorised staff
- sensors on the reject bin which are linked to the conveyor drive which stops the belt when the reject bin is full and when the reject door has been left open for more than a pre-set time (maximum 30 seconds)
- a tunnel guard which encloses the line between the search head and the reject bin. If an access point is required, this shall be either lockable or interlocked to the conveyor drive
- an air pressure monitor linked to the conveyor drive which stops the belt when it detects a fault with the air pressure for the reject mechanism
- a photogating sensor to facilitate the correct timing of the reject system
- a reject confirmation system linked to the conveyor drive which stops the belt if contaminated product has not successfully entered the reject bin.

Belt stop metal detectors shall only be used for large products not capable of being automatically rejected from the line. There shall be suitable storage facilities (i.e. lockable cages) located near the metal detector for holding rejected material pending further inspection. Only authorised personnel shall have access to rejected product.

Where the detector equipment is used for simultaneously testing products in a multi lane format across a production line, a retracting belt rejection system shall be used. A Reject Activation Confirmation, which confirms that the belt has fully retracted may be used instead of a reject confirmation system.

4.10.2.5 Metal detector test pieces shall contain a metal sphere of a known diameter. Each test piece shall comprise of a ferrous, non-ferrous or stainless steel (316 grade) ball. For products with foil packaging, only a ferrous test piece is required however this shall be tested twice at the leading and trailing edges of product.

X-ray test pieces shall consist of a stainless-steel ball, stainless steel wire (both 316 grade) and 2 other materials reflective of the production environment (e.g. glass, PTFE, ceramic).

All test pieces shall be of a known dimension and/or density with a corresponding certificate of conformance against a recognised quality standard and shall match the limits specified by the site's procedures. Test pieces shall be colour coded by material type to assist with correct identification.

The site shall have suitable controls in place for test pieces, this shall include:

- test pieces to be stored securely to prevent unauthorised access when not in use
- documented test piece integrity checks shall be completed at the start and end of shift
- the serial number for each test piece shall be recorded when used. This may be recorded once at the start of shift however records must be able to show when test pieces have been replaced during a shift.

4.10.2.6 Where test packs are used, these shall be:

- initially put through the detector to verify that they are free from contamination
- clearly labelled and controlled to ensure that they are not inadvertently returned to the production line. Test packs shall be labelled with product, date, time, test piece type and size
- made up at a frequency which reflects the nature, durability and shelf life of the product and temperature controlled where required.

The location of the piece of metal within the test piece shall be considered when generating test packs. For metal detectors, test pieces shall be positioned so that they travel as close as possible to the centre of the aperture.

For pipeline detector equipment, where placement of the test pieces into the product flow is not possible, the test pieces shall be placed between the pipeline and the test head. Test pieces shall be inserted in the direction of the product flow.

4.10.2.7 A routine consecutive test to verify that the sensitivity and rejection mechanism is working effectively under normal working conditions shall be completed at the following frequency:

- at the start of each production shift
- at the end of each production shift
- at the start of a production run defined by a change to recipe, customer or size of pack
- at the end of a production run defined by a change to recipe, customer or size of pack
- every hour
- after a break and/or emergency evacuation
- following the identification and rectification of a fault.

The site may reduce the frequency of testing for bulk products where a change in product does not require a change of programme to the metal detector however this shall be adequately risk assessed.

Test packs shall be passed through the equipment at normal spacing/standard line speed. For metal detectors, test pieces shall be placed on test packs at the leading edge, centre and trailing edge and passed through the detector in that order.

If any test is unsuccessful, the following actions shall be taken:

- stop production immediately and call an engineer
- identify all product packed since last successful test and place on-hold
- prior to starting the line, carry out all routine and failsafe tests to verify all functions are working correctly
- all product on-hold to be re-tested.

4.10.2.8 The following are considered to be fail safe functions. Where fitted, these shall be verified to be operational at the start and end of each production shift:

- reject confirmation
- reject bin full
- reject bin door status monitor
- air pressure failure
- photogating sensor and reject check sensor
- memory test.

Additionally, a large ferrous metal test (20mm minimum) shall be completed for metal detectors at the start and end of shift.

If any test is unsuccessful then the necessary corrective action shall be taken. All failsafe tests shall be repeated to verify the fault has been rectified.

4.10.2.9 The site shall have procedures in place that explain how to perform the tests to the frequencies defined and corrective action to be taken in accordance with this standard. Additionally, procedures shall include:

- the principles of metal detection and/or x-ray
- limits for each test piece and product type
- responsibility for completing tests/checks and inspection of rejected product
- reason for test/checks (i.e. start/end of run, hourly, after a fault/maintenance)
- recording of test results, number of rejects and number of foreign body finds
- reconciliation of the number of rejects in the bin against the number rejected packs by the detection equipment
- all system checks to be completed after a fault or significant maintenance work
- the production line to be stopped where multiple/high level of failures occur which indicate an issue with either the product or equipment
- identification and secure storage of product pending investigation and/or re-testing.

4.10.2.10 Rejected product shall be handled in accordance with the relevant customer requirements.

For customers that do not allow rejected product to be packed, the site shall label all rejected packs once they are removed from the reject bin. Product shall be re-tested through the detection equipment a minimum of 3 times in different orientations, including once in the original orientation. If the product successfully passes on all occasions, then no further action is required and the product must be disposed of immediately. Any product rejected on further testing shall be inspected to locate the contaminant.

For customers that allow rejected product to be packed after further inspection, the site shall re-test the product through the detection equipment a minimum of 3 times in different orientations, including once in the original orientation. If the product successfully passes on all occasions, then this can be placed back onto the production line. Any product rejected on further testing shall be labelled as a rejected pack for further inspection to locate the contaminant.

Product that requires further inspection shall be stored in a secure location to prevent unauthorised access. Inspections shall take place away from the packing line and be performed within the same production shift and prior to the product leaving site.

Contaminant finds shall be fully investigated to identify the origin with appropriate corrective actions to prevent recurrence. The site shall use a multidisciplinary team where appropriate.
The investigation and actions taken shall be documented. Where required by the customer, they shall be notified of 2 or more foreign body finds (internal finds or rejected product) of the same nature from the same production line in a 24-hour period.

The site shall trend internal contaminant finds to provide an indication as to which products/ production lines are susceptible to risk and establish actions to reduce occurrence.

4.10.3 FILTERS AND SIEVES

CLAUSE	REQUIREMENTS
4.10.3.1	Filters and sieves used for foreign-body control shall be of a specified mesh size or gauge and designed to provide the maximum practical protection for the product. The mesh shall be constructed from non-corrodible metal (e.g. chrome steel) or where nylon is used, this shall be coloured to contrast with the product being sieved. Sieves shall be labelled with a unique identification number and the mesh size. A record of all filters and sieves on site shall be maintained including the location, size, material and required inspection frequency. Filters and sieves shall be included on the planned preventative maintenance schedule and replaced when necessary.
4.10.3.2	Filters and sieves shall be regularly inspected or tested for damage at a documented frequency based on risk, as a minimum these checks shall be completed at the start and end of shift. Where a filter or sieve is defined as a CCP in the process, the frequency of checks shall meet the requirements in clause 4.10.2.7. Records of the checks shall be maintained. Where defective filters or sieves are identified this shall be recorded and the potential for contamination of products investigated with appropriate action taken.
4.10.3.3	Sieves/filters must completely cover the container that the ingredient is being sieved into. ingredients shall not be sieved directly over production equipment (i.e. hoppers, kettles). Liquids shall pass through filters at the last possible point before use. Sieved ingredients shall be identifiable from un-sieved ingredients. A record of ingredients sieved shall be kept, detailing the name, batch number, quantity sieved, date and time, mesh size used and details of any contaminants found. Foreign matter shall be identified as soon as practical and investigations as to potential sources initiated immediately.
4.10.3.4	The site shall risk assess all dry ingredients and liquids to determine if sieving or filtration is required taking into consideration if the action of sieving causes the ingredient particles to fractionate. The site shall refer to customer guidance where specific mesh sizes are required.

4.10.4 MAGNETS

CLAUSE	REQUIREMENTS
4.10.4.1	The type, location and strength of magnets shall be fully documented. Procedures shall be in place for the inspection, cleaning, strength testing and integrity checks. Records of all checks shall be maintained. Magnets shall be included on the planned preventative maintenance schedule. Records of foreign matter found shall be kept. Foreign matter shall be identified as soon as practical and investigations as to potential sources initiated immediately.

4.10.5 OPTICAL SORTING EQUIPMENT

CLAUSE	REQUIREMENTS
4.10.5.1	Each unit shall be checked in accordance with the manufacturer's instructions or recommendations. Checks shall be documented.

4.10.6 CONTAINER CLEANLINESS, GLASS JARS, CANS AND OTHER RIGID CONTAINERS

CLAUSE	REQUIREMENTS
4.10.6.1	Based on risk assessment, procedures shall be implemented to minimise foreign-body contamination originating from the packaging container (e.g. jars, cans and other pre-formed rigid containers). This may include the use of covered conveyors, container inversion and foreign-body removal through rinsing with water or air jets.
4.10.6.2	The effectiveness of the container-cleaning equipment shall be checked and recorded during each production run. Where the system incorporates a rejection system for dirty or damaged containers, the check shall incorporate a test of both the detection and effective rejection of the test container.

4.10.7 OTHER FOREIGN BODY DETECTION

CLAUSE	REQUIREMENTS
4.10.7.1	Other foreign-body detection and removal equipment, such as gravity separation, fluid bed technology or aspirators, shall be checked in accordance with the manufacturer's instructions or recommendations. Checks shall be documented

4.11 HOUSEKEEPING AND HYGIENE

FUNDAMENTAL: Housekeeping and cleaning systems shall be in place which ensure appropriate standards of hygiene are maintained at all times and the risk of product contamination is minimised.

CLAUSE	REQUIREMENTS
4.11.1	The premises and equipment shall be maintained in a clean and hygienic condition. A clean as you go approach shall be in place.
4.11.2	There shall be a suitably competent person responsible for overseeing all cleaning functions and the standards achieved. The site shall work in collaboration with its hygiene supplier to establish and validate an effective cleaning programme to meet the requirements of this standard. The validation process shall be reviewed If there are changes to the process, cleaning methods, chemicals, equipment, addition or removal of products containing allergens or different species, or significant change to personnel. The cleaning programme shall identify areas where dry cleaning is optimal and where water/hoses can be used on equipment. There shall be evidence that the cleaning programme is continually monitored to verify its efficacy and identify where additional resource may be required. Post clean inspections, swab results and environmental monitoring data shall be considered as part of this review.
4.11.3	Cleaning personnel shall be trained in the use and handling of chemicals, microbiological contamination controls and against cleaning procedures. Personnel carrying out cleaning shall be effectively trained and supervised to ensure that cleaning instruction cards (CIC) are being followed and jobs are being carried out correctly, they shall not be permitted to carry out unsupervised cleaning until relevant training has been completed. Training shall be carried out by a suitably competent person with records that include the CIC trained against (including version number), length of time spent training and a competency check to verify that the cleaning standards required can be achieved by the operative. The necessary resources and time must be set aside to ensure the cleaning schedule can be completed in full and without posing a risk of contamination to nearby product. Resources include personnel, cleaning equipment, chemicals and protective clothing. Hygiene window requirements shall be known, documented and planned into production capacity.
4.11.4	Where it is necessary to dismantle equipment for cleaning purposes or to enter large equipment for cleaning, this shall be appropriately scheduled and, where necessary, planned for non-production periods. Cleaning staff shall be adequately trained or engineering support provided where access within equipment is required for cleaning.
4.11.5	Cleaning schedules and methods shall be determined by risk assessment and be in place across all areas. The risk assessment must be used to determine the level and frequency of cleaning and the equipment required. Cleaning methods shall be developed in a way that prevents the risk of contamination to exposed product, packaging and equipment (e.g. equipment cleaned off the floor, bucket stands, appropriate water pressure to reduce aerosols). High pressure hoses shall not be used where there is a risk of contamination to product. Cleaning schedules shall be fully integrated into the production scheduling and planning process.

4.11.5	The cleaning programme shall include, as a minimum, cleaning of: • equipment • production/packing, storage and maintenance areas (including walls and ceilings) • employee facilities • drains • external areas. The site shall have emergency cleaning procedures in place for blocked drains and bodily fluid spillages.
4.11.6	Where panels on equipment are only accessible by engineering e.g., electrical and pneumatic panels, there shall be a documented schedule in place for cleaning and this shall be based on risk assessment and shall consider the time it takes for build-up of food debris, water ingress, mould, yeasts and pathogens.
4.11.7	Cleaning procedures for the processing equipment and food contact surfaces shall, as a minimum, include: • responsibility for cleaning • item/area to be cleaned • frequency of cleaning • method of cleaning, including dismantling equipment for cleaning purposes where required (photographs shall be used where relevant e.g. for more complex items of equipment). • cleaning chemicals and concentrations • cleaning materials to be used • key inspection points for final inspection • PPE • cleaning records and responsibility for verification. The procedures shall be implemented to ensure appropriate standards of cleaning are achieved.
4.11.8	Cleaning schedules and records shall make reference to the relevant CIC card for each task. Documented records of cleaning shall include: • date the cleaning task was carried out • sign off by the person who carried out the cleaning • sign off by an authorised supervisor or manager • where issues have been found, additional cleaning shall be recorded • explanation for cleaning tasks not completed along with a new timescale for completion.
4.11.9	Limits of acceptable and unacceptable cleaning performance shall be defined for food contact surfaces and processing equipment. These limits shall be based on the potential hazards relevant to the product or processing area (e.g. microbiological, allergen, foreign-body contamination or product-to-product contamination). Therefore, acceptable levels of cleaning may be defined by visual appearance, ATP bioluminescence techniques (see glossary), microbiological testing, allergen/species testing or chemical testing as appropriate. The site shall define the corrective action to be taken when monitored results are outside of the acceptable limits. Where cleaning procedures are part of a defined prerequisite plan to control the risk of a specific hazard, the cleaning and disinfection procedures and their frequency shall be validated and records maintained. This shall include the risk from cleaning chemical residues on food contact surfaces.

4.11.10 The cleanliness of equipment shall be checked before equipment is released back into production.

The results of checks on cleaning, including visual, analytical and microbiological checks, shall be recorded and used to identify trends in cleaning performance and to instigate improvements where required.

4.11.11 Automated cleaning equipment, for example tray washes, shall operate at the correct temperature with correct chemical type and dilution rate. There shall be an initial validation study to establish that the microbiological load is reduced to an acceptable level and that there is no carry over from allergens or species. The validation shall also identify the point at which rinse water, where re-circulated, needs to be changed.

Equipment shall be monitored and verified. The frequency and type of checks shall be determined by risk assessment based on the validation study.

Cleaning areas shall be designed to ensure a flow of equipment from dirty to clean and have adequate drainage to prevent contamination of clean equipment. There shall be sufficient extraction to minimise condensation build up.

4.11.12 Clean equipment shall be stored in manner which prevents re-contamination. Where product contact equipment has been stored (even if visually clean) but is not in daily use, it must be re-sanitised prior to use.

4.11.13 Cleaning equipment shall be:

- hygienically designed and fit for purpose
- suitably identified for intended use (e.g. colour-coded or labelled)
- cleaned and stored in a hygienic manner to prevent contamination.

The site shall ensure that there is dedicated equipment for:

- food-contact surfaces
- non-food contact surfaces
- drains
- ancillary areas (with separate equipment for toilets)
- external areas.

Equipment used for cleaning in high-care and high-risk areas shall be visually distinctive and dedicated for use in that area.

Hoses shall be clean, well maintained and stored off the floor. Where hoses are stored on overhead lines or reels, they shall be kept clean to prevent the risk of contamination to product.

4.11.14 Cleaning chemicals shall be stored securely with bulk containers stored on bunded pallets. Chemicals shall be suitable for the production environment, equipment and their intended use. Phenolic or scented products shall not be used.

4.11.15 Chemical dosing/dilution shall be controlled, monitored and recorded by trained personnel. Equipment used for chemical dosing shall be verified as accurate to within specified tolerances on a scheduled basis.

Residual levels of chemicals shall comply with relevant legislation. Residual chemical testing to be completed annually, where required.

4.11.15 CLEANING IN PLACE (CIP)

CLAUSE	REQUIREMENTS

4.11.15.1 All CIP equipment shall be designed by specialist food engineers and constructed to ensure effective operation. This shall include:

- spray balls and rotating devices capable of providing full surface coverage of vessels that are periodically inspected for blockages
- adequate separation from active lines to safeguard against cross contamination
- validation confirming the correct design and operation of the system prior to commissioning
- training of staff on the CIP system by a suitably competent individual
- an up to date schematic diagram of the layout of the CIP system
- where rinse solutions are recovered and reused, an assessment of the risk of cross contamination (e.g. due to the re-introduction of allergen).

Alterations or additions to the CIP system shall be authorised by a suitably competent individual before changes are made. A record of the changes shall be maintained.

The effectiveness of the CIP system shall be verified annually (either by internal review or independent audit).

The system shall be re-validated at a frequency based on risk and following any alterations or additions.

4.11.15.2 Limits of acceptable and unacceptable performance for key process parameters shall be defined to ensure the removal of target hazards (e.g. soil, allergens, micro-organisms, spores). As a minimum these parameters shall include:

- times for each stage
- detergent concentrations
- flow rate and pressure
- temperatures.

These shall be validated and records of the validation maintained.

Measuring devices used to monitor and verify the correct application of the CIP process shall be subject to formal calibration.

4.11.15.3 The CIP equipment shall be maintained by suitably trained staff to ensure effective cleaning is carried out. This shall include:

- detergent concentrations shall be checked routinely
- recovered post-rinse solutions shall be monitored for build-up of carry-over from the detergent tanks
- filters, where fitted, shall be cleaned and inspected at a defined frequency
- where used, flexible hoses shall be stored hygienically when not in use, and inspected at a defined frequency to ensure that they are in good condition.

4.11.15.4 CIP facilities, where used, shall be monitored at a defined frequency based on risk. This may include:

- monitoring of process parameters defined in clause 4.11.7.2
- ensuring correct connections, piping and settings are in place
- confirming the process is operating correctly (e.g. valves opening/closing sequentially)
- ensuring effective completion of the cleaning cycle
- monitoring for effective results, including draining where required.

Procedures shall define the action to be taken if monitoring indicates that processing is outside the defined limits.

4.11.16 ENVIRONMENTAL MONITORING

Risk-based environmental monitoring programmes shall be in place for pathogens or spoilage organisms. As a minimum, these shall include all production areas with open and ready-to-eat products.

CLAUSE	REQUIREMENTS
4.11.16.1	The design of the environmental monitoring programme shall be based on risk, and as a minimum include: • sampling protocol • identification of sample locations • frequency of tests • target organism(s) (e.g. pathogens, spoilage organisms and/or indicator organisms) • test methods (e.g. settle plates, rapid testing and swabs) • recording, evaluation and trending of results. • proactive pathogen plan for looking for pathogens within the environment and rectifying issues The programme and its associated procedures shall be documented.
4.11.16.2	Appropriate control limits shall be defined for the environmental monitoring programme. The site shall document the corrective action to be taken when monitored results indicate a failure to meet a control limit, or when monitored results indicate an upward trend of positive results.
4.11.16.3	The site shall review the environmental monitoring programme at least annually and whenever there are: • changes in processing conditions, process flow or equipment • new developments in scientific information • failures of the programme to identify a significant issue (e.g. regulatory authority tests identifying positive results which the site programme did not) • product failures (products with positive tests) • consistently negative results (e.g. a site with a long history of negative results should review its programme to consider whether the correct parts of the factory are being tested, that testing is being conducted correctly, whether the tests are for the appropriate organisms, etc.).

4.12 WASTE/WASTE DISPOSAL

Waste disposal shall be managed in accordance with legal requirements and to prevent accumulation, risk of contamination and the attraction of pests.

CLAUSE	REQUIREMENTS
4.12.1	Where licensing is required by law for the removal of waste, it shall be removed by licensed contractors and records of removal shall be maintained and available for audit.
4.12.2	Waste disposal shall be managed to ensure that it does not compromise food safety. The site shall complete a risk assessment that considers the movement and flow of waste and waste containers in all open product areas. For example, waste bins shall be dedicated to either high-risk or high-care areas and not be moved between different production risk zones.

4.12.3	Waste shall be suitably segregated and managed in such a way that it does not present a risk to product. Internal and external waste collection containers and rooms housing waste facilities shall be: • clearly identified • designed for ease of use and effective cleaning • well maintained to allow cleaning and, where required, disinfection • emptied at appropriate frequencies. Where recyclable materials are collected, the site shall have a system in place to ensure effective segregation. External waste containers shall be covered or doors kept closed as appropriate.
4.12.4	If unsafe products or substandard trademarked materials are transferred to a third party for destruction or disposal, that third party shall be a specialist in secure product or waste disposal and shall provide records which include the quantity of waste collected for destruction or disposal.

4.13 MANAGEMENT OF SURPLUS FOOD AND PRODUCTS FOR ANIMAL FEED

Effective processes shall be in place to ensure the safety and legality of by-products of the primary processing activity of the site.

CLAUSE	REQUIREMENTS
4.13.1	Surplus customer-branded products shall be disposed of in accordance with customer-specific requirements. Customer brand names shall be removed from packed surplus products under the control of the factory before the product enters the supply chain, unless otherwise authorised by the customer.
4.13.2	Where customer-branded products which do not meet specifications are sold to staff or passed on to charities or other organisations, this shall be with the prior consent of the brand owner and in accordance with their requirements. The site shall maintain traceability for these products in the event this information is needed for mass balance exercises or in the event of a product recall/withdrawal. Staff sale products shall be stored in appropriate conditions (e.g., perishable food shall be stored in controlled and adequate temperature condition). Staff sales products shall be adequately labelled and comply with legal requirements. The site shall comply with the Group Food Donations and Surplus Stock Procedure GTC-197 to ensure that all products are fit for consumption and meet legal requirements.
4.13.3	By-products and downgraded/surplus products intended for animal feed shall be segregated from waste and protected from contamination during storage. Products for animal feed shall be managed in accordance with the relevant legislative requirements.

4.14 PEST MANAGEMENT

The whole site shall have an effective preventive pest management programme in place to minimise the risk of infestation and resources shall be available to respond rapidly to any issues which occur to prevent risk to products.

Pest management programmes shall comply with all applicable legislation.

CLAUSE	REQUIREMENTS
4.14.1	If pest activity is identified, it shall not present a risk of contamination to products, raw materials or packaging. The presence of any infestation on site shall be documented in pest management records and be part of an effective pest control programme to eliminate or manage the infestation so that it does not present a risk to products, raw materials or packaging.
4.14.2	There shall be a pest management programme in place that has been contracted to a pest management organisation registered with the British Pest Control Association (BPCA). The site shall take overall responsibility for preventive pest control with a nominated person having accountability for pest control matters and to act as the point of contact for the pest control provider. The scope of the service shall be clearly defined and include as a minimum: • details of pests covered by routine inspections, elimination methods, treatments used and pests excluded from the contract • number of routine inspections and visits • follow up arrangements for any pest activity identified • responsibilities for the contractor and site • call out arrangements including out of hours emergency numbers • responsibility for removing old evidence of pests (e.g. droppings) • confirmation that the contract complies with current legislation • confirmation that the contract complies with this standard, the current BRCGS Global Standard for Food Safety and other relevant retailer requirements. Technicians carrying out pest control activities shall be trained to Level 2 in Pest Management as a minimum. Field Biologists carrying out in-depth inspections shall additionally have a Level 3 in Food Hygiene and are, or working towards being, a Certificated Field Biologist.
4.14.3	The pest management provider shall complete a risk assessment of the site to determine the frequency of inspection. The risk assessment shall consider: • site location and surrounding environment • building design and construction • raw materials and products on site • historical pest issues • seasonal variations. As a minimum, the following number of visits shall be scheduled: • 8 routine inspections • 4 Electronic Fly Killing unit (EFK) inspections including catch tray analysis • 4 Biologist Inspections, one of which shall be completed by a contractor independent from the pest management programme. The risk assessment shall be reviewed annually as a minimum or when there are changes to the building or processes which could have an impact on the pest management programme or if there has been a significant pest issue.

4.14.4	The pest management programme shall be documented and include as a minimum: • details of the pest control provider including BPCA membership • emergency call out details • contract specification (4.14.2) • site risk assessment (4.14.3) • evidence of pest control technician and field biologist training • frequency and type of inspections, with a minimum of 8 general inspections a year • schedule of planned visits • EFK tube change frequency • a site plan identifying the area covered by the programme and pest monitoring points (e.g. toxic/nontoxic baits) • the pest management methods and procedures in use • material safety data sheets for all chemicals used • details of how the programme complies with all local legislation (e.g. bird control, traps etc). Where electronic/paperless systems are in operation, the designated site contact and their nominated deputy shall have access to the system and be fully conversant with its use.
4.14.5	Bait stations or other rodent monitoring or control devices shall be appropriately located and maintained to prevent contamination risk to product. Toxic rodent baits are not allowed, except as an infestation tool control. All baits shall be secured. Any missing bait stations shall be recorded, reviewed and investigated.
4.14.6	EFKs, pheromone traps and/or other insect monitoring devices shall be appropriately sited and operational. The positioning and choice of unit type shall ensure that dead or dying insects do not cause any contamination of food.
4.14.7	EFKs shall be serviced at least once a year with an associated bulb change for each device. The bulb change shall take place at an appropriate point within the year to ensure optimum efficiency at times of peak insect populations. Specification limits shall be established for EFK catch tray analysis.
4.14.8	The site shall have adequate measures in place to prevent birds from entering buildings or roosting above loading or unloading areas.
4.14.9	In the event of infestation, or evidence of pest activity, immediate action shall be taken to identify at-risk products and to minimise the risk of product contamination. Any potentially affected products should be subject to the non-conforming product procedure. Where required, the customer shall be notified of an infestation that may affect any products.

4.14.10 Records of pest management inspections, pest proofing and hygiene recommendations and actions taken shall be maintained. Records shall cover all areas covered by the pest management programme. Each inspection report shall include the following as minimum:

- all observations of pests or evidence of pest activity
- recommendations for action by the site
- details of any chemicals used
- pest proofing requirements
- poor hygiene/housekeeping conditions
- any difficulties accessing or lost/missing pest monitoring devices
- name of site representative accompanied on the inspection
- signature from a site representative.

A representative from the site shall accompany at least part of each inspection.

It shall be the responsibility of the site to ensure that all of the relevant recommendations made by its contractor or consultant are agreed with appropriate timescales. Actions shall be carried out in a timely manner and signed off on completion.

4.14.11 An in-depth, documented pest management survey shall be undertaken quarterly, by a suitably qualified Field Biologist (see clause 4.14.2). These surveys shall not coincide with routine visits by the Technician. The survey shall:

- provide an in-depth inspection of the facility for pest activity
- review the existing pest management measures in place and make any recommendations for change
- meet the requirements of clause 4.14.10.

At the end of the survey a review meeting shall take place with the site representative to discuss findings and agree timescales for any actions required.

One of the surveys shall be completed by a consultant independent from the pest management contractor. Any findings from this survey shall be discussed with pest management contractor, ideally this should be done during annual review. If any concerns arise after the consultant inspection, this shall be discussed with pest management contractor as soon as possible.

4.14.12 The site shall hold an annual review with the contractor's Senior Technical Representative, this may need to occur at a greater frequency where required by the customer.

The review shall be used to discuss service standards and the support given by the factory in achieving the pest prevention objectives. Results of pest management inspections shall be assessed and analysed for trends. The analysis shall include results from trapping and monitoring devices to identify problem areas. The review shall be used as a basis for improving the pest management procedures.

Where appropriate, the site shall review trends in the event of an infestation.

4.14.13 Employees shall understand the signs of pest activity and be aware of the need to report any evidence of pest activity to a designated manager.

4.15 STORAGE FACILITIES

All facilities used for the storage of raw materials, packaging, in-process products and finished products shall be suitable for purpose and do not pose a food safety risk to product, materials and packaging.

CLAUSE	REQUIREMENTS
4.15.1	Procedures to maintain product safety and quality during storage shall be developed on the basis of risk assessment, understood by relevant staff and implemented accordingly. These may include, as appropriate: • managing chilled and frozen product transfer between temperature-controlled areas • segregation of products where necessary to avoid cross-contamination (physical, microbiological or allergens) or taint uptake • storing materials off the floor and away from walls • specific handling or stacking requirements to prevent product damage.
4.15.2	Where appropriate, packaging shall be stored away from other raw materials and finished product. Any part-used packaging materials suitable for use shall be effectively protected from contamination and clearly identified to maintain traceability before being returned to an appropriate storage area.
4.15.3	Where temperature control is required (e.g. for raw materials, semi-finished materials or final products), the storage area shall be capable of maintaining product temperature within specification and operated to ensure specified temperatures are maintained. Temperature recording equipment with suitable temperature alarms shall be fitted to all storage facilities or there shall be a system of recorded manual temperature checks, typically on at least a 4-hourly basis or at a frequency which allows for intervention before product temperatures exceed defined limits for the safety, legality or quality of products. Procedures shall be in place to take appropriate corrective actions when the environment is found to be outside the specified tolerance.
4.15.4	Freezer areas shall be adequately maintained to prevent excessive build-up of ice on walls, floors and ceilings.
4.15.5	Where controlled atmosphere storage is required, the storage conditions shall be specified and effectively controlled. Records shall be maintained of the storage conditions.
4.15.6	Where external storage units such as demountable trailers, mobile chiller units and similar are utilised, these shall be managed to ensure that all relevant sections of this standard, including calibration, hygiene, pest control and security are demonstrably in place. The site shall ensure that all such units are included in the internal audit and GMP inspection programmes at appropriate risk assessed frequencies.
4.15.7	Where storage outside is necessary, items shall be protected from contamination and deterioration. Items shall be checked for suitability before being brought into the factory.
4.15.8	The site shall facilitate correct stock rotation of raw materials, intermediate products and finished products in storage and ensure that materials are used in the correct order in relation to their manufacturing date and within the prescribed shelf life. All materials shall be clearly labelled with internal use/process by dates, time and product details. Materials shall be protected from contamination.
4.15.9	Where products are received fresh and subsequently frozen, the site shall ensure freezing is legally permitted and the product is suitable for freezing. Details are to be included in the raw material specification and the site shall ensure it meets shelf life requirements prior to freezing.

4.15.10 There shall be validated procedures in place for freezing and defrosting of product intended to be sold under chilled conditions. This shall include controls ensuring the product is assigned the correct shelf life and has been restored to an appropriate temperature prior to dispatch.

4.16 DISPATCH AND TRANSPORT

Procedures shall be in place to ensure that the management of despatch and of the vehicles and containers used for transporting products from the site do not present a risk to the safety, security or quality of the products.

CLAUSE	REQUIREMENTS
4.16.1	Procedures to maintain product safety and quality during loading and transportation shall be developed and implemented. These may include, as appropriate: • controlling temperature of loading dock areas and vehicles • the use of covered bays for vehicle loading or unloading • securing loads on pallets to prevent movement during transit • inspection of loads prior to dispatch.
4.16.2	All vehicles or containers used for the transport of raw materials and the dispatch of products shall be fit for purpose. This shall ensure that they are: • in a clean condition • free from strong odours which may cause taint to products • in a suitable condition to prevent damage to products during transit • designed to hold materials securely to prevent unauthorised access/tampering • equipped to ensure any temperature requirements can be maintained throughout transportation. Records of inspections shall be maintained.
4.16.3	Where temperature control is required, the transport shall be capable of maintaining product temperature within specification, under minimum and maximum load. Temperature data-logging devices which can be interrogated to confirm time/temperature conditions or a system to monitor and record at predetermined frequencies the correct operation of refrigeration equipment shall be used and records maintained.
4.16.4	Maintenance systems and documented cleaning procedures shall be available for all vehicles and equipment used for loading/unloading. There shall be records of the measures taken.
4.16.5	The site shall have procedures for the transport of products, which shall include: • any restrictions on the use of mixed loads • requirements for the security of products during transit, particularly when vehicles are parked and unattended • clear instructions in the case of vehicle breakdown, accident or failure of refrigeration systems, which ensure that the safety of the products is assessed and records maintained.
4.16.6	Where the site employs third-party hauliers, all the requirements specified in this section shall be clearly defined in the contract or terms and conditions and verified, or the contracted company shall be certificated to the Global Standard for Storage and Distribution or similar GFSI-recognised scheme. Additionally, the haulier shall be approved by GTS in accordance with Group Suppplier Approval Procedure.

5 PRODUCT CONTROL

5.1 PRODUCT DESIGN/DEVELOPMENT

Product design and development procedures shall be in place for new products or processes and any changes to existing product, packaging or manufacturing processes to ensure that safe, legal and quality products are produced.

CLAUSE	REQUIREMENTS
5.1.1	The site shall have procedures for new/existing product development and process change to control the introduction of hazards which would be unacceptable to the site or customers (e.g. the introduction of allergens and new product claims, glass packaging or, chemical, physical or microbiological risks). This shall include the following: • restrictions to the scope of new product developments (e.g. the introduction of allergens, glass packaging or microbiological risks) • product development process that complies with relevant customer requirements • identification of samples, WIP material and product from trials stored in production areas to prevent accidental use • identification of any variability in raw materials • consideration for variability of raw materials that impact product quality/composition. • a handover process to ensure that the first product/packing run achieves the product specification • consideration to health, safety and environmental hazards for new processes or changes to existing processes • review of HACCP plan (see clause 5.1.2). • artwork approval and sign-off Product packaging and transit packaging shall be an integral part of the process to ensure it is fit for purpose. The site shall record and retain all development reviews, factory trial documentation and product testing, which meet the requirements of this section, and collate these into a launch pack for review.
5.1.2	All new products and changes to product formulation, packaging or methods of processing shall be adequately assessed against the HACCP plan and formally approved by the HACCP team leader or authorised HACCP team member. This review shall ensure that hazards have been assessed relating to food safety, quality and legality and that suitable controls, identified through the HACCP system, are implemented. An assessment of the impact to existing finished product claims shall also be documented. This assessment and subsequent approval shall be granted before factory trials are undertaken.
5.1.3	The site shall identify where variability in raw materials may significantly impact the safety, authenticity, legality and quality of the final product during the development stage. Parameters shall be established and agreed with the supplier (see clause 3.5.1). Where necessary, the site shall verify that that these parameters can be achieved by the supplier prior to launch. There shall be a nutritional testing protocol in place to evaluate variables and known tolerances within raw materials and processes that would significantly impact on the nutritional composition of a finished product. This shall be completed during the development stage with results used to establish ongoing monitoring procedures for: • raw materials (see clause 3.4.2.2) • processes (see clause 6.1.3) • finished products (see clause 5.6.1.1).

5.1.3 Relevant legal requirements related to product content or nutritional testing (e.g., Fat, Sugar, and Salt content) shall be included in the testing protocol.

5.1.4 Trials using production equipment shall be carried out where it is necessary to validate that product formulation and manufacturing processes are capable of producing a safe product of the required quality.

All variations to the process such as freezing of product to build stock for peak periods shall be captured during the trial stage. Where variations to the process of existing products are being considered, a trial to validate the safety and quality of the product shall be completed prior to the production of finished products.

5.1.5 Initial shelf-life trials shall be undertaken using documented protocols that reflect conditions expected during manufacture, storage, transport/distribution, use and handling to determine product shelf life.

Results shall be recorded and retained and shall confirm compliance with the relevant microbiological, chemical and organoleptic analysis.

The shelf life trial shall include a summary report which contains the following information:

- date of production and packing
- packaging format used
- transit trial (where required)
- storage conditions at each stage of analysis (including storage temperature and date pack was opened)
- freezer trials (where the product is suitable for freezing, the shelf life stated on pack shall be validated)
- the date and result of microbiological, chemical and organoleptic analysis
- conclusion of results and sign off by an authorised person.

Laboratory certificates of analysis and sensory panel documents shall be included with the summary report.

The site shall ensure that shelf-life testing meets any additional customer requirements where required.

Where shelf-life trials prior to production are impractical, for instance for some long-life products, a documented science-based justification for the assigned shelf life shall be produced.

5.1.6 Where Cranswick have responsibility for the brand, products with 'free from' or 'made without', 'reduced amount of ingredient' claims design shall be visibly different (e.g., coloured packaging, product shape and size).

5.1.7 Product claims related to food safety, dietary or nutritional needs / benefits (e.g., 'free from') validation results shall be in place for the finished product within the NPD process. Validation results shall be provided and certified by external, approved laboratory.

Validation shall be completed before artwork approval and sign-off.

5.1.8 Where cooking instruction / method is provided, cooking instruction shall be fully validated within the NPD process to ensure that, when the product is cooked according to the instructions, a safe, quality, ready-to eat product is consistently produced.

Validation results shall be documented.

Validation shall be completed before artwork approval and sign-off.

5.1.9 Where any other instruction or suitability is provided (e.g., suitable for freezing, defrosting method) on the packaging, this shall be validated within the NPD process and ensure product safety and quality is not compromised.

Validation shall be appropriately documented and completed before artwork approval and sign-off

5.2 PRODUCT LABELLING

Product labelling shall comply with the appropriate legal requirements for the designated country of sale and contain information to enable the safe handling, display, storage and preparation of the product within the food supply chain or by the customer.

CLAUSE	REQUIREMENTS
5.2.1	All products shall be labelled to meet legal requirements in the country of sale and adhere to customer labelling policies. Product labels shall include information to allow the safe handling, display, storage, preparation and use of the product within the food supply chain or by the consumer. There shall be a process to verify that ingredient and allergen labelling is correct based on the product recipe and ingredient specifications. Site shall follow group artwork sign off procedure.
5.2.2	There shall be effective processes in place to ensure that labelling information is reviewed whenever changes occur to: • product recipe • process • raw materials • the supplier of raw materials • the country of origin of raw materials • legislation.
5.2.3	Where a product is designed to enable a claim to be made to satisfy a consumer group (e.g. a nutritional claim, reduced sugar), the site shall ensure that the product formulation and production processes are fully validated to meet the stated claim (see clause 5.1.6). The site shall comply with customer requirements with regards to the frequency of testing and shall report any adverse results to the relevant customer.
5.2.4	Where the label information is the responsibility of a customer or a nominated third party, the site shall provide information: • to enable the label to be accurately created • whenever a change occurs which may affect the label information.
5.2.5	All cooking instructions and methods detailed on pack shall be validated. (see clause 5.1.7)

5.3 MANAGEMENT OF ALLERGENS AND SPECIATION CONTROL

FUNDAMENTAL: The site shall have a system for the management of allergenic materials which minimises the risk of allergen contamination of products and meets legal requirements for labelling in the country of sale.

There shall be systems in place to ensure that speciation risks are managed and that the substance and nature of products are maintained.

CLAUSE	REQUIREMENTS
5.3.1	The site shall have the Group Allergen Policy displayed within staff areas. All staff shall have allergen awareness training as part of their induction and shall be trained against site procedures. Controls shall be in place to ensure all visitors and contractors are aware of and comply with allergen management procedures.
5.3.2	The site shall identify and list allergen-containing materials handled on site. This shall include raw materials, processing aids, intermediate and finished products.
5.3.3	A documented risk assessment shall be carried out to identify routes of contamination from allergenic material to assist with developing policies and procedures for handling raw materials and intermediate and finished products to ensure cross-contamination (cross-contact) is avoided. For sites that handle multiple species and/or make vegetarian claims on pack, an additional risk assessment shall be carried out to identify routes of contamination from meat and other materials that are derived from the carcass of a species not declared on pack (e.g. meat preparations, gelatines, casings, animal fats, bouillons). The risk assessment shall be used to assist with developing policies and procedures for handling raw materials and intermediate and finished products to ensure cross-contamination (cross-contact) is avoided. Where sites make vegan claims, the site shall consider the risk from other products of animal origin (POAO) not derived from the carcass of animals (e.g. milk and milk products, eggs, honey). Group allergen and species risk assessment template is recommended to use, where this is not practicable, site shall cover all principles from the group template in their own format to ensure consistent standards. The risk assessment must be reviewed at least annually, six monthly when handling multiple allergens and/or species, quarterly when making a 'Free From' claim and if new allergens, species, products, processes (or any other changes like new cleaning chemicals) are introduced, that could affect the sites allergen status and/or effectiveness of current allergen and species controls.
5.3.4	Procedures shall be established to ensure the effective management of materials of concern. These are allergenic materials, different species of meat and other products of animal origin (where appropriate) that require controls to prevent cross contamination (cross-contact) of products where these materials are not declared on pack. These shall include, as appropriate: • identification of allergen/species status for raw materials on receipt • clear identification of allergenic materials, meat and other products of animal origin during storage, processing, any intermediary steps, and packing • suitable controls for the storage of materials of concern to prevent cross-contamination (cross-contact) • physical or time segregation where materials of concern are processed or packed • the use of separate or additional protective overclothing when handling materials of concern • use of identified, dedicated equipment and utensils for processing • scheduling of production to reduce changes between products containing different allergens/species

5.3.4	• systems to restrict the movement of airborne dust containing allergenic material • waste handling and spillage controls (see clause 5.3.5) • restrictions on food brought onto site by staff, visitors and contractors and for catering purposes. Personnel involved with production schedules shall be aware of site procedures and scheduling requirements to minimise the number of interim clean downs required. Schedules shall allocate sufficient time to perform clean downs when they are required. Site procedures shall be in line with Group Allergen Management Procedure and Group Species Control Procedure.
5.3.5	A spillage control procedure shall be in place to ensure all spilled allergenic material is disposed of in such a manner to prevent cross contamination of other material, surfaces or equipment. The spillage procedure shall be available for reference in the event of a spillage. Specific measures shall include: • review of spillage area by a nominated person to determine the contamination risk to equipment and product • isolation of the spillage area to prevent further spread of allergens, this shall include a restriction of movement within the isolated area • dedicated, colour-coded cleaning equipment which is subject to disposal or a verified cleaning process after use • change of protective clothing after cleaning has taken place • inspection of footwear and cleaning/changing if necessary • sign off by a nominated person to confirm adherence to the spillage procedure and that finished cleaning standards are satisfactory. A record of each spillage incident shall be logged.
5.3.6	Where rework is used, or reworking operations are carried out, procedures shall be implemented to ensure any rework containing a specific allergen or species is not used in products that do not declare these ingredients.
5.3.7	Where a justified, risk-based assessment demonstrates that the nature of the production process is such that cross-contamination (cross-contact) from an allergen cannot be prevented, a warning should be included on the label. Statements shall comply with customer requirements and any legal requirements in the country of sale and customer labelling policies.
5.3.8	Where a claim is made regarding the suitability of a food for allergy or food sensitivity sufferers, the site shall ensure that the production process is fully validated to meet the stated claim and the effectiveness of the process is routinely verified, this shall be documented.
5.3.9	Equipment or area-cleaning procedures shall be designed to remove or reduce to acceptable levels any potential cross-contamination (cross-contact) by an allergen/species. The cleaning method and verification procedure shall be validated using a quantitative method by a UKAS accredited laboratory in addition to visual inspection. The test method used shall be included in the laboratory's UKAS Schedule of Accreditation. Group allergen and species validation format shall be used. Where the site handles multiple allergens and/or species and choses to validate cleaning methods with one test then consideration must be given to worst-case products/allergen/species (i.e. most difficult to clean, amount of allergen within product, lowest detection level, severity of allergen). The justification must be documented and agreed with relevant customer. Validation shall be performed at least annually, and if new allergens, species or processes are introduced, that could affect the sites allergen status, species segregation or effectiveness of current controls and clean downs including cleaning method or detergent change. The site's validation programme shall comply with customer requirements with regard to surface and/or product testing, and the number of tests required to be considered successful.

5.3.10 Cleaning equipment used to clean allergenic materials shall either be identifiable and specific for allergen use, single use, or effectively cleaned after use. The method of cleaning shall also be validated to ensure it is effective.

The site may also consider the use of dedicated cleaning equipment for free-from products, this shall be based on risk.

5.3.11 The site shall have an ongoing cleaning verification procedure in place which includes the following;

- visual inspection of key inspection points confirming that equipment is free from debris
- swabbing of key inspection points using an industry recognised rapid test kit where available
- prevention of any production until verification has been confirmed
- corrective action in the event of a failure.

Verification must be completed by a suitably trained, competent employee independent from the hygiene and operation team.

The visual inspection and swab results shall be recorded along with checks to confirm that dedicated PPE has been cleaned or disposed of prior to re-starting production.

5.3.12 Results from verification swabbing shall be reviewed regularly. Where trends indicate a potential issue with the cleaning programme, this shall be formally reviewed, and risk assessed to determine whether another cleaning validation is required.

5.3.13 Allergen Management System shall be reviewed at least annually, six monthly when handling multiple allergens, quarterly when making a 'Free From' claim and if new allergens, products, processes (or any other changes like new cleaning chemicals) are introduced, that could affect the sites allergen status and/or effectiveness of current allergen and controls. The site shall complete a review of its allergen management system following a recall of a similar product that involves undeclared allergens (see clause 2.12.1).

The review shall consider audit outcomes, customer complaints and other incidents of non-conformance relating to the allergen management programme to look for trends and to ensure that appropriate and effective corrective action have been instigated.

5.4 PRODUCT VULNERABILITY, AUTHENTICITY, CLAIMS & CHAIN OF CUSTODY

Systems shall be in place to minimise the risk of purchasing fraudulent or adulterated food raw materials and to ensure that all product descriptions and claims are legal, accurate and verified.

CLAUSE	REQUIREMENTS
5.4.1	Where personnel are engaged in vulnerability assessments, the individual or team responsible shall understand potential food fraud risks. This shall include knowledge of raw materials used by the site and the principles of vulnerability assessment.
5.4.2	Site shall demonstrate that they have access to the Group Vulnerability Assessment and dedicated personnel are competent to interpretate the information provided in the assessment.
5.4.3	Where raw materials are identified as being at particular risk of adulteration or substitution, site shall determine adequate controls at the intake.
5.4.4	The site shall have processes in place to access information on historical and developing threats to the supply chain which may present a risk of adulteration or substitution of raw materials. Such information may come from;

5.4.4

- trade associations
- government sources
- private resource centres
- GTS alerts/notifications
- Activities completed for clause 1.1.5

The site shall provide evidence that alerts from these sources are being reviewed and that appropriate actions have been taken where necessary.

5.4.5 The site shall identify all claims made on finished packs which are dependent on the status of a raw material (see list above), this shall also include any process performed by the site which may be declared on pack.

The site shall risk assess each raw material/process carrying the claim to identify if a visually similar raw material and/or WIP material is present on site which does not carry that claim and therefore have the potential to be substituted. The site shall risk assess each process step for that raw material to identify potential areas where loss of claim may occur. Appropriate controls shall be established for each step identified as a risk to ensure the integrity of product claims.

When developing new products, the site shall consider the impact to existing raw materials carrying a claim as well as new raw materials. A documented assessment shall be carried out to determine if any additional controls are required to mitigate new risks which may be present by introducing a new raw material to site. This shall be done alongside the HACCP review which is completed prior to factory trials taking place (see clause 5.1.2).

The risk assessment shall be reviewed and updated prior to the launch of new products.

Sites handling more than one species shall acknowledge speciation risks within a separate risk assessment. The controls shall meet the requirements outlined in section 5.3.

5.4.6 Where a product is designed to enable a claim to be made, site shall ensure that all claims are substantiated, and product formulation and the production process are fully validated to meet the stated claim and any legal requirements (in country of intended sale) relating to the claim. The process flow (see clause 2.3.1) for the production of products where claims are made shall be documented and potential areas for contamination or loss of identity identified.

Appropriate controls shall be established to ensure the integrity of the product claims.

5.4.7 Where products are labelled or claims are made on finished packs which are dependent on the status of a raw material, the status of each batch of the raw material shall be verified. These claims include:

- specific provenance or origin
- breed/varietal claims
- assured status/welfare status (e.g., Red Tractor, RSCA assured, outdoor bred, free-range)
- genetically modified organism (GMO) status
- identity preserved
- named specific trademarked ingredients (e.g., RSPO).

The site shall maintain purchasing records, traceability of raw material usage and final product packing records to substantiate claims. The site shall undertake a documented mass balance tests at a frequency to meet the particular scheme requirements or at least every 6 months in the absence of a scheme-specific requirement.

5.4.8 Where claims are made about the methods of production (e.g., organic, halal, kosher) the site shall maintain the necessary certification status in order to make such a claim and ensure certified material is effectively segregated from non-certified material.

5.4.9 Where RSPO Palm Oil is used on site, the site shall follow the Group RSPO procedures and be listed on the Group RSPO membership.

5.5 PRODUCT PACKAGING

Product packaging shall be appropriate for the intended use and shall be stored under conditions to prevent contamination and minimise deterioration.

CLAUSE	REQUIREMENTS
5.5.1	When purchasing or specifying primary packaging, the supplier of packaging materials shall be made aware of any particular characteristics of the food (e.g. high fat content, pH, usage conditions such as microwaving, other packaging used on the product) which may affect packaging suitability. Certificates of conformity or other evidence shall be available for primary packaging to confirm it complies with applicable food safety legislation and is suitable for its intended use.
5.5.2	Product liners and bags purchased by the site for use in direct contact with ingredients, or work in process, shall be appropriately coloured (e.g. contrasting colour to the product) and resistant to tearing to prevent accidental contamination.
5.5.2	Site shall have a procedure to manage obsolete packaging (including labels) This shall include: • mechanisms to prevent accidental use of obsolete packaging • control and disposal • appropriate procedures for the disposal of obsolete printed materials (e.g., rendering trademarked materials unusable)

5.6 PRODUCT INSPECTION AND LABORATORY TESTING

The site shall undertake or subcontract inspection and analyses which are critical to confirm product safety, legality, integrity and quality, using appropriate procedures, facilities and standards.

5.6.1 PRODUCT INSPECTION AND TESTING

CLAUSE	REQUIREMENTS
5.6.1.1	The site shall have a scheduled programme of product testing which may include microbiological, chemical, physical and organoleptic testing based on risk and in accordance with customer requirements. The methods, frequency and specified limits shall be documented.
5.6.1.2	Test and inspection results shall be recorded and reviewed regularly by the site to identify trends. The significance of external laboratory results shall be understood and acted upon accordingly. Where legal and scheme limits apply, these shall be understood, and appropriate action taken promptly to address any exceedance of these limits. In the event of unsatisfactory results or where trends have been identified, an investigation shall be completed to identify root cause with appropriate actions taken to prevent recurrence. Where applicable, the measurement of uncertainty associated with laboratory test results shall be considered. The site shall activate product withdrawal/recall procedure Group Business Continuity & Crisis Manual where there is a significant risk to product safety or quality.
5.6.1.3	The site shall ensure that a system of validation and ongoing verification of the shelf life is in place. This shall be based on risk but completed annually as a minimum. The tests shall include sensory analysis and, as applicable, microbiological testing and relevant chemical factors such as pH and aw. Records and results from shelf-life tests shall verify the shelf-life period indicated on the product. The site shall notify the relevant customer of any adverse results. Records and results from shelf-life tests shall verify the shelf-life period indicated on the product.

5.6.2 LABORATORY TESTING

CLAUSE	REQUIREMENTS
5.6.2.1	All microbiological and chemical testing (this includes the environmental monitoring programme) shall be contracted to an external UKAS accredited laboratory. The laboratory shall be UKAS accredited for the test method being completed. The site shall have records of the laboratory's UKAS accreditation and, where relevant, customer accreditation. Some testing suites may be subcontracted to another laboratory, the site shall have records of the relevant accreditation for the subcontractor.
5.6.2.2	Specified acceptable and action limits for each test suite shall be communicated to the laboratory. These limits shall meet the relevant customer requirements.
5.6.2.3	The site shall ensure that the external laboratory has the most up-to-date contact information to ensure that alerts for unsatisfactory results are received by the necessary personnel on site.
5.6.2.4	Procedures shall be in place to ensure reliability of laboratory results. These shall include: • use of recognised test methods, where available • documented testing procedures • ensuring staff are suitably qualified and/or trained and competent to carry out the analysis required • use of a system to verify the accuracy of test results (e.g., proficiency testing where applicable) • use of appropriately calibrated and maintained equipment. (See clause 3.4.3.2)

5.6.3 ORGANOLEPTIC ASSESSMENT

CLAUSE	REQUIREMENTS
5.6.3.1	The site shall have procedures in place to ensure that organoleptic assessments are conducted in a standardised manner. This shall include: • a designated Taste Panel Co-ordinator with responsibility for ensuring that the requirements in this clause are met. The Taste Panel Co-ordinator shall have completed the Campden BRI Sensory Evaluation course or equivalent. • a designated assessment area for conducting taste panel reviews (this area should be controlled during assessments to ensure a quiet environment which is free from distractions) • a minimum of 3 assessors in attendance for each taste panel review. Each assessor shall be screened for their sensory capabilities, records of screening shall be kept on file. The screening of assessors shall be completed by the Taste Panel Co-ordinator (or nominated deputy if they have completed the appropriate sensory evaluation course) • each product shall be scored separately on appearance, aroma, flavour, texture with an overall score • the individual scores as well as the final product score shall be recorded • finished product specifications and/or internal QAS documents shall be displayed for reference during each product assessment. Actions for unsatisfactory results shall meet the requirements within clause 5.6.1.2.

5.6.4 BENCHMARKING

CLAUSE	REQUIREMENTS
5.6.4.1	The site shall have a benchmarking schedule in place for comparing its products against a competitive set of products. The site shall be aware of and comply with its customer's benchmarking requirements for frequency of testing. Where there are no benchmarking requirements, the site shall complete benchmarking sessions for each product category annually as a minimum.

5.6.4.2 Benchmarking sessions shall be conducted in a standardised manner (see clause 5.6.3.1). Additional requirements include:

- products used in the assessment shall be purchased from store and not pre-selected
- products to be tested with the same (if not similar) number of days' life remaining
- assessors not informed of any information that may influence their assessment (i.e. brand or product claims that suggest it is a premium product)
- all related packaging to be out of sight from panellists during the organoleptic assessment and comparison so as not to influence the outcome
- products to be ranked by preference from each assessor and then establish the average placing of each product
- packaging for each product to be examined after the organoleptic assessment with notes on packaging materials (e.g. fit for purpose, recyclability) and any claims, declarations or certifications (e.g. British, Reduced Fat, RSPCA Assured).

Results of benchmarking assessments shall be recorded and discussed with the relevant customer where appropriate.

5.7 PRODUCT RELEASE

The site shall ensure that finished product is not released unless all agreed procedures have been followed.

CLAUSE	REQUIREMENTS
5.7.1	Where products are held either on or off site awaiting positive release, there shall be documented procedures indicating the process for release and who is responsible for it. The procedures shall be sufficiently robust to ensure that accidental release cannot occur.

6 PROCESS CONTROL

6.1 CONTROL OF OPERATIONS

FUNDAMENTAL: The site shall operate to procedures and/or work instructions that ensure the production of consistently safe and legal product with the desired quality characteristics, in full compliance with the HACCP food safety plan.

CLAUSE	REQUIREMENTS
6.1.1	Documented process specifications and work instructions/procedures shall be available for the key processes in the production of products to ensure product safety, legality and quality. The specifications/procedures as appropriate shall include: • recipes – including identification of any allergens • mixing instructions, speed, time • equipment process settings • cooking times and temperatures • cooling times and temperatures • labelling instructions • coding and shelf-life marking • storage conditions (e.g., storage temperatures) • any additional critical control points identified in the HACCP or food safety plan. Process specifications shall be in accordance with the agreed finished product specification The site shall review the process specifications and work instructions/procedures prior to any changes which may affect food safety, legality and quality. All food safety procedures (e.g., cooking, cooling) shall be validated and demonstrate compliance to relevant food safety requirements (e.g., minimum safe cook temperature/time has been achieved). (see clause 2.6.3)
6.1.2	Where equipment settings are critical to the safety or legality of the product, changes to the equipment settings shall only be completed by trained and authorised staff. Where applicable, controls shall be password-protected or otherwise restricted. Where access levels are password-protected, the user shall ensure that they have logged out before leaving the equipment to prevent unauthorised adjustments. Where possible, passwords shall be timed out after a set time, but no more than 60 seconds inactivity.
6.1.3	Process monitoring, such as of temperature, time, pressure and chemical properties, shall be implemented, adequately controlled and recorded to ensure that product is produced within the required process specification.
6.1.4	In circumstances where process parameters or product quality are controlled by in-line monitoring devices, these shall be linked to a suitable failure alert system that is routinely tested.
6.1.5	Where variation in processing conditions may occur within equipment critical to the safety or quality of products, the processing characteristics shall be validated and verified at a frequency based on risk and performance of equipment (e.g. heat distribution in retorts, ovens and processing vessels; temperature distribution in freezers and cold stores).
6.1.6	In the case of equipment failure or deviation of the process from specification, procedures shall be in place to establish the safety status and quality of the product to determine the action to be taken.
6.1.7	Where a site handles products or materials (e.g., by-products from production processes) that are outside the scope of the audit, these shall be controlled to ensure that they do not create a product safety, authenticity, quality or legality risk to products within the scope.

6.2 CCP PROCESS CONTROL

FUNDAMENTAL: The site shall implement a planned set of checks or measurements for each CCP to ensure that the parameters established during the validation for each process are being achieved. The monitoring system shall be documented both in terms of the procedures to be followed and the results obtained.

6.2.1 DEEP CHILL

CLAUSE	REQUIREMENTS
6.2.1.1	Deep chill storage areas shall be equipped with accurate and calibrated continuous time/temperature monitoring equipment. The monitoring equipment shall have an alarm system in place to alert personnel if critical air or product temperatures are exceeded. There shall be evidence that appropriate action has been taken in a timely manner where alerts have been sent. Deep chill storage areas shall be set at a temperature that maintains the core temperature of product below 3°C but not at a temperature that causes the formation of ice crystals within product/materials.
6.2.1.2	The site shall have systems and procedures in place to control the storage requirements for products in deep chill. This shall include: • manual air and product temperature monitoring checks • verification checks on entry and exit from deep chill to verify that the product core temperature is below 3°C • corrective action to be taken for readings out of specification (this shall meet the requirements set out in clause 2.9.1) • packaging integrity checks on entry and exit.

6.2.2 COOKING AND COOLING

CLAUSE	REQUIREMENTS
6.2.2.1	All cooking and cooling equipment shall have accurate and calibrated temperature monitoring equipment to ensure it is operating with an accuracy of ±0.5°C of master/reference probe temperature. The accuracy shall be taken into account when determining limits.
6.2.2.2	The site shall have a procedure for determining the heat distribution profile of cooking and cooling equipment to determine where hot and cold spots are located. Once identified, this shall be the placement for any in-cook probes. This must be reviewed annually as a minimum or after any significant engineering work which may affect the heat distribution of equipment.
6.2.2.3	Cooked products must be cooled rapidly to less than 5°C within 4 hours or alternatively have staged temperature reduction with at risk times & temperatures identified and controlled. (An example of this approach can be found in the Campden BRI guideline 'Identification and prevention of hazards associated with slow cooking of hams and other large cooked meats and meat products')
6.2.2.4	Where process parameters that affect the performance of cooking programmes have been identified, there shall be checks in place to ensure these have not been exceeded. These checks may include, where relevant; • temperature checks • weight/size of product • configuration of product on trays/trolleys against photographic standards • cooker programme/temperature setting.

6.2.2.4	Cooker programme/temperature and product temperature/configuration checks shall be completed prior to loading. Weight/size parameters shall also be completed prior to cooker loading unless this has been checked on intake for the relevant parameters and product has not undergone further size reduction before cooking (e.g. whole chicken breast/thigh). When a continuous oven or fryer is being used, raw material variation and presentation to the oven must be controlled through the use of a height bar. This height bar can either be physical or electronic however it must be alarmed, be placed as close as possible to oven entry and should have a belt stop system. The bar must cover the full width of the belt. The height of this bar must be established through reviewing raw product with adjustment only allowed by authorized individuals. If the alarm has been activated, it must not be deactivated by line staff and a clear procedure to manage this event must be in place. The height of the bar and belt speed shall be checked at the start of production and after adjustments for changes in product.
6.2.2.5	There shall be procedures in place for the monitoring of critical limits, this shall include: • manual checks to be taken from the high-risk side • where measurements are to be taken in relation to the position of product on the trolley/belt • measurements to be taken from the core of the product, where this is not possible, equivalent validated test shall be appointed effective cleaning of insertion probes before and after use. • effective cleaning of insertion probes before and after use For cookers that have an internal probe to control the cooking process, the probe shall be placed in the slowest heating point identified during the cold spot analysis (see clause 6.2.2.2). Where practical, the largest unit shall be placed at this position. There shall be systems in place to manually check the online system and the core temperature of the product to verify that the critical limit has been achieved.
6.2.2.6	Equipment that controls the cooking and cooling process shall be designed to ensure that the cooling phase of the process does not start until the critical limit of the cooking phase has been achieved. There shall be manual checks of the online system to verify that the critical limits for both phases have been achieved prior to release from the area.
6.2.2.7	For blast chillers that have an internal probe to monitor the cooling process, the probe shall be placed in the slowest cooling point identified during the cold spot analysis. Where practical, the largest unit shall be placed at this position. Cooling times shall be calculated from the time product has exited the oven until the point at which the core temperature of the product has achieved <5°C.
6.2.2.8	Alarm systems associated with cooking and cooling equipment shall be verified daily. The tolerances of the alarm system shall be known and taken into consideration when setting limits. A log of all the alarm activations shall be documented with details of the corrective action taken.
6.2.2.9	Where a secondary cooking process takes place (i.e. roasting), the secondary cooling phase shall be calculated and added to the initial cooling phase. The total cooling phase shall not exceed the critical limit. The total cooling phase shall be verified prior to further processing of product.
6.2.2.10	A procedure must be in place which details the definition of an undercooked incident, the management of the event and includes the escalation process. All undercook incidences must be investigated thoroughly to root cause. Where an undercook has occurred resulting in raw product (product that hasn't achieved the CCP), the procedure shall include, as a minimum, the following principles:

6.2.2.10

- Stop production immediately and quarantine the affected area – this may need to extend to any blast chills or continuous chills that the product has come into contact with.
- Site escalation procedure to be followed (Technical manager should be informed immediately)
- Reject all suspected production and return to low risk either through a return hatch which is only used for that purpose or a waste route.
- The oven exit area and any other areas that the product has come into contact with, should be subject to a deep clean and only released back into production once the root cause has been identified and the cleaning has been signed off by the member of the technical management team.

6.2.3 MODIFIED ATMOSPHERE PACKING & VACUUM PACKING

CLAUSE	REQUIREMENTS
6.2.3.1	The site shall have monitoring procedures in place for vacuum packing. Controls shall include checks to confirm sufficient removal of air from the pack has taken place and that the seal integrity is sufficient to maintain the vacuum environment. Checks shall be completed at the following frequency: • start of run • hourly • after any adjustment • end of run. Ideally seal integrity shall be checked by using a pressure chamber. Where this is not possible and checks rely on visual inspection, visual standards shall be available at the point of inspection.
6.2.3.2	The site shall have monitoring procedures in place for MAP packing. Controls shall include checks to confirm that the correct mixture of gases are introduced to packs and that seal integrity is sufficient to maintain the modified atmosphere. Checks shall be completed at the following frequency: • start of run • hourly • after any adjustment • end of run. Ideally seal integrity shall be checked by using a pressure chamber or submerging the pack underwater. Where this is not possible and checks rely on visual inspection, visual standards shall be available at the point of inspection. Gas mixture ratio shall be included in on line QAS, where this is not practical, gas mixture ratio shall be specified on the separate document. The document shall be available to staff responsible for gas checks and gas settings on the production line.

6.3 LABEL AND PACK CONTROL

FUNDAMENTAL: The management controls of product labelling activities shall ensure that products will be correctly labelled and coded.

CLAUSE	REQUIREMENTS
6.3.1	The site shall have documented and detailed procedures outlining how to operate manual and automated systems for the verification of packaging and date codes and actions to take in the event of deviation from the norms.
6.3.2	All printed packaging (i.e. labels, trays, films, cartons, sleeves, promotional labels) for finished products that display product information shall be subject to the intake checks stated in clause 3.4.2.1. There shall be additional procedures in place for the control of printed packaging. This shall include: • a sample from each batch to be cross-referenced against a master copy of the artwork to confirm that the information is correct and the quality of the print is acceptable prior to release • where label information on printed packaging is not checked at the time of acceptance on intake, it shall be placed on hold to prevent unauthorised use. Cross-reference checks shall include: • product name, legal description and ingredient claims • ingredients declaration (including QUID and relevant allergens highlighted) • nutritional information • health mark • weight • barcodes (including 2D when used to control different packaging versions) • price and promotional information (where applicable). Where barcodes are placed on backing paper or reels for automated systems to verify, these shall be included in the printed packaging checks.
6.3.3	There shall be procedures in place to manage obsolete printed packaging. This shall include: • mechanisms to prevent accidental use of obsolete packaging • control and disposal of obsolete packaging • appropriate procedures for the disposal of obsolete printed materials (e.g. rendering trademarked materials unusable). Non-conforming and obsolete packaging shall be adequately labelled to prevent accidental use.
6.3.4	Storage areas for printed packaging shall be adequately controlled to prevent unauthorised access. Each packaging format shall have a dedicated location in the storage area to prevent mixing of packaging. Where possible, visually similar packaging shall not be stored together. Non-conforming and obsolete shall be adequately labelled to prevent accidental use. Packaging that is transferred to a secondary location (i.e. label room) shall be adequately controlled to prevent unauthorised use. Where packaging such as film, labels, sleeves, promotional stickers, have been decanted from their cardboard outers, adequate signage and/or shelf labelling shall be used to clearly indicate the exact storage location. There shall be adequate controls in place to prevent mixing of packaging.
6.3.5	There shall be a coding schedule that states the date code in exactly the same format as the code expected to appear on product packaging along with any price or promotional information. The site shall define when coding date changes occur (i.e. midnight). Price and promotional information with a clear indication of the start and end date for promotional labels shall be agreed, verified and signed off by an authorised member of the Commercial and Technical Teams as a minimum. All date coding and price/promotional information shall be made available to the relevant personnel (i.e. label room operatives, QAs, line leaders) with previous versions being removed/destroyed.

6.3.6

The issue and return of packaging (including outer case labels) to production lines shall be adequately controlled to prevent more than one set of packaging being issued to a line at any one time. All packaging issued to the line shall be checked to verify that the correct packaging has been issued. A record of all checks on relevant aspects of packaging information (i.e. version, price/promotional information, pre-printed information, allergen information/declaration) shall be kept along with batch information and quantity issued.

Price and promotional labels shall be checked against the coding schedule.

Returned packaging shall be reconciled against the quantity issued with consideration of the number of units packed and any waste generated.

Any left-over pre-coded/labelled packaging or outer case labels shall be destroyed after they have been returned to the controlled area. The amount shall be recorded and the disposal documented.

6.3.7

There shall be controls in place for the date coding/promotional labelling of packaging off-line. This shall include:

- setting and amendments to printer parameters to be completed by authorised personnel only
- documented checks against the coding schedule
- labelling area to be free from other packaging items
- records of quantity printed/labelled
- pre-coded/labelled packaging to be segregated from original packaging storage location.

Staff responsible for adjusting equipment settings and completing verification checks shall be trained against the site's labelling procedure.

Price and promotional labels shall be checked against the coding schedule.

Where coding/labelling activity takes place away from the controlled storage area, packaging shall be subject to the same issue and return controls defined in clause 6.3.5.

A full visual check of all relevant aspects of product packaging information (i.e. correct packaging/version, date/Julian code, price/promotional information) shall be completed at the same frequency defined in clause 6.3.9.

6.3.8

There shall be documented checks of the packing lines before commencing production and following changes of product. This shall include confirmation that the line has been cleared of product and packaging from the previous run.

Documented checks shall be completed before requesting packaging for the next production run.

6.3.9

Reference samples, or photographic standards, of the packaging being used for each production run shall be available to those staff carrying out checks. The samples shall include key distinguishing information with points of difference clearly highlighted.

6.3.10

There shall be procedures in place to ensure that all products are packed with the correct packaging components and are correctly labelled (this includes outer case labels). For every production run, a full visual check of all relevant aspects of product packaging information (e.g. product name, date/Julian code, price/promotional information) shall be completed at the following frequency:

- first pack produced in the run
- last pack produced in the run
- first pack produced after any break in production (i.e. box/carton change, film/label reel change staff breaks and fire alarms)
- first pack produced after a change in a batch of raw material that would require a Julian code change
- after any adjustment/maintenance to coding/labelling equipment
- every 30 minutes during the production run.

Visual checks shall be carried out by the person responsible for running the packing line. The following information on packaging shall be verified, as appropriate:

- correct packaging (including version)

6.3.10

- date code and format match the coding schedule
- price and promotional information match the coding schedule
- Julian code/batch information
- Other information printed during packing (i.e. country of origin, allergen information, quantity, barcode).

Any information printed during packing shall be checked to verify that the print quality is acceptable.

Staff responsible for adjusting equipment settings and completing verification checks shall be trained against the site's labelling procedure.

The results of checks shall be recorded and signed off. The sample used for inspection shall be retained with the paperwork. Where this is not possible, photographic records may be used however all relevant aspects of the packaging shall be captured.

In the event that a labelling or date coding error has been identified, there shall be documented corrective actions in place to identify and quarantine all affected products with adequate controls for rectifying or disposing of non-conforming products and packaging.

The site shall complete root cause analysis to prevent reoccurrence (see clause 3.6.4).

6.3.11

Where finished packed products are not coded at the time of packing, there shall be suitable documented procedures in place to ensure full control of these products. In these instances, each product shall be labelled with a batch code or Julian code for ease of traceability and stock control.

There shall be systems in place to ensure that the use of an external contractor to apply labelling and coding comply with the labelling requirements set out in this standard.

6.3.12 AUTOMATED LABEL VERIFICATION EQUIPMENT

CLAUSE	REQUIREMENTS
6.3.12.1	Automated label verification and date coding system shall have a controlled database which is stored on suitable equipment in a secure location (e.g., a server). There shall be sufficient back up system in place to prevent/minimise loss of data in event of a system failure. The information with the database shall be maintained to ensure is always accurate. Access to the database shall be appropriately controlled to avoid unauthorised amendments. System shall be able to maintain amendments log which indicates the originator, approver, date of change and the change in the product details.
6.3.12.2	The site shall complete the packaging verification decision tree for each packaging type / product group to determine if automated label verification and date coding equipment is required asper Control and Verification of Packaging, Labelling and coding of Product Group Procedure.
6.3.12.3	Where the site has an automated system installed, it shall comply with GTC-198 Control and Verification of Packaging, Labelling and Coding of Product. A documented gap analysis shall be completed against the requirements to demonstrate compliance with GTC-198.
6.3.12.4	Once the site is satisfied that the system is functioning correctly and can demonstrate all relevant staff are suitably trained and competent, the site can remove the 30-minute check or reduce the frequency. The following checks shall still be completed: • first pack produced in the run • last pack produced in the run • first pack produced after any break in production (i.e. box/carton change, film/label reel change staff breaks and fire alarms) • after any adjustment/maintenance to coding/labelling equipment.

6.3.12.4 Any information printed during packing shall be checked to verify that the print quality is acceptable if this is not already monitored by a vision system.

The results of checks shall be recorded and signed off. The sample used for inspection shall be retained with the paperwork. Where this is not possible, photographic records may be used however all relevant aspects of the packaging shall be captured.

6.3.12.5 In the event of the automated packaging/date code verification system being partially or completely disabled, there shall be a documented procedure to immediately implement manual systems in order to cover the function(s) that have been disabled. The manual system shall comply with clause 6.3.9.

The site shall notify the relevant customer where required. Manual checks shall continue until the automated system is fully functional.

6.4 QUANTITY - WEIGHT, VOLUME AND NUMBER CONTROL

FUNDAMENTAL: The site shall operate a quantity control system which conforms to legal requirements in the country where the product is sold and any additional industry sector codes or specified customer requirements.

CLAUSE	REQUIREMENTS

6.4.1 The site shall complete a risk assessment to determine the controls required to ensure that products packed meet the legal requirements in the intended country of sale.

The risk assessment shall consider:

- nature of the product (e.g. dripless, moisture loss)
- location (i.e. airflow, vibrations), accuracy and acceptable tolerances of measuring equipment
- the frequency of checks required based on the site's ability to identify affected packs in the event of an issue.

The site shall have documented procedures for the management of weight, volume, size and/or count for each product manufactured to ensure conformance to the legal requirements in the intended country of sale.

6.4.2 Where products declare a number of units (or minimum quantity), there shall be procedures in place to ensure that the correct number of items are present in the pack. Documented checks of product count shall be completed at the following frequency:

- start of run
- hourly
- following a break in production (i.e. staff break, emergency evacuation)
- end of run.

6.4.3 The zone of indecision (ZOI) shall be calculated via the Weight and Quantity Control app for each dynamic catch weight and check-weighing system to establish if the variability could lead to non-compliant batches of product. The zone of indecision shall be calculated across all weight bands (prescribed by current legislation) which are packed on that line.

The zone of indecision shall be re-evaluated when:

- a new product launches that falls outside the weight range of the existing ZOI evaluation
- a new product launches that significantly deviates from existing packaging formats that may affect the performance of equipment (i.e. wider/narrower or shorter/longer tray)
- where significant maintenance has taken place
- when equipment has been relocated.

In addition, the ZOI shall be completed annually on calibration by the service provider for a minimum of one product on each automatic check-weigher/catch-weigher. Where any issues are highlighted, corrective actions shall be made.

6.4.4 The tare weight of each packaging format shall be established for all catch weight and average weight systems. This shall include all components that do not contribute to the declared weight (e.g. drip pads, forks, skewers). The outcome shall determine the significance of any variation in packaging weight and frequency of monitoring checks required for packaging tare weights.

6.4.5 Procedures shall be in place to ensure that catch weight products are labelled with the correct weight. Catch weight equipment shall be verified daily for their accuracy using externally calibrated weights across the normal operating range.

At the start of each production run, catch weight equipment shall be verified that it rejects and/or does not label underweight packs.

The verification method shall reflect how the equipment takes weight measurements (i.e. static or dynamic). Where dynamic measurements are taken, the test pack shall be in same packaging format as the product.

6.4.6 Where automatic check-weighers are used, they shall have the following functionality:

- calculate the tolerable negative error (TNE) for a given nominal weight in accordance with current legislation
- produce a batch with a mean weight not less than the nominal quantity
- produce a batch with no more than 2.5% of packs that are less than the nominal weight minus the TNE (t1)
- reject all packs containing less that twice the TNE (t2)
- display the statistical data for each batch.

The site shall have a system in place to retrieve and archive all statistical data. Where the equipment does not store this data, the site shall maintain a manual record for each production run. All records shall be kept for a minimum of 2 years (see clause 3.2.3).

6.4.7 Procedures shall be in place to ensure that check-weighers are operating within specified tolerances. Check-weighers shall be verified daily for their accuracy across the normal operating range. The equipment shall take a dynamic measurement of externally calibrated weights to verify that the equipment is operating within the specified tolerance.

6.4.8 Procedures shall be in place to ensure that check-weighing equipment is operating within specified tolerances and that the rejection mechanisms are operating correctly. This shall be completed at the start of each product run.

The test pack used shall be 1g below the acceptable minimum weight. This is generally set at t2 however some systems may be programmed to reject at t1.

The test shall confirm that test packs have successfully rejected from the line or that the finished product label has not been applied.

Test packs shall be in same packaging format as the product.

6.4.9 The statistical data for each batch shall be analysed and verified at the end of the packing run by an authorised person. Where data is not stored on the equipment, the relevant data shall be manually recorded.

6.4.10 Underweights/rejected packs shall be suitably controlled to prevent accidental release. Reject bins, where used, shall be locked at all times and shall not be used for foreign body detection rejects.

6.4.11 When exporting outside the UK or where required by the customer, products weighed using an average weight system shall carry the 'e' mark to indicate compliance with the average weight system.

The local authority shall be notified of the manufacture of products intended for export to the EU.

6.4.12 Test packs shall be checked weekly against a calibrated scale to verify they are the weight declared.

6.4.13 The site shall have a clear procedure for handling rejected packs to ensure they do not re-enter the product flow.

Rejected packs shall be reworked in a controlled manner to ensure product safety, legality, quality and authenticity has not been compromised.

6.5 CALIBRATION AND CONTROL OF MEASURING AND MONITORING DEVICES

The site shall be able to demonstrate that measuring equipment is sufficiently accurate and reliable to provide confidence in measurement results.

CLAUSE	REQUIREMENTS
6.5.1	The site shall identify and control measuring equipment used to monitor critical control points and product safety, legality and quality, this also includes reference measuring equipment (e.g. master probe, multi-port calibrator). Acceptable tolerances shall be established with consideration given to acceptable tolerances for equipment used to monitor critical control points. Equipment shall be calibrated to National Standards at a frequency determined by risk assessment. calibration certificates and records shall be made available and up to date. Calibration shall be conducted across the normal operating range and reflect the method used by the equipment to take measurements (i.e. dynamic or static). Each piece of equipment shall be logged onto a master list which includes the following, as a minimum: • serial number and/or ID number of equipment • frequency of calibration • date of last calibration • date next calibration is due • frequency of internal verification • acceptable tolerances. • location of equipment (e.g., area, line, office) • if mobile equipment, specify in which area / areas mobile equipment is used (e.g., low risk area, high risk area)
6.5.2	The site shall have procedures in place to control measuring equipment defined in clause 6.5.1. These procedures shall include the following: • measures to prevent adjustment by unauthorised staff • protection from damage, deterioration or misuse • defined method and frequency for verification (see clause 6.5.3) to be completed by authorised personnel • corrective action to be taken when equipment is found to be operating outside acceptable tolerances including action to ensure that at-risk product is identified and placed on hold for assessment • contingency or back-up device to be available for when equipment is sent for external calibration

6.5.3	A system of interim verification shall be in place for all identified measuring devices, including new equipment and include checks: • at a predetermined frequency based on risk assessment to ensure the equipment is operating within prescribed parameters, to be completed weekly as a minimum • against equipment calibrated traceable back to a recognised National Standard where available. Records shall be maintained of such checks including: • target levels and tolerances • reference / ID of the testing equipment used • corrective action taken when equipment is found to be outside acceptable tolerances • assessment of results to identify any potential issues or long-term drift. Equipment shall be readable and of a suitable accuracy for the measurement it is required to perform. All measuring equipment used to monitor critical control points shall be checked for accuracy at the start of shift as a minimum.
6.5.4	Measuring equipment used to monitor a CCP shall be verified daily against a master reference unit or other valid method of verification.

7 PERSONNEL

7.1 TRAINING: RAW MATERIAL HANDLING, PREPARATION, PROCESSING, PACKING AND STORAGE AREAS

FUNDAMENTAL: The company shall ensure that all personnel performing work that affects product safety, legality and quality are demonstrably competent to carry out their activity, through training, work experience or qualification.

CLAUSE	REQUIREMENTS
7.1.1	The site shall have a documented and effective training and development programme in place to ensure that all employees including agency/temporary staff, are fully trained and competent to carry out their role.
7.1.2	All food handlers; both permanent and agency staff; shall undergo an induction process prior to commencing work. These shall cover staff safety, food safety, product quality, legality, food fraud and product integrity, environmental and energy. Staff shall be adequately supervised throughout the working period Food handlers training programme shall cover as minimum: • Site products and hazards • All procedures related to CCP control • Specific product risks e.g., microbiological, foreign matter and allergens, and the role of the personnel in preventing product contamination • Requirements of the site food safety and quality management systems (e.g., cleaning, metal detection, personal hygiene) • Pest awareness Any changes to the content of the programme shall be communicated to staff. The site's training programme shall ensure that anyone who is working or potentially could work in food handling areas including agency/temporary staff, undergo Food Safety Level 2 training within 12 weeks of employment and that all staff undergo Health and Safety Level 2 training within 12 weeks of employment.
7.1.3	Site shall ensure training of their staff, including temporary/agency staff and contractors in all appropriate safe methods that are relevant to the job and task they do. Training shall be done against updated version of the relevant procedures.
7.1.4	The site shall put in place documented programmes covering the training needs of relevant personnel. These shall include, as a minimum: • identifying the necessary competencies for specific roles • providing training or other action to ensure staff have the necessary competencies • reviewing the effectiveness of training • delivery of training in the appropriate language of trainees. All training shall be delivered by competent trainers or personnel. Competency assessments must be in place for trainers. Level of language understanding shall be established prior to commencement of training with consideration to the use of translators where resources are not available in a particular language. Information, instruction, supervision and training shall be in an understandable format and structure for all employees. Language understanding shall be appropriate to the roles being undertaken.

7.1.5	Staff training shall include a documented assessment of understanding and competence before the trainee engages in activities that place reliance on the training. Where such assessment identifies incorrect answers have been given, then the trainer shall discuss the correct answers with the trainee and satisfy themselves that the required level of understanding has been achieved then, assessment shall be repeated. Where repeated assessment identifies incorrect answers once again, formal discussion shall take place to discuss suitability for the role. All such training reviews shall be formally recorded. The site shall routinely review the competencies of its staff, where appropriate, it shall provide additional training. This may be in the form of, refresher training, coaching, mentoring or on-the-job experience. The frequency of refresher training, where required, shall be defined within the site's training procedure.
7.1.6	All training undertaken shall be documented and filed either in hard copies or online systems. These records shall be available for inspection. Records shall include as a minimum: • name of the trainee • date of training • title or course contents, procedures covered • duration of training • training provider • evidence of successful training • copies of certificates may form part of these records (e.g. Safe Use of Chemicals) • signature of trainer • signature of trainee indicating that they understood and agree to follow the procedures and systems of work being trained against • when refresher training is due, where required.
7.1.7	Training matrix shall be in place and demonstrate the following information: • who has been trained • who is currently under training • who can train others • refresher training
7.1.8	All relevant personnel, including engineers, agency-supplied staff, temporary staff and contractors, shall receive general allergen awareness training and be informed of the allergens permitted on site. Where entering production areas, they shall be trained in the site's allergen-handling procedures.
7.1.9	Where personnel are engaged in activities relating to critical control points, relevant training and competency assessment shall be in place.
7.1.10	All relevant personnel (including relevant agency-supplied staff, temporary staff and contractors) shall receive training on the site's labelling and packing processes which are designed to ensure the correct labelling and packing of products.
7.1.11	Site PPE and personal hygiene training shall include all requirements specified in clause 7.2.2 and as minimum: • PPE changing procedure • the use of company issued metal detectable pens. • Eating is not allowed other than in canteens and designated quality control tasting areas (after which hands must be washed.) Eating in offices can be permitted when offices are located away from the production areas and do not pose risk to product safety.

7.1.12 Additional training shall be implemented for staff working in High Care or High Risk areas, including all procedures related to these areas such as barrier control and cross contamination prevention.

7.1.13 Training programme shall be reviewed and updated regularly, based on needs identified during internal / external audits, competency assessment, after site KPI reviews related to Food Safety, Legality and Quality (e.g., customer complaints) if adverse trend identified and any process, procedural or product changes.

7.2 PERSONAL HYGIENE: RAW MATERIAL HANDLING, PREPARATION, PROCESSING, PACKING AND STORAGE AREAS

The site's personal hygiene standards shall be developed to minimise the risk of product contamination from personnel, be appropriate to the products produced and be adopted by all personnel, including agency-supplied staff, contractors and visitors to the production facility.

CLAUSE	REQUIREMENTS
7.2.1	Personal hygiene rules/standards shall be documented and adopted by all personnel at all times, including visitors and contractors.
7.2.2	The requirements for personal hygiene shall be documented and communicated to all personnel. These shall include, as a minimum, the following: • watches shall not be worn • jewellery shall not be worn, with the exception of a plain wedding ring, wedding wristband, medical alert jewellery or religious jewellery • prescription glasses frames (where worn) shall be plain with no stones attached • rings, tongue piercings and studs in exposed parts of the body, such as ears, noses and eyebrows, shall not be worn • fingernails shall be kept short, clean and unvarnished • false fingernails/nail art and false eyelashes shall not be permitted • hairclips shall be plain and kept underneath hair coverings • excessive perfume or aftershave shall not be worn. • Nose blowing to be conducted away from open product, disposal of tissues on each occasion followed by hand washing and disinfection. Compliance with the requirements shall be checked routinely. The site shall maintain a list of those permitted to wear religious and medical alert jewellery in production areas. Visitors and contractors shall be made aware of site personal hygiene requirements prior to entering production areas.
7.2.3	Hand-washing and/or sanitising shall take place whenever required. The site shall define handwashing frequencies which shall include as a minimum: • on entry to production/storage areas • after touching any part of the head or face • after sneezing, coughing or blowing nose • after handling waste, contaminated product and chemicals • before and after cleaning procedures • after handling allergens (and products of animal origin where required for speciation control) • after visiting the toilet • after smoking • after contact with pests or pest control equipment.

7.2.4	All cuts and grazes on exposed skin shall be covered by an appropriately coloured plaster that is not flesh coloured and is different from the product colour (preferably blue) and containing a metal detectable strip where foreign body detectors are used. These shall be site issued by an authorised person with records kept of when plasters have been issued, all plasters issued shall be monitored and accounted for at the end of the shift. There shall be procedures in place to identify if a plaster is lost and ensure there is no risk to the product. Where applied to hands, in addition to the plaster, a glove should be worn.
7.2.5	Personal items (e.g. keys (except locker keys), watches, personal mobile phones, coins and medicines) shall not be taken into production/packing or storage areas however agency workers are permitted to carry ID cards issued by the agency. Company-issued (Cranswick) mobile phones may be permitted however this shall be at the discretion of the site and based on risk assessment. Processes and written instructions for staff shall be in place to control the use and storage of personal medicines, so as to minimise the risk of product contamination. Where approval has been granted to carry personal medicines in production areas, the site shall maintain a list of the personnel and the medicines authorised.
7.2.6	Procedures shall be in place for the breakage/loss of glasses and contact lenses.
7.2.7	The site shall have procedures in place for the entry of emergency personnel (e.g. paramedics, fire personnel) and include any cleaning requirements prior to re-starting production/ packing requirements. There shall be procedures in place for the exit and re-entry into production areas in the event of an emergency evacuation.
7.2.8	Smoking including e cigarettes shall only be allowed in designated areas with adequate ventilation, which comply with local legislation. (see clause 4.8.10)

7.3 MEDICAL SCREENING

The site shall have procedures in place to ensure that employees, agency staff, contractors or visitors are not a source of transmission of food-borne diseases to products.

CLAUSE	REQUIREMENTS
7.3.1	The site shall comply with group visitors and contractors control procedure and shall use group medical questionnaires.
7.3.2	The site shall make employees aware of the symptoms of infection, disease or condition which would prevent a person working with open food. The site shall have a procedure which enables notification by employees, including temporary employees, contactors and visitors to the site, of any relevant symptoms, infection, disease or condition with which they may have been in contact or be suffering from. A procedure for returning to work after illness shall be in place. A risk assessment shall be completed prior to employees returning to work after illness based on the nature of the symptoms suffered during absence. This shall be overseen by a suitably competent person. Employees shall notify the site of any travel taken outside of the UK in the previous 3 months.

7.3.3 Where there may be a risk to product safety, visitors and contractors shall be made aware of the types of symptoms, infection, disease or condition which would prevent a person visiting areas with open food.

Visitors shall be required to complete a health questionnaire to confirm that they are not suffering from any symptoms which may put product safety at risk, prior to entering the raw material, preparation, processing, packing and storage areas.

Visitors shall be requested to notify the site of any travel taken outside of the UK in the previous 3 months.

7.3.4 There shall be procedures for employees, contractors and visitors relating to action to be taken where they may be suffering from or have been in contact with an infectious disease. Expert medical advice shall be sought where required.

7.4 PROTECTIVE CLOTHING: EMPLOYEES OR VISITORS TO PRODUCTION AREAS

Suitable site-issued protective clothing shall be worn by employees, contractors or visitors working in or entering production areas.

CLAUSE	REQUIREMENTS
7.4.1	There shall be adequate provision of suitable Personal protective equipment, to be used by all staff, contractors, and visitors, which is put on in accordance with the site's documented changing procedure. Personal protective equipment (PPE) must be in good condition, fit for purpose and used where necessary to minimise foreign body risks associated with people.
7.4.2	The company shall document and communicate to all employees (including agency and temporary personnel), contractors or visitors, the rules regarding the wearing of protective clothing in specified work areas (e.g. low risk areas, workshops, high-risk areas, high-care areas etc.). This shall also include policies relating to the wearing of protective clothing away from the production environment (e.g. removal before entering toilets, and use of canteen and smoking areas). There shall be clear instructions for the order of changing into and out of dedicated protective clothes to prevent contamination of clean clothing.
7.4.3	Protective clothing shall be available that: • is provided in sufficient numbers for each employee • is of suitable design to prevent contamination of the product (as a minimum containing no external pockets above the waist or sewn-on buttons) • fully contains all scalp hair to prevent product contamination • includes snoods for beards and moustaches, where required, to prevent product contamination.
7.4.4	Laundering of protective clothing shall take place by an approved contracted or in-house laundry using defined criteria to validate the effectiveness of the laundering process. The laundry must operate procedures which ensure: • adequate segregation between dirty and cleaned clothes • effective cleaning of the protective clothing • protective clothing for high-risk/high-care areas is commercially sterile following the washing and drying process • cleaned clothes are supplied protected from contamination until use (e.g. by the use of covers or bags).

7.4.4	Washing of protective clothing by the employee is exceptional but shall be acceptable where the protective clothing is to protect the employee from the products handled and the clothing is worn in enclosed product or low-risk areas only.
7.4.5	Protective clothing shall be changed at an appropriate frequency, based on risk. For high-risk and high-care areas the protective clothing shall be changed at least daily.
7.4.6	If gloves are used, they shall be replaced regularly. Where appropriate, gloves shall be suitable for food use, of a disposable type, of a distinctive colour (blue where possible), be intact and not shed loose fibres. Gloves shall be latex free (unless no other suitable option is available for safety reasons)
7.4.7	Where items of personal protective clothing that are not suitable for laundering are provided (such as chainmail gloves and aprons), these shall be cleaned and sanitised by an appropriate method that has been validated. The frequency of cleaning shall be based on risk. These items shall be subject to recorded inspections for damage on a scheduled basis.
7.4.8	Pre-shift start-up checks shall be in place to ensure the correct wearing of PPE and constant vigilance must be maintained throughout production.
7.4.9	All employees should be issued with suitable site-dedicated footwear to protect the manufacturing environment from sources of external contamination. Footwear should be kept clean and in good repair. Where in low-risk sites, factory footwear is site dedicated rather than only worn inside the factory. In such cases, it is permissible to wear factory footwear in the site boundary, but it is suggested that sticky mats or alternatives are put in place on entry to production areas. In wet operations, boot washing, and appropriate storage facilities should be provided. If boot or shoe washers/dips are in place they should be maintained in a clean condition and chemical additives should be monitored.
7.4.10	Plastic washable aprons should be cleaned and disinfected daily. When wear or cut damage is evident, aprons should be renewed.

8 PRODUCTION RISK ZONES - HIGH RISK, HIGH CARE AND AMBIENT HIGH CARE

Where a site produces product where the production process or part of it requires high-risk, high-care production and ambient high-care zones, all the relevant requirements from section 1-7 of the CMS shall be fulfilled in addition to the requirements in this section.

The site shall be able to demonstrate that product facilities and controls are suitable to prevent pathogen contamination of products.

8.1 LAYOUT, PRODUCT FLOW AND SEGREGATION IN HIGH-RISK, HIGH-CARE AND AMBIENT HIGH-CARE ZONES

FUNDAMENTAL: The factory layout, flow of process and movement of personnel shall be sufficient to prevent the risk of pathogen contamination of products.

CLAUSE	REQUIREMENTS
8.1.1	The design, layout and construction of the premises shall ensure there is effective separation between Low-Risk and High-Risk / High-Care areas, to prevent the risk of cross contamination from the Low-Risk environment. This includes movement of ingredients / raw materials, packaging, equipment, personnel, water, air, drainage, chemicals and disposal of waste.
8.1.2	Suitable facilities shall be in place for the controlled return of equipment e.g., racks from High-Risk / High-Care, back to Low Risk. These facilities and the flow of the equipment shall be designed to ensure the integrity of the manufacturing area is not compromised.
8.1.3	When finished products are returned into a High-Risk / High-Care area, site shall ensure that the return process prevents both cross-contamination of Work in Progress (WIP) components and raw materials in High-Risk / High-Care from finished product packaging. It shall also avoid the temperature abuse of product. The method shall be documented, and its operation recorded to demonstrate that it is being followed when needed.
8.1.4	The map of the site shall include the location of the pathogen control steps. (see clause 4.3.1)

8.1.4.1 HIGH RISK AND HIGH CARE BARRIER CONTROLS

There shall be a robust barrier control system to prevent risk of pathogen contamination from low-risk areas.

CLAUSE	REQUIREMENTS
8.1.4.1.1	The site shall complete a risk assessment for all transfer points and methods of all people, food items (e.g., raw materials, returned finished products), packaging, waste, drainage, equipment and utilities that enter and exit high-risk and high-care areas in order to maintain and improve the integrity of the production zone. The assessment shall be formally reviewed and signed by the HACCP team. The risk assessment shall undertake the following steps; **1. Identification and Assessment of Transfer Points**

8.1.4.1.1 The site shall complete a risk assessment for all transfer points and methods of all people, food items (e.g., raw materials, returned finished products), packaging, waste, drainage, equipment and utilities that enter and exit high-risk and high-care areas in order to maintain and improve the integrity of the production zone. The assessment shall be formally reviewed and signed by the HACCP team.

The risk assessment shall undertake the following steps;

1. Identification and Assessment of Transfer Points

The site shall Identify all transfer points used for people, food items (e.g., raw materials, returned finished products), packaging, waste, drainage, equipment and utilities that enter and exit high-risk and high-care areas ensuring that emergency exits have also been considered. There shall be a plan of the site showing the relevant high-risk and high-care areas and the location and reference number of each transfer point.

The barrier integrity for each transfer point shall be risk assessed with consideration to the following;

- minimisation of open hatches
- elimination of physical gaps, where elimination cannot be achieved justifications shall be detailed within the assessment
- alarming of doors
- installation of airlocks/interlocks
- other physical methods or systems and procedures in place

2. Assessment of Controls for Each Transfer Point

The site shall identify all items that enter and exit the high-risk/high-care area, these shall be appropriately grouped according to type with utilities listed individually. The following shall be considered as a minimum;

- people
- cooked product
- feedstock and other ingredients
- packaging
- air
- water and waste water
- finished product
- waste
- equipment
- tools and other portable devices
- cleaning equipment
- chemicals

The site shall identify the transfer points associated with each item and document the transfer process or procedures used to control microbiological hazards. The verification methods for each transfer process or procedure shall be documented.

The HACCP team shall determine if any further action is required for each transfer point.

8.1.4.1.2	Product/ingredients/packaging transferred to a high care/risk area within chilled or frozen operations, shall be subject to a validated heat treatment or non-heat treatment method to ensure their effective decontamination. (see blueprint document) Correct application of the validated process shall be monitored, recorded and subject to verification. Where an ingredient is to be used without heat treatment in High Risk / Care, there needs to be a C of A present. Where portable equipment is used in high care/risk areas (including equipment used by visitors and auditors), these items shall be visually distinctive and dedicated for use in that area or be subject to a sufficient treatment method to ensure their effective decontamination.

SECTION 8: PRODUCTION RISK ZONES – HIGH RISK, HIGH CARE AND AMBIENT HIGH CARE

8.1.4.1.3 Where cookers form the barrier between high care/risk and low risk chilled or frozen operations, these shall be fitted with interlocks to prevent both doors being opened at the same time. Any other transfer area between low and high care/risk areas shall also be controlled so that doors, hatchways or similar cannot be opened simultaneously. For in-line cooking operations with access hatches in low risk, these shall be locked off and secured during production. (see blueprint document)

Interlocks shall be checked daily to confirm they are working correctly.

8.1.4.1.4 Where conveyorised belts are used to transfer product from high-risk/high-care, these shall not extend beyond the transfer point.

8.1.4.1.5 The microbiological cleanliness of food contact packaging for high-risk/high-care products shall be routinely monitored. The frequency shall be based on risk assessment.

8.1.4.1.6 High-risk and high-care areas shall have their own cleaning facilities. Equipment shall not be returned to low risk for cleaning, unless the equipment goes through a heating or disinfection process on return to high-risk/high-care (e.g., through a heating cycle in an oven).
Cleaning facilities shall have sufficient hot water for cleaning.

8.1.4.1.7 High care/risk areas shall maintain positive pressure such that all air movements are in the direction from high care/risk to low risk. Positive air pressure shall be demonstrable at all product and equipment transfer points and a system shall be in place to regularly monitor and record positive pressure.

8.1.4.1.8 The site shall include air quality within its environmental monitoring programme and shall be subject to the requirements set out in 4.11.16.

8.1.4.1.9 All personnel working in High-Risk / High-Care area (Operations, Quality, Engineering, Hygiene, long term contactors etc.) shall be trained to understand the detail and importance of high care controls, basic knowledge of listeria and the consequences of poor standards.

CLAUSE	REQUIREMENTS
8.1.5	Where ambient high-care areas are required a documented risk assessment shall be completed to determine the risk of cross-contamination with pathogens. The risk assessment shall take into account the potential sources of microbiological contamination and include: • the raw materials and products • flow of raw materials, packaging, products, equipment, personnel and waste • airflow and air quality • utilities (including drains). Effective processes shall be in place to protect the final product from this contamination. These processes may include segregation, management of process flow or other controls.

8.2 BUILDING FABRIC IN HIGH-RISK AND HIGH-CARE ZONES

CLAUSE	REQUIREMENTS
8.2.1	Where site include high-risk or high-care facilities, there shall be a map of the drains for these areas which shows the direction of flow and the location of any equipment fitted to prevent the backup of waste water. The flow from drains shall not present a risk of contamination to the high-risk/care area.

8.2.2 Air conditioning/ventilation for high care/risk operations shall incorporate high efficiency filtration which is appropriately maintained. Air handling requirements shall be risk assessed with the following criteria considered;

- the source of air. The air inlet needs to be located to minimise intake of contaminated air (at a minimum, it should be upwind of potential contaminants such as dust and chemical vapours)
- the specification of filter used
- the frequency of air changes (see Blueprint document)
- the frequency of replacement of filters (see Blueprint document)

At a minimum, filtration should comply with (expressed in accordance with EN779 and EN1822) F7 - F9 for high care and F9-H12 for high risk.

8.2.3 Where sites include removable walls as part of the design of the high-risk or high-care area (e.g., to allow occasional movement of large items or specialist maintenance equipment), procedures shall be in place to ensure:

- removable walls are tight fitting
- their use is managed
- movement of the walls is authorised and is completed only by trained and authorised staff
- cleaning and reconditioning procedures are in place and completed prior to production.
- area is released following inspection by competent member of management team

8.2.4 Any intended changes to the factory structure shall be carefully considered, and planned in advance, to assess the impact on the High-Risk / High- Care area's integrity. A method of documenting these changes shall be in place.

8.2.5 All surfaces of rooms, e.g., walls, ceilings and floors must be sealed and impervious to water and must be capable of being cleaned and being kept in a dry condition.

Any panels (e.g., back splash panels), signs and equipment mounted to the wall shall be appropriately sealed to avoid accumulation of dirt, condensation and mould growth, micro harbourage under the item.

8.2.6 Fabrication audits shall be done at least every three months to check for damage and the potential for micro harbourage. Depending on the age of the factory, its throughput, and its condition, these audits may need to be more regular.

8.2.7 Fire shutters - need to be appropriately constructed to allow for cleaning and not painted. All barrier points, including cookers, high risk / care to be physically sealed from the roof void.

Where cookout from products may occur, this is to be disposed on in the High Risk side only via drains which are specifically installed by the oven door for this purpose. There shall be no gaps between any metal plating, the floor and the drains or these plates need to be fully removable and cleaned on a daily basis.

8.2.8 All drains need to be accessible for cleaning and not located under equipment which prevents access or cannot be moved.

8.2.9 Fabrication repair/improvement action list shall be adequately managed to ensure correct prioritisation and completion in timely manner. (see appendix 5) (this clause shall be incorporated with clause 4.4.14)

8.3 MAINTENANCE IN HIGH-RISK AND HIGH-CARE ZONES

CLAUSE	REQUIREMENTS
8.3.1	Maintenance activities undertaken in high-risk and high-care areas shall respect the segregation requirements of the area. Wherever possible tools and equipment shall be captive to the area with toolboxes and their contents subject to the requirements of clause 4.7.9. Specific barrier transfer controls shall be in place for non-captive portable items. Where portable maintenance equipment is used in high care/risk areas (including contractors), these items shall be visually distinctive and dedicated for use in that area or be subject to a sufficient, validated treatment method to ensure their effective decontamination.
8.3.2	Where equipment is removed from the high-risk and high care area, the site shall have a procedure to ensure the cleanliness and removal of contamination hazards before the equipment is accepted back into the area. Records of acceptance back into the area shall be maintained.
8.3.3	Separate engineering and storage facilities for both Low Risk and High-Risk/High-care sides of the premises shall be in place wherever possible. Engineering staff working in High-Risk / High-Care areas must be provided with appropriate protective clothing and footwear, dedicated to High-Risk / High-Care areas. (see clause 8.7)

8.4 STAFF FACILITIES FOR HIGH-RISK AND HIGH-CARE ZONES

CLAUSE	REQUIREMENTS
8.4.1	Personnel shall enter and exit high-risk or high-care area via a specially designated changing facility at the entrance to the area. The changing facilities shall incorporate the following: • clear instructions for the order of changing into and out of dedicated protective clothes to prevent the contamination of clean clothing • protective clothing that is visually distinct from that worn in other areas and which shall not be worn outside the area • a hand-washing routine during the changing procedure to prevent contamination of the clean clothing (i.e., handwashing after hair covering and footwear have been put on, but before handling clean protective clothing) • handwashing and disinfection facilities that shall, as a minimum be situated: prior to entry into high-risk/care areas and on entry into high-risk/care areas • dedicated site footwear that is provided by the site and which shall not be worn outside the factory • an effective control of footwear to prevent the introduction of pathogens into the area. Control may be by segregation and a controlled change of footwear before entering the area (such as a barrier or bench system) or by the use of controlled and managed boot-wash facilities where these demonstrably provide an effective control of footwear to prevent the introduction of pathogens into the area. A programme of environmental monitoring shall be used to assess the effectiveness of footwear controls.
8.4.1	Hand washing shall be routinely monitored in person. This shall be scheduled and documented. Where adverse trends are seen, corrective action shall be taken.

8.5 HOUSEKEEPING AND HYGIENE IN HIGH-RISK AND HIGH-CARE ZONES

FUNDAMENTAL

CLAUSE	REQUIREMENTS
8.5.1	Environmental cleaning procedures in high-care/high-risk areas shall consider the different microbiological risks associated with each production risk zone. The frequency and methods of cleaning shall be based on risk, and the procedures shall be implemented to ensure that appropriate standards of cleaning are achieved.
8.5.2	Microbiological limits for acceptable and unacceptable cleaning performance shall be defined for high-risk/high-care production risk zones. These limits shall be based on the potential hazards relevant to the product or processing area. Therefore, acceptable levels of cleaning shall be defined, for example, by visual appearance, ATP bioluminescence techniques, microbiological testing, or chemical testing as appropriate. The site shall define the corrective action to be taken when monitored results are outside of the acceptable limits. Where cleaning and disinfection procedures are part of a defined prerequisite plan to control the risk of a specific hazard, the procedures and frequencies shall be validated and records maintained. This shall include the risk from cleaning chemical residues on food contact surfaces.
8.5.3	The calculated cleaning window and resourcing shall be strictly adhered to. Any deviations shall be reported to senior management and tracked as a site Key Performance Indicator (KPI) (see clause 1.3)
8.5.4	High-Risk / High-Care production areas shall be kept as dry as possible to reduce the transfer of Listeria.
8.5.5	Changeover cleans and break time cleaning methods shall be validated, effective, and controlled so that they achieve their purpose while also preventing excess wetting of the production environment and reducing the risk of cross-contamination to other lines or equipment.
8.5.6	High pressure (>145psi) cleaning is strictly prohibited in any High-Risk / High-Care environment. Nozzles which produce excessive aerosolization shall not be used.
8.5.7	Equipment used for cleaning in high-care and high-risk areas shall be: • visually distinctive and dedicated for use in that area • hygienically designed and fit for purpose • cleaned and stored in a hygienic manner to prevent contamination (for example, storing equipment in designated locations, off the floor, when not in use).
8.5.8	Where the site uses CIP equipment, either this shall be for a specific area only (i.e., separate equipment for high-risk, high-care and other production areas) or the CIP system shall be designed and controlled so that it does not present a risk of contamination to the high-risk/high-care area (i.e., controlling direction of flow from high-risk/high-care to low-risk areas, preventing the recycling or re-use of rinse solutions from one area to another).
8.5.9	Hygiene inspections for all High Care/High Risk areas shall be completed monthly.

8.6 WASTE AND WASTE DISPOSAL IN HIGH-RISK, HIGH-CARE ZONES

CLAUSE	REQUIREMENTS
8.6.1	Waste disposal systems shall ensure that the risk of contamination of products is minimised through the control of potential cross-contamination. Risk assessment shall consider the movement and flow of waste and waste containers. (See clause 4.12.2 and 8.1.4.1.1)

8.7 PROTECTIVE CLOTHING IN HIGH-RISK AND HIGH-CARE ZONES

CLAUSE	REQUIREMENTS
8.7.1	Laundering of protective clothing for high-risk and high-care areas shall be done by an approved and audited by GTS contractor. The laundry must operate procedures which ensure: • adequate segregation between dirty and cleaned clothes • adequate segregation between clothes for high-risk, high-care and low-risk areas etc. • effective cleaning of the protective clothing • commercial sterilisation of the protective clothing following the washing and drying process • protection of the cleaned clothes from contamination until use. The site shall monitor the performance of cleaning through the micro swabbing programme at a pre-determined frequency.
8.7.2	Transfer of laundered protective clothing shall be controlled to minimise risk of contamination, as minimum high-risk / high-care overalls shall be stored in the protective bag applied at the laundry stage until transferred into high-risk / high-care changing area.
8.7.3	Protective clothing for use in high-risk and high-care areas shall be changed at an appropriate frequency based on risk and at a minimum daily.

9 REQUIREMENTS FOR TRADED PRODUCTS

9.1 SPECIFICATIONS

FUNDAMENTAL: The site shall have a fully implemented and effective HACCP plan incorporating Codex Alimentarius HACCP principles.

CLAUSE	REQUIREMENTS
9.1.1	The company shall either: • have a HACCP or food safety plan specifically for the traded products handled on site, or • incorporate the traded products into its existing HACCP or food safety plans (see section 2). The scope of traded products HACCP or food safety plan shall include the products and the processes for which the site is responsible. At a minimum, this shall include goods receipt, storage, and dispatch.

9.2 APPROVAL AND PERFORMANCE MONITORING OF MANUFACTURERS/ PACKERS OF TRADED FOOD PRODUCTS

The site shall operate procedures for approval of the last manufacturer or packer of food products which are traded to ensure that traded food products are safe, legal and manufactured in accordance with any defined product specifications.

CLAUSE	REQUIREMENTS
9.2.1	The approval process of suppliers and the manufacturer/processor of each product traded shall be managed in accordance with GTS-027 Supplier Approval Procedure. Approval shall give consideration to; • the nature of the product and associated risks • customer-specific requirements • legislative requirements in the country of sale or importation of the product • source or country of origin • potential for adulteration or fraud • potential risks in the supply chain to the point of receipt of the goods by the company • the brand identity of products (i.e. customer own brand or branded product).
9.2.2	Ongoing performance review of manufacturers of products shall be managed by GTS, this shall meet the requirements outlined within GTS-027. The site shall ensure that all issues identified on receipt of raw materials and packaging (see 3.4.2.1) are communicated to GTS so that the information may be used as part of the supplier review.
9.2.3	Records shall be maintained of the manufacturer's/packer's approval process, including audit reports or verified certificates confirming the product safety status of the manufacturing/packing sites supplying the products traded. There shall be a process of review and records of follow-up of any issues identified at the manufacturing/packing sites with the potential to affect food products traded by the company.

9.2.4 ● There shall be a process for the ongoing review of manufacturers/packers, based on risk and using defined performance criteria, which may include complaints, results of any product tests, regulatory warnings/alerts, customer rejections or feedback. The process shall be fully implemented.

Where approval is based on questionnaires, these shall be reissued at least every 3 years and suppliers shall be required to notify the site of any significant changes in the interim, including any change in certification status.

Records of the review shall be kept.

9.3 SPECIFICATIONS

Specifications or information to meet legal requirements and assist customers in the safe usage of the product shall be maintained and available to customers.

CLAUSE	REQUIREMENTS
9.3.1 ●	Specifications shall be available for all products. These shall either be in the agreed format as supplied by the customer or, where this is not specified, include key data to meet legal requirements and assist the customer in the safe usage of the product. Specifications may be in the form of a printed or electronic document, or part of an online specification system.
9.3.2 ●	The site shall seek formal agreement of the specifications with relevant parties. Where specifications are not formally agreed, the site shall be able to demonstrate that it has taken steps to ensure formal agreement is in place.
9.3.3 ●	The site shall operate demonstrable processes to ensure that any customer-specified requirements are met. This may be by inclusion of customer requirements within buying specifications or by undertaking further work on the purchased product to meet the customer's specification (e.g. sorting or grading of product).
9.3.4 ●	Specification review shall be sufficiently frequent to ensure that data is current or at a minimum every 3 years, taking into account product changes, suppliers, regulations and other risks. Reviews and changes shall be documented.

9.4 PRODUCT INSPECTION

The site shall operate processes to ensure that the products received comply with buying specifications and that the supplied product is in accordance with any customer specification.

CLAUSE	REQUIREMENTS
9.4.1 ●	The site shall have a product sampling or assurance programme to verify that the products are in accordance with buying specifications and meet legal and safety requirements. Where verification is based on sampling, the sample rate and assessment process shall be risk-based. Records of the results of assessments or analysis shall be maintained.
9.4.2 ●	Where verification of conformity is provided by the supplier (e.g. certificates of conformity or analysis), the level of confidence in the information provided shall be supported by commissioning periodic independent product analysis.

9.4.3 ●	Where claims are made about the products being handled, including the provenance, chain of custody and assured or 'identity preserved' status of a product or raw materials used, supporting information shall be available from the supplier or independently to verify the claim.
9.4.4 ●	Where the site undertakes or subcontracts analyses which are critical to product safety or legality, the laboratory or subcontractors shall have gained recognised laboratory accreditation or operate in accordance with the requirements and principles of ISO 17025. Documented justification shall be available where non-accredited test methods are used.
9.4.5 ●	Test and inspection results shall be retained and reviewed to identify trends. Appropriate actions shall be implemented promptly to address any unsatisfactory results or trends.

9.5 PRODUCT LEGALITY

The company shall have processes in place to ensure that the food products traded comply with the legal requirements in the country of sale where known.

CLAUSE	REQUIREMENTS
9.5.1 ●	The site shall have documented processes to verify the legality of products which are traded. These processes shall include as appropriate: • labelling information • compliance with relevant legal compositional requirements • compliance with quantity or volume requirements. Where such responsibilities are undertaken by the customer, this shall be clearly stated in contracts.

9.6 TRACEABILITY

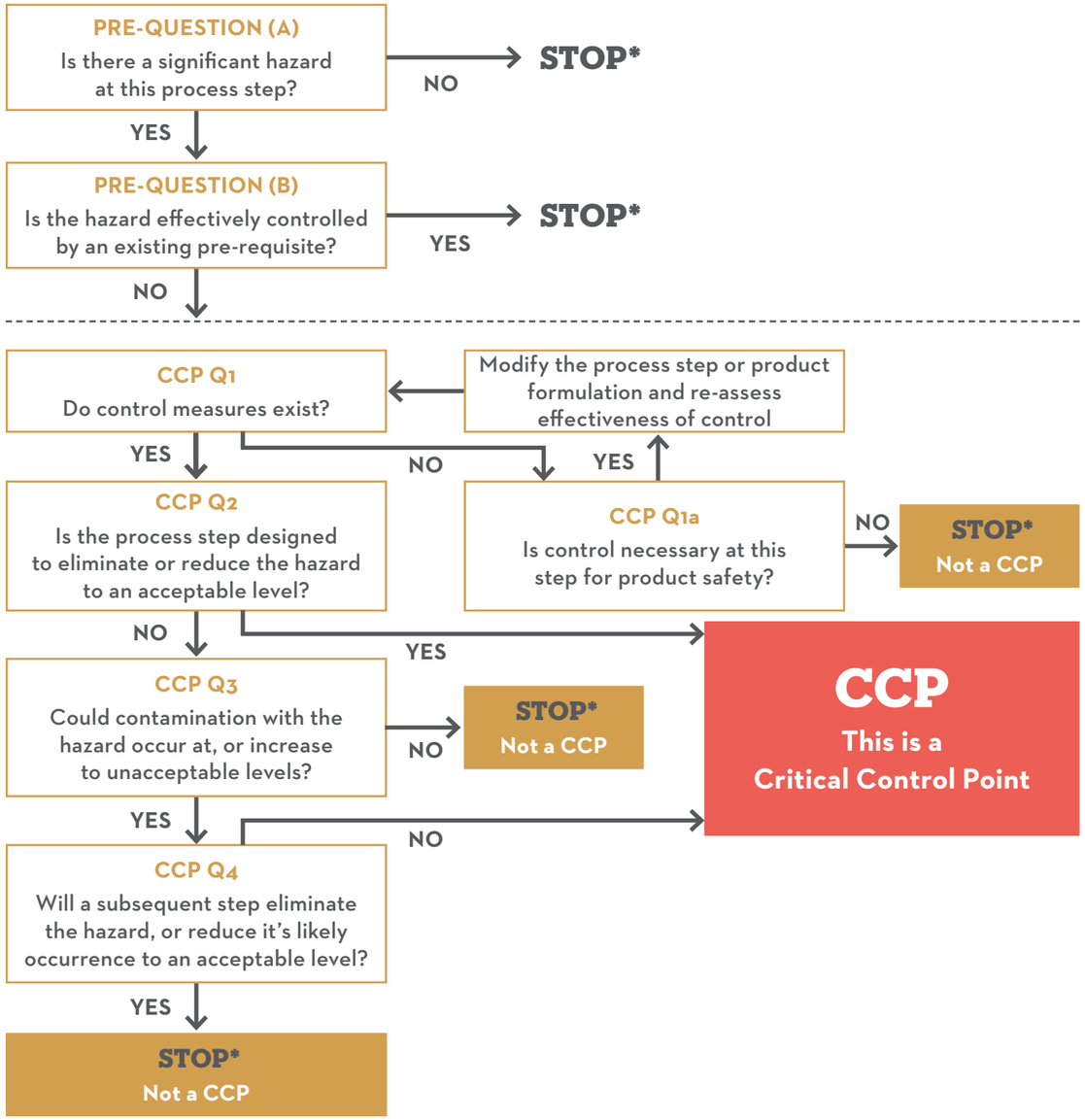
The company shall be able to trace all product lots back to the last manufacturer and forward to the customer of the company.

CLAUSE	REQUIREMENTS
9.6.1 ● ●	The site shall maintain a traceability system for all batches of product which identify the last manufacturer or, in the case of primary agricultural products, the packer or place of last significant change to the product. Records shall also be maintained to identify the recipient of each batch of product from the company.
9.6.2 ●	The site shall test the traceability system at least annually to ensure that traceability can be determined back to the last manufacturer and forward to the recipient of the product from the company. This shall include identification of the movement of the product through the chain from the manufacturer to receipt by the company (e.g. each movement and intermediate place of storage).
9.6.3 ●	The traceability test shall include the reconciliation of quantities of product received by the site for the chosen batch or product lot. Traceability should be achievable within 4 hours (1 day when information is required from external parties).

APPENDIX 1

NOTE: The Pre-Questions (A) & (B) are NOT included in the CODEX standard decision tree. There are however still important question's to ask when considering CCP's. All questions below the dotted line form the CODEX CCP decision tree.

CRANSWICK CCP DECISION TREE



***STOP - DOCUMENT THE DECISIONS AND PROCEED ON TO THE NEXT PROCESS STEP**

APPENDIX 2

EXAMPLES OF GROUP AUDITS, SITE BASED AUDITS AND SITE BASED INSPECTIONS

Group Internal Audits	Food Safety Audit
	Training & Medical Verification Audit
	Ethical Verification Audit
	Hygiene Verification Audit
	Integrity Audit
	Environmental Controls
	Validation Verification Audit
	Pest Management, External, Security Verification Audit
Site Based Internal Audits	Storage, Despatch & Transport and Product Release Verification Audit
	Control of NC product, Doc Control & Record Completion Verification Audit
	NPD Verification Audit
	Relevant BRCGS Modules
	Relevant BMPA Modules
	Relevant Retailers Modules
	Staff Facilities, Personal Hygiene Verification Audit
Site Based Documented Inspections	Building and Fabrication
	Foreign Body Control
	Personal Hygiene
	Allergen Control
	Pest Activity
	Traceability and Identification of Materials

APPENDIX 3

SERVICES PROVIDERS APPROVAL

Suppliers of services managed by GTS

The approval and monitoring, of the following service providers shall be managed by GTS team in accordance with Group Supplier Approval Procedure:

Offsite storage facilities of meat, ingredients, packaging, or finished products (Any processing conducted by the Coldstore on behalf of a Cranswick site will be looked after by site technical.

- Laundry
- Sandwich suppliers to vending machines at Cranswick sites
- Food contact gases
- Transport for distribution

Suppliers of services not managed by GTS

Such services may include, as appropriate:

- pest control
- contracted cleaning
- contracted services and purchasing & maintenance of equipment
- providers of PPE, metal detectable plasters and stationery for factory use. E.g., metal detectable pens.

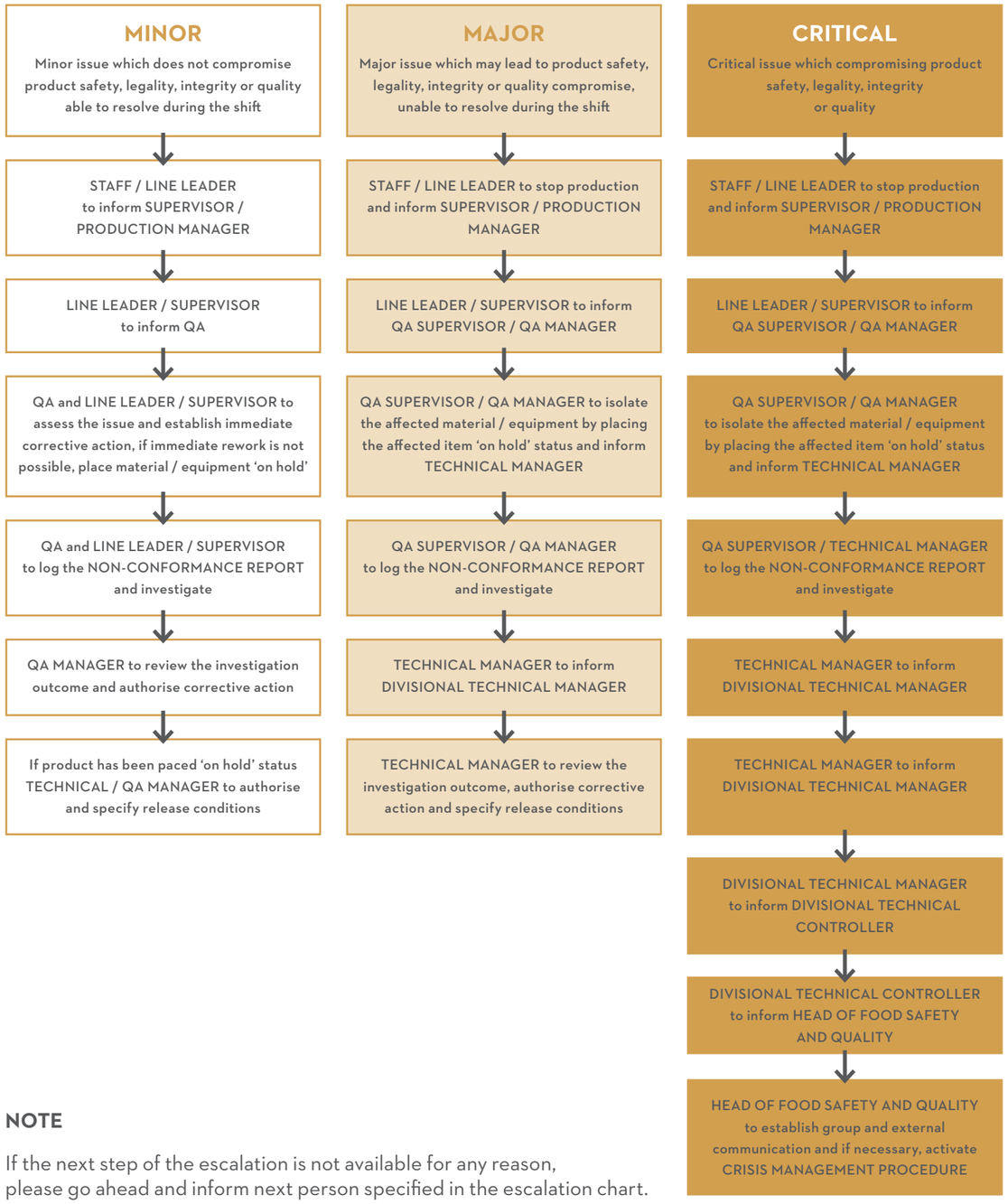
Pallets & Dolavs including plastic freezer sheets.

- off-site packing, labelling and processing of products
- laboratory testing
- vending machine providers (excluding sandwich manufacturers which are managed by GTS)
- onsite catering contractors
- waste management.
- providers of product safety training
- product safety consultants
- cleaning chemical providers e.g., Holchem

Any outsourced or subcontracted process (i.e., any activity that is part of the production process but does not occur on site)

APPENDIX 4

FOOD SAFETY NON-CONFORMANCE ESCALATION AND ACTION CHART



NOTE

If the next step of the escalation is not available for any reason, please go ahead and inform next person specified in the escalation chart.

APPENDIX 5

FABRICATION REPAIR / IMPROVEMENT PRIORITISATION

1. LOW RISK AREA

Priority Rating	Definition		Immediate action time sale (Control the risk within temporary measures)	Long (Preventative) action time scale*
HIGH	Direct food contact and indirect food contact (e.g., ceiling above open product production line)	• Direct food contamination risk (Foreign body, chemical, allergen, micro) • Immediate hygiene issue • Risk to CCP control • Direct risk of pest infestation/harborage • Cross-contamination risk (Foreign body, chemical, allergen, micro)	Immediately after report	Within 2 weeks from report
			Within 1 day from report	Within 4 weeks from report
MEDIUM	Open food area but not indirect food contact	• Increased Hygiene maintenance risk (e.g., dirt, residues harbourage) • Increased risk of pest access to the area / site • No risk of product contamination or cross-contamination (Foreign body, chemical, allergen, micro)		
LOW	Enclosed product area	• Minimal risk to hygiene maintenance • No risk to pest control	Within 5 days from report	Within 12 weeks from report

* Where long (Preventative) action time scale is not achievable, temporary measures shall be monitored for effectiveness. Senior management team shall agree safe and practicable time scale. Agreement shall be documented, including agreed action evidence (e.g., capex sign off, contact agreement, item purchase confirmation).

2. HIGH-RISK / HIGH CARE AREA

Priority Rating	Definition			Immediate action time sale (Control the risk within temporary measures)	Long (Preventative) action time scale*
		Foreign body, chemical, allergen	Pathogen (micro)		
HIGH	Direct food contact and indirect food contact (e.g., ceiling above open product production line)	• Direct food contamination risk • Risk to CCP control • Direct risk of pest infestation/harborage	• Direct food contamination risk • Immediate hygiene issue • Direct risk of growth and pathogen harbourage • Risk to CCP control • Risk to barrier control • High usage item (e.g., automatic door opening button, film holder etc.)	Immediately after report	Within 2 weeks from report
				Within 1 day from report	Within 4 weeks from report
MEDIUM	Open food area but not indirect food contact	• Cross-contamination risk • Increased risk of pest access to the area / site	• Cross-contamination risk (high activity area where process flows daily) • Increased hygiene risk which may support pathogen growth and dirt harborage		
LOW	Enclosed product area	• No risk of product contamination or cross-contamination • No risk to pest control	• No risk of product contamination or cross-contamination • Minimal risk to hygiene maintenance	Within 5 days from report	Within 12 weeks from report

* Where long (Preventative) action time scale is not achievable, temporary measures shall be monitored for effectiveness. Senior management team shall agree safe and practicable time scale. Agreement shall be documented, including agreed action evidence (e.g., capex sign off, contact agreement, item purchase confirmation).

APPENDIX 6

GROUP DOCUMENTS LIST

DOCUMENT REF No.	DOCUMENT TITLE
GTC001	Group Document Index Record
GTC029	Group Internal Audit Procedure
GTC037	Group Internal Audit report
GTS072	Group Visitor and Contractor Controls Procedure
GTS073	Group Visitor Questionnaire
GTS074	Group Artwork Approval Procedure
GTS075	Group Pack Copy and Artwork Check Sheet
GTS076	Site Visitor Privacy Notice (GDPR)
GTC077	Group Internal Audit Schedule - Safe and Legal Storage & Distribution supplier
GTC081	Ethical Verification Audit Template
GTC082	Supplier & Internal Auditor Verification Record
GTC085	Employee Training Record
GTC103	Data Protection Waiver
GTC115	CMS - Group Technical Services Verification Audit Template
GTC117	Equal Opps Survey Template
GTC121	Cranswick Group Crisis Simulation report
GTC126	Energy ISO50001 Internal audit report
GTC127	Ethical Verification Audit - Farms
GTC139	Labour Provider Approval Procedure
GTC141	Approved List of Internal auditors
GTC142	Group Induction Questionnaire
GTC142a	Group Induction Questionnaire - Answer Sheet
GTC142b	Group Refresher Induction Questionnaire
GTC142c	Group Refresher Induction Questionnaire - Answer Sheet
GTC146	SEDEX Supplier Control Procedure
GTC148	End of Line Testing Questionnaire

APPENDIX 6

DOCUMENT REF No.	DOCUMENT TITLE
GTC148a	End of Line Testing Questionnaire - Answer Sheet
GTC152	Label Verification Protocol
GTC155	Cranswick Stakeholder Map for Human Rights
GTC155a	Cranswick Stakeholder Engagement Plan for Human Rights
GTC182	Stakeholder Map for Interested Parties for Environmental, Energy and H&S
GTC182a	Stakeholder Engagement Plan for Interested Parties for Environmental, Energy and H&S
GTC184	List of Approved Supplier Auditors
GTC185	Site Verification Audit
GTC186	Farm Verification Audit
GTC190	Group Organisational Structures
GTC195	Group Training Procedure
GTC195a	Group Training Competency Management System
GTC196	Site Induction Audit Programme - Feedback Form
GTC197	Group Foods Donation / Surplus Stock Procedure
GTC198	Cranswick LV Protocol
GTC199	GC Task Risk Assessment
GTC200	Risk Incident Procedure
GTC201	Group Internal Training Quality Assessment Procedure
GTC202	Group Internal Training Appeals Procedure
GTS041	Group Document Control Procedure
GTS042	Secuirty Provider Supplier Audit Form
GTS043	Adhoc Labour Provider Approval Questionnaire
GTC203	Remote Auditing Risk Assessment
GTC204	Coronavirus Visitor Questionnaire
GTC206	Group Training Platform Procedure
GTC209	Zone of Indecision Template
GTC210	Tare Weight Determination Template

APPENDIX 6

DOCUMENT REF No.	DOCUMENT TITLE
GTC211	CMS - Allergen & Integrity Verification Audit Template
GTC212	CMS - Contamination Control Verification Audit Template
GTC213	CMS - Labelling & Pack Controls Verification Audit Template
GTC214	CMS - Quantity and Weight Control Verification Audit Template
GTC215	Group Management of Change Procedure
GTC215a	Group Change Management Record
GTC216	Group Coronavirus Decision Trees
GTC217	Group Face Covering and Visor Procedure
GTC218	CMS - Internal Audit Verification Audit Template
GTC219	CMS - Group Systems Verification Audit
GTC220	HACCP VA Audit Template (Cranswick Standard)
GTC221	Cranswick Manufacturing Standard Gap Tool
GTC222	Covid Lateral Flow Testing Procedure
GTC223	Master Risk Assessment
GTC224	COVID Immunisation and Testing Policy
GTC225	ISO Master Risk Assessment
GTC226	CMS Gap Analysis
GTC227	Group Concessions Procedure
GTC228	CMS - Traded Goods Verification Audit Template
GTC229	CMS - Control of Non-Conforming Product Verification Audit Template
GTC230	CMS - NPD Verification Audit Template
GTC231	CMS - Hygiene Verification Audit Template
GTC232	CMS - Training & Medical Verification Audit Template
GTC233	CMS - Validation Verification Audit Template
GTC234	CMS - HACCP Verification Audit Template
GTC235	CMS - Pest Control, External, Security, Waste Verification Audit Template
GTC236	CMS - Staff Facilities, Personal Hygiene Verification Audit Template

APPENDIX 6

DOCUMENT REF No.	DOCUMENT TITLE
GTC237	CMS - Storage Facilities, Dispatch & Transport, Product Release and Supplier Approval Verification Audit Template
GTC238	CMS - Derrogations Form
GTC239	Annual Internal Audit Risk Assessment
GTC240	Group Complaints Procedure
GTC241	Group Average Weight Control Procedure
GTC242	Group Principles of Check Weighing Procedure
GTC243	Legal Temperature Requirements
GTC244	Group Compliance UV Torch Risk Assessment
GTC245	UV Torch Procedure
GTC246	UV Light Hazard Awareness
GTC247	CMS – SEDEX and Labour Provider Remote Verification Audit
GTC248	Cranswick Weight Control Competancy Review
GTC249	BRC Module 11 Audit Template
GTC250	Risk Assessment Decision Matrix
GTC251	Group Process Control Audit Procedure
GTC252	Environmental Verification Audit Template
GTC253	Cranswick CCP Decision Matrix
GTC254	Group RSPO Procedure
GTC254a	RSPO Group Membership Declaration
GTC255	Group RSPO Site Requirements Procedure
GTC255a	Group Monthly Supplier RSPO Certification Check Record
GTC 256	High Fat, Sugar and Salt (HFSS) Legislation overview

GLOSSARY OF TERMS

TERM	DEFINITION
Accreditation	The procedure by which an authoritative body gives formal recognition of the competence of a certification body to provide certification services against a specified standard.
Adulterant	An undeclared material added into a food item or raw material for economic gain.
Adulteration	The addition of an undeclared material into a food item or raw material for economic gain.
Agent	A company that facilitates trade between a site or company and their raw material or packaging suppliers or their customers through the provision of services but does not at any point own or take title to the goods.
Allergen	A known component of food which causes physiological reactions due to an immunological response (e.g. nuts and others identified in legislation relevant to the country of production or sale).
Ambient high care	An ambient area designed to a high standard where practices relating to personnel, ingredients, equipment, packaging and environment aim to minimise potential product contamination by pathogenic micro-organisms.
Announced audit	An audit where the company agrees the scheduled audit day in advance with the certification body.
Annual	Within 12 months since the action was last conducted.
Assured status	Products produced in accordance with a recognised product certification scheme, the status of which needs to be preserved through the certified production facility (e.g. Global G.A.P.).
ATP bioluminescence techniques	A rapid test for cleanliness of surfaces based on ATP (adenosine triphosphate) – a substance used in energy transfer in cells and therefore present in biological material.
Audit	A systematic examination to measure compliance of practices with a predetermined system, and whether the system is implemented effectively and is suitable to achieve objectives, carried out by certified bodies.
Auditor	A person possessing the appropriate competence and skills to carry out an audit.
Authenticity/authentic product	Food authenticity is ensuring that food or raw materials purchased and offered for sale, are of the nature, substance and quality expected.
Batch	The quantity of material prepared or required for one production operation.
Brand owner	The owner of a brand logo or name who places the said logo or name onto retail products.
Branded product	Products bearing the logo, copyright or address of a company that is not a retailer.
Broker	A company which purchases or 'takes title to' products for resale to businesses (e.g. manufacturers, retailers or food service companies) but not to the ultimate consumer.

GLOSSARY OF TERMS

TERM	DEFINITION
Calibration	A set of operations that establish, under specified conditions, the relationship between values of quantities indicated by a measuring instrument or measuring system, or values represented by a material measure or reference material, and the corresponding values realised by standards.
Certificate suspension	Revocation of certification for a given period, pending remedial action on the part of the company.
Certificate withdrawal	Where certification is revoked. Certification may only be regained following successful completion of the full audit process
Certification	The procedure by which an accredited certification body, based on an audit and assessment of a company's competence, provides wrien assurance that a company conforms to a standard's requirements.
Certification body	Provider of certification services, accredited to do so by an authoritative body (e.g. BRC Global Standards, ISO).
Clause	A specific requirement or statement of intent that a site must comply with in order to achieve certification.
Cleaning in place (CIP)	The process of cleaning and sanitising food-processing equipment in its assembled position without the need for dismantling and cleaning the individual parts.
Codex Alimentarius Commission	A body responsible for establishing internationally recognised standards, codes of practice and guidelines, of which HACCP (hazard analysis and critical control points) is one standard.
Competence	Demonstrable ability to apply skill, knowledge and understanding of a task or subject to achieve intended results.
Compliance	Meeting the regulatory or customer requirements concerning product safety, legality and quality.
Consumer	The end-user of the finished product, commodity or service.
Contamination	Introduction or occurrence of an unwanted organism, taint or substance to packaging, food, raw material or the food environment. Contamination includes physical, chemical, radiological, biological and allergen contamination.
Contract packer	A company that packages the final product into consumer packaging.
Contractor or supplier	A person or organisation providing services or materials.
Control	To manage the conditions of an operation to maintain compliance with established criteria, and/or the state wherein correct procedures are being followed and criteria are being met.
Control measure	Any action or activity that can be used to prevent or eliminate a product safety hazard or reduce it to an acceptable level.
Controlled document	A document which is identifiable and for which revisions and removal from use can be tracked. The document is issued to specified individuals and their receipt of the document is recorded.

GLOSSARY OF TERMS

GLOSSARY OF TERMS

TERM	DEFINITION
Cook	A thermal process designed to heat a food item to a minimum of 70°C for 2 minutes or equivalent (see Appendix 3). Alternative cooking processes may be accepted or required where these meet recognised national guidelines and are validated by scientific data.
Correction	Action to eliminate the cause of a detected non-conformity.
Critical control point (CCP)	A step at which control can be applied and is essential to prevent or eliminate a food or product safety hazard or reduce it to an acceptable level.
Cross-contamination/ cross-contact	The transfer of any material from one surface or food to another.
Cross-docking	Material is unloaded at distribution premises, and handled, but not formally put away into storage. This may be a staging area where inbound materials are sorted, consolidated and temporarily stored until the outbound shipment is complete and ready to ship.
Customer	A business or person to whom a service or product has been provided, either as a finished product or as a component part of the finished product.
Customer focus	A structured approach to determining and addressing the needs of an organisation to which the company supplies products and which may be measured by the use of performance indicators.
Despatch/dispatch	The point at which the product leaves the factory site or is no longer the responsibility of the company.
Distribution	The transportation of goods within any container (goods on the move) by road, rail, air or ship.
End-consumer	The ultimate consumer of a foodstuff, who will not use the food as part of any food business operation or activity.
Enclosed product area	An area of the factory where all products are fully enclosed and therefore not vulnerable to environmental contamination.
Flow diagram	A systematic representation of the sequence of steps or operations used in the production or manufacture of a particular food item.
Food defence	Procedures adopted to ensure the safety of raw materials and products from malicious contamination or theft.
Food fraud	Fraudulent and intentional substitution, dilution or addition to a product or raw material, or misrepresentation of the product or material, for the purpose of financial gain, by increasing the apparent value of the product or reducing the cost of its production.
Food handler	Anyone who handles or prepares food, whether open (unwrapped) or packaged.
Food integrity	Products that are of the nature, substance and quality expected (e.g. not substituted, diluted, adulterated or misrepresented).
Food raw materials	Food ingredients, additives and processing aids used in the manufacture of a product.
Food safety	Assurance that food will not cause harm to the consumer when it is prepared and/or eaten according to its intended use.

GLOSSARY OF TERMS

TERM	DEFINITION
Food safety culture	The attitudes, values and/or beliefs which are prevalent at the site, relating to the importance of product safety and the confidence in the product safety systems, processes and procedures used by the site.
Food security	Procedures adopted to ensure the continued availability of raw materials and products.
Fundamental requirement	A requirement of the Standard that relates to a system which must be well established, continuously maintained and monitored by the company as absence or poor adherence to the system will have serious repercussions on the integrity or safety of the product supplied.
Genetically modified organism (GMO)	An organism whose genetic material has been altered by the techniques of genetic modification so that its DNA contains genes not normally found there.
Good hygiene practice	The combination of process, personnel and/or service control procedures intended to ensure that products and/or services consistently achieve appropriate levels of hygiene.
Good manufacturing practice (GMP)	Implemented procedures and practices undertaken using best-practice principles.
Hazard	An agent of any type with the potential to cause harm (usually biological, chemical, physical or radiological).
Hazard Analysis and Critical Control Point (HACCP)	A system that identifies, evaluates and controls hazards which are significant for food safety.
High-care area	An area designed to a high standard where practices relating to personnel, ingredients, equipment, packaging and environment aim to minimise product contamination by pathogenic micro-organisms.
High-care product	A product that requires chilling or freezing during storage, is vulnerable to the growth of pathogens, has received a process to reduce the microbiological contamination to safe levels (typically 1–2 log reduction) and is ready to eat or heat.
High-risk area	A physically segregated area, designed to a high standard of hygiene, where practices relating to personnel, ingredients, equipment, packaging and environment aim to prevent product contamination by pathogenic micro-organisms.
High-risk product	A chilled or frozen ready-to-eat/ready-to-heat product or food where there is a high risk of growth of pathogenic micro-organisms.
Identity preserved	A product which has a defined origin or purity characteristic which needs to be retained throughout the food chain (e.g. through traceability and protection from contamination).
Importer	A company facilitating the movement of products across an international border. Usually the first recipient of the products in that country.
Incident	An event that has occurred that may result in the production or supply of unsafe, illegal or nonconforming products.
Inspection	Targeted verification (often a visual check against a 'tick list' for fabrication, environment and equipment) to ensure operation to safe expected levels.

GLOSSARY OF TERMS

TERM	DEFINITION
Integrity	See food integrity.
Internal audit	General process of audit, for all the activity of the company. Conducted by or on behalf of the company for internal purposes.
Job description	A list of the responsibilities for a given position at a company.
Key staff	Those staff whose activities affect the safety, legality, integrity and quality of the finished product.
Legality	In compliance with the law in the place of production and in the countries where the product(s) is/ are intended to be sold.
Lot	See batch.
Low-risk area	An area where the processing or handling of foods presents minimum risk of product contamination or growth of micro-organisms, or where the subsequent processing or preparation of the product by the consumer will ensure product safety.
Malicious contamination	Deliberate contamination of a product or raw material with the intention to cause harm to the consumer or damage to the company or brand owner.
Manufacturer	A company that produces product from raw materials and/or components and packs the product into retail units or supplies product in bulk to a packing company that packs the product into retail units. A packer that packs product into retail units from bulk-supplied material can also be classed as a manufacturer.
May	Indicates a requirement or text which provides guidance but is not mandatory for compliance with the Standard.
Mitigation strategies	Controls to remove, or reduce to an acceptable level, an identified risk, vulnerability or threat. It is often used in food defence where controls are needed to prevent potential threats from occurring.
Monitoring	A planned sequence of observations or measurements of defined control parameters to assess whether predefined limits are being met.
Non-conformity	The non-fulfilment of a specified product safety, legal or quality requirement or a specified system requirement.
Office Operations	An office within production area where product is checked/tested as per scheduled production check and then re-enter the production flow
Open product area	An area in which product is open to the environment (i.e. not fully enclosed in packaging or within equipment/pipes).
Outer packaging	Packaging which is visible when the product is released from the site. For example, a cardboard box could be considered outer packaging even if wrapped in clear film.
Outsourced processing (subcontracted processing	Outsourced processing is where an intermediate production process or step in the manufacture of a product is completed at another company or site.
Ownership (change of company ownership)	A change of ownership occurs when the title is transferred from one individual or entity to another and results in a change of control of the organisation.
Performance indicators	Summaries of quantified data that provide information on the level of compliance against agreed targets (e.g. customer complaints, product incidents, laboratory data).

GLOSSARY OF TERMS

TERM	DEFINITION
Positive release	Ensuring a product or material is of an acceptable standard prior to release for use.
Potable water	Water that is safe to drink, free from pollutants and harmful organisms, and conforms to local legal requirements.
Premises	A physical building or place owned by the company and audited as part of a site.
Pre-packaged products	Products in their final packaging that is designed for sale to the consumer.
Prepared primary product	A food product which has undergone a washing, trimming, size-grading or quality-grading process and is pre-packed.
Prerequisite	The basic environmental and operational conditions in a food business that are necessary for the production of safe food. These control generic hazards covering good manufacturing and hygiene practices, and shall be considered within the HACCP study.
Preventive action	Action to eliminate the fundamental, underlying cause (root cause) of a detected non-conformity and prevent recurrence.
Primary packaging	The packaging that constitutes the unit of sale to the consumer or customer (e.g. bole, closure and label of a retail pack or a raw material bulk container).
Procedure	Agreed method of carrying out an activity or process which is implemented and documented in the form of detailed instructions or process description (e.g. a flowchart).
Processed food	A food product which has undergone any of the following processes: aseptic filling, baking, battering, blending, bottling, breading, brewing, canning, coating, cooking, curing, cuing, dicing, distillation, drying, extrusion, fermentation, freeze drying, freezing, frying, hot filling, irradiation, microfiltration, microwaving, milling, mixing, being packed in modified atmosphere, being packed in vacuum packing, packing, pasteurisation, pickling, roasting, slicing, smoking, steaming or sterilisation.
Processing aid	Any substance not consumed as a food by itself, intentionally used in the processing of raw materials, foods or their ingredients to fulfil a certain technological purpose during treatment or processing, and which may result in the unintentional but technically unavoidable presence of the residues of the substance or its derivatives in the final product – provided that these residues do not present any health risk and do not have any technological effect on the finished product.
Product recall	Any measures aimed at achieving the return of an unfit product from customers and final consumers.
Product withdrawal	Any measures aimed at achieving the return of out-of-specification or unfit products from business customers, but not from final consumers.
Protective clothing	Clothing designed to protect the product from potential contamination by the wearer.
Provenance	The origin or the source of food or raw materials.
Quality	Meeting the customer's specification and expectation.
Quantity check/mass balance	A reconciliation of the amount of incoming raw material against the amount used in the resulting finished products, which also takes into account process waste and rework.
Quantity control	A check on the amount of product in the pack. May be related to weight, volume, number of pieces, size etc.

GLOSSARY OF TERMS

TERM	DEFINITION
Quarantine	The status given to any material or product set aside while awaiting confirmation of its suitability for its intended use or sale.
Raw material	Any base material or semi-finished material used by the organisation for the manufacture of a product. Raw materials include food ingredients, packaging materials, additives, processing aids etc.
Ready-to-cook food	Food designed by the manufacturer to require cooking or other processing to effectively eliminate, or reduce to an acceptable level, micro-organisms of concern.
Ready-to-eat food	Food designed by the manufacturer to be suitable for direct human consumption without the need for cooking. The heating of the product is intended to make the product more palatable.
Recognised laboratory accreditation	Laboratory accreditation schemes that have gained national and international acceptance, have been awarded by a competent body, and are recognised by government bodies or users of the Standard (e.g. ISO/IEC 17025 or equivalents).
Reference sample	Agreed product or components for referral by the manufacturer for production.
Requirement	Those statements comprising a clause with which compliance will allow sites to be certificated.
Retail brand	A trademark, logo, copyright or address of a retailer.
Retailer	A business selling products to the public by retail.
Retailer-branded products	Products bearing a retailer's logo, copyright, address or ingredients used to manufacture within a retailer's premises. These are products that are legally regarded as the responsibility of the retailer.
Retained production sample	Representative product or components taken from a production run and securely held for future reference.
Risk	The likelihood of occurrence of harm from a hazard.
Risk analysis	A process consisting of three components: risk assessment, risk management and risk communication.
Risk assessment	The identification, evaluation and estimation of the levels of risk involved in a process to determine an appropriate control process.
Root cause(s)	The underlying cause(s) of a problem, which, if adequately addressed, will prevent a recurrence of that problem.
Sampling plan	A documented plan defining the number of samples to be selected, the acceptance or rejection criteria and the statistical confidence of the result.
Satellite depot	A warehouse/distribution site receiving products only from another site within the same company.
Schedule	A tabulated statement giving details of actions and/or timings.
Seasonal production site	A site that is opened specifically to harvest and process a product for the duration of the short term of that harvest (typically 12 weeks or less) during a 12-month cycle.
Secondary packaging	Packaging that is used to collate and transport sales units to the retail environment (e.g. corrugated case).

GLOSSARY OF TERMS

TERM	DEFINITION
Senior management	Those with strategic/high-level operational responsibility for the company and the capability to authorise the financial or human resources necessary for the implementation of the Standard.
Shall	Signifies a requirement to comply with the contents of the clause.
Should	Signifies that compliance with the contents of the clause or requirement is expected or desired.
Site	A unit of a company; the entity which is audited and which is the subject of the audit report and certificate.
Specification	An explicit or detailed description of a material, product or service.
Specifier	A company or person requesting the product or service.
Standard, the	The Cranswick Manufacturing Standard.
Supplier	The person, firm, company or other entity to which a site's purchase order to supply is addressed.
Suspension	Where certification is revoked for a given period, pending remedial action on the part of the company.
Threat assessment	A risk assessment designed to examine site processes for potential product security and food defence issues.
Traceability	Ability to trace and follow raw materials, components and products, through all stages of receipt, production, processing and distribution both forwards and backwards.
Traded goods/products	Goods that are not manufactured or further processed on site but bought from an outside supplier, stored at the site and sold on.
Trend	An identified pattern of results.
Unannounced audit	An audit undertaken on a date unknown to the site in advance.
Utilities	Commodities or services, such as electricity or water, that are provided by a public body.
Validation	Obtaining evidence through the provision of objective evidence that a control or measure, if properly implemented, is capable of delivering the specified outcome.
Vehicle	Any device used for the conveyance of product that is capable of being moved upon highways, waterways or airways. Vehicles can be motorised (e.g. a lorry) or non-motorised (e.g. container or rail truck).
Verification	The application of methods, procedures, tests and other evaluations, in addition to monitoring, to determine whether a control or measure is or has been operating as intended.
Vulnerability assessment	A risk assessment designed to examine processes and supply chains for potential food fraud. BRC Global Standards has developed a guideline to assist sites with vulnerability assessments.

GLOSSARY OF TERMS

TERM	DEFINITION
Where appropriate	In relation to a requirement of the Standard, the company will assess the need for the requirement and, where applicable, put in place systems, processes, procedures or equipment to meet the requirement. The company shall be mindful of legal requirements, best-practice standards, good manufacturing practice and industry guidance, and any other information relating to the manufacture of safe and legal product.
Work in progress/work in process	Partially manufactured products, intermediates or materials waiting for completion of the manufacturing process.
Workwear	Company-issued or authorised clothing designed to protect the product from potential contamination by the wearer.

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